2. Math Toolkit I: Duality, LMIs & the S-Procedure

Lagrangian duality, semidefinite programming, Schur complements, S-lemma, quadratic constraints, dissipativity

BEFORE YOU START

This module assumes:

- Positive definiteness, quadratic forms, matrix square roots, and ellipsoids (Primer A)
- Convex sets and functions, affine functions, and optimality conditions (Primer B)
- Infimum, supremum, argmin, and attainment of an optimum (Primer B)
- Constraints, multipliers, projections, and barrier methods (Primer B)
- KKT conditions, constraint qualifications, and approximate KKT points; gradients and the chain rule (Primer B)
- Block matrices, trace identities, and determinants (Primer A)
- Spectral norms, singular values, and bounds on stretching (Primer A)
- Linear-system stability and quadratic Lyapunov functions (Primer D)
- Transfer functions, frequency response, and signal-energy gains (Primer D)
- Neural-network layers, activations, and their slopes (Primer E)

HOW TO STUDY THIS MODULE

Core reading. Work through one scalar Lagrangian, PSD matrices and the Lyapunov LMI, the definite-corner Schur complement, and one-constraint S-procedure. Then connect the sector inequality to the walkthrough, and read the elementary energy balance in dissipativity.

Practice route. Try the three readiness checks. If a skill is rusty, use its exact review link, then return. Work through Easy P1–P4, Medium P5–P8, and Hard P9–P12; each has a separate hint and a full worked solution. Reattempt a missed problem later without opening the answer.

Optional research reading. The supporting-hyperplane and Fenchel–Moreau detours, interior-point complexity, full KYP equivalence, and dynamic IQC refinements are research extensions. Learn what each certificate proves before returning to how the general theorem is established.

Readiness check — Three prerequisite skills ▼

Try these before opening the answer. The review links lead to earlier material.

1. What is the derivative of $(x-2)^2$? Review gradients.
2. For $P = \mathrm{diag}(2,3)$, what is $x^\top P x$ at $x = (1,-1)^\top$? Review quadratic forms.
3. Does $x = 0$ satisfy the constraint $x - 1 \le 0$ strictly? Review feasibility.

▼ Show readiness answers

The derivative is $2(x-2)$. The quadratic form is $2(1)^2 + 3(-1)^2 = 5$. At $x = 0$, $x - 1 = -1 < 0$, so the constraint is strictly satisfied. These are the operations underneath the first duality and matrix certificates.

Almost every guarantee in this section is produced by the same four moves: *dualise* a constrained problem, write the certificate as a *matrix inequality*, use a *Schur complement* to make a quadratic term linear, and use the *S-procedure* to trade a "for all signals satisfying a constraint" statement for a single semidefinite condition. The Lagrangian methods of Modules 8–9 are the first move applied to a CMDP; LipSDP, the Sandwich layer, LipKernel, the Zames–Falb and Yin–Seiler–Arcak certificates of Modules 12–14 are the other three applied to a neural network. This page builds the tools slowly enough that those derivations can later be read line by line.

NOTATION ON THIS PAGE

Optimisation: decision variable $x \in \mathbb{R}^n$, Lagrange multipliers $\lambda \in \mathbb{R}^m_+$ (inequalities) and ν (equalities). Matrices: $\mathbb{S}^n$ symmetric $n \times n$ matrices, $\mathbb{S}^n_+$ the positive semidefinite (PSD) cone; $M \succeq 0$, $M \succ 0$, $M \preceq 0$, $M \prec 0$ are matrix (Loewner) inequalities, never elementwise. Control: state x_t, input or disturbance w_t, output z_t (see Primer D), storage function V, supply rate $s(w,z)$, ℓ_2 gain γ (see Primer D; in the RL modules and in the CMDP remarks of Section 1 γ is the discount, elsewhere on this page it is a gain). Neural networks: weights W_i, activation φ (see Primer E) slope-restricted in $[\alpha, \beta]$ (this β is unrelated to the GP scaling β_t of Modules 3–6), diagonal multiplier $T = \mathrm{diag}(\lambda_1, \ldots, \lambda_n) \succeq 0$. The clash between λ (Lagrange multiplier) and λ_i (entries of T) is deliberate: Section 4 shows that the entries of T *are* Lagrange multipliers, one per neuron. LipSDP writes $\rho = L^2$ for the squared Lipschitz bound; LipKernel writes $\rho = L$. We follow LipSDP and say so whenever we switch. A few letters are reused locally and always re-introduced: f_i, h_j are objective and constraint functions in Section 1 but f, h are dynamics and output maps in Section 6 (neither is the barrier h of Module 10); A, B, C are generic matrix blocks in Sections 3–4 and system matrices in Section 6; $\rho(A)$ is the spectral radius of A (see Primer A), unrelated to LipSDP's ρ.

1. Lagrangian Duality & KKT

INTUITION — A STARTING EXAMPLE

Start with one price. In $\min (x-2)^2$ subject to $x \leq 1$, the unconstrained best point 2 is excluded, so the answer is $x = 1$ with value 1. Write the constraint as $x - 1 \leq 0$ and attach a price $\lambda \geq 0$: $L = (x-2)^2 + \lambda(x-1)$. At a feasible point the added term is nonpositive, which is why minimizing this relaxed expression gives a *lower* bound. The sign and the order of the minimization/maximization are the main ideas to track; the graded problems compute the entire example.

In plain words. A constrained minimisation can be hard to solve directly. Duality turns it into an easier question. Attach a price $\lambda_i \geq 0$ to each constraint, let the optimiser violate constraints if it pays the price, and see how low the cost can then go. Whatever the prices, this relaxed problem can only do better than the original one, so its value is a *lower bound* on the true optimum (weak duality). The dual problem looks for the prices that give the best lower bound. When that bound is exact (strong duality) and some prices attain it, those best prices, together with a primal optimum, are KKT multipliers in the sense of Primer B (for differentiable problems on an open domain). Exactness alone is not enough: the example $\min x$ s.t. $x^2 \leq 0$ below has zero gap but no finite optimal price.

SYMBOLS IN THIS SECTION

- $p^\star$: the optimal value of the original (primal) problem; f_0, $f_i \leq 0$, $h_j = 0$: objective and constraints, as in Primer B.
- $L(x, \lambda, \nu)$: the Lagrangian, the cost plus price times violation of each constraint; $\lambda_i \geq 0$ prices inequalities, ν_j (any sign) prices equalities.
- $g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$: the dual function, the lowest cost reachable when constraints may be violated at their price.
- $d^\star = \sup_{\lambda \geq 0, \nu} g(\lambda, \nu)$: the best lower bound; $p^\star - d^\star \geq 0$ is the duality gap.
- $\inf$ and $\sup$: greatest lower bound and least upper bound, which need not be attained (Primer 0).

Duality is the art of bounding a hard minimisation from below by an easy maximisation. Nothing in this subsection assumes convexity until strong duality; the first two results hold for *any* problem, which is why they survive the non-convex policy spaces of Module 8.

DEFINITION — PRIMAL PROBLEM, LAGRANGIAN, DUAL FUNCTION

$$p^\star = \inf_{x \in \mathcal{D}} f_0(x) \quad \text{s.t.} \quad f_i(x) \le 0,\ i = 1, \dots, m, \qquad h_j(x) = 0,\ j = 1, \dots, p.$$

Here f_0 is the objective, $\mathcal{D}$ is the set on which all f_i, h_j are defined, and a point $x \in \mathcal{D}$ satisfying every constraint is *feasible* (see Primer B).

The **Lagrangian** attaches a price to each constraint,

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{j=1}^{p} \nu_j h_j(x), \qquad \lambda \in \mathbb{R}^m_+,\ \nu \in \mathbb{R}^p.$$

The **dual function** is its infimum over the domain without the original constraints, $g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu)$ (possibly $-\infty$; see Primer B), and the **dual problem** is $d^\star = \sup_{\lambda \ge 0,\ \nu} g(\lambda, \nu)$.
We define $p^\star$ by an infimum and $d^\star$ by a supremum and write min or max only when an optimiser exists: in the example $\min x$ s.t. $x^2 \le 0$ below, $d^\star = \sup_{\lambda > 0} \big(-1/(4\lambda)\big) = 0$, but no finite λ attains it. An empty feasible set gives $p^\star = +\infty$.

THEOREM — CONCAVITY OF THE DUAL AND WEAK DUALITY

For any primal problem (convex or not): (i) g is concave (see Primer B) on $\mathbb{R}^m \times \mathbb{R}^p$; (ii) $g(\lambda, \nu) \le p^\star$ for every $\lambda \ge 0$ and every ν, hence $d^\star \le p^\star$. The number $p^\star - d^\star \ge 0$ is the **duality gap**. *In words:* every nonnegative price vector yields a lower bound on the optimal cost, and the best bound is found by maximising a concave function, however hard the primal is. The one assumption, $\lambda \ge 0$, is what makes a priced constraint penalise violation instead of rewarding it.

Proof sketch — Both statements in five lines each ▾

(i) Concavity. For fixed x the map $(\lambda, \nu) \mapsto L(x, \lambda, \nu)$ is affine (it is $f_0(x)$ plus a linear function of (λ, ν)). Take $\theta \in [0, 1]$ and two multiplier pairs $\mu = (\lambda, \nu)$, $\mu' = (\lambda', \nu')$. Then

$$g(\theta\mu + (1-\theta)\mu') = \inf_x \big[\theta L(x, \mu) + (1-\theta) L(x, \mu')\big] \ \ge\ \theta \inf_x L(x, \mu) + (1-\theta) \inf_x L(x,$$

because the infimum of a sum is at least the sum of the infima (each term is bounded below by its own infimum). A pointwise infimum of affine functions is concave; no property of f_i was used.

(ii) Weak duality. Let $\tilde{x}$ be primal feasible. Since $\lambda_i \geq 0$ and $f_i(\tilde{x}) \leq 0$ we have $\sum_i \lambda_i f_i(\tilde{x}) \leq 0$, and $h_j(\tilde{x}) = 0$ kills the equality terms, so $L(\tilde{x}, \lambda, \nu) \leq f_0(\tilde{x})$. Therefore

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \;\leq\; L(\tilde{x}, \lambda, \nu) \;\leq\; f_0(\tilde{x}).$$

Taking the infimum over feasible $\tilde{x}$ gives $g(\lambda, \nu) \leq p^\star$; maximising over $\lambda \geq 0, \nu$ gives $d^\star \leq p^\star$. The only place the sign $\lambda \geq 0$ enters is the step $\lambda_i f_i(\tilde{x}) \leq 0$, which is why dual variables of inequalities live on the nonnegative orthant while those of equalities are free. ■

CONNECTION TO MODULE 8

Constrained RL uses the mirror image (maximisation): $\max_\pi J_r(\pi)$ s.t. $J_c(\pi) \leq d$, Lagrangian $J_r(\pi) - \lambda\,(J_c(\pi) - d)$, dual function $D(\lambda) = \max_\pi L(\pi, \lambda) \geq p^\star$, which is *convex* in λ, and dual problem $\min_{\lambda \geq 0} D(\lambda)$. Every inequality above flips, nothing else changes. See Module 8 for the CMDP version and its primal-dual algorithms.

Strong duality and Slater's condition

THEOREM — STRONG DUALITY UNDER SLATER'S CONDITION

Suppose the problem is convex ($f_0, \ldots, f_m$ convex, h_j affine, $\mathcal{D}$ convex) and **Slater's condition** holds: there is a point $\bar{x}$ in the relative interior of $\mathcal{D}$ (see Primer B) with $f_i(\bar{x}) < 0$ for all non-affine f_i (affine inequalities only need $f_i(\bar{x}) \leq 0$) and $h(\bar{x}) = 0$. Then $p^\star = d^\star$, and if $p^\star > -\infty$ the dual optimum is attained by some $(\lambda^\star, \nu^\star)$ (Boyd & Vandenberghe 2004, Sec. 5.2.3).

The *affine hull* of a set is the smallest affine subspace containing it; the *relative interior* is the interior measured inside that subspace. A line segment in $\mathbb{R}^2$ has empty ordinary interior, but every point except its endpoints lies in its relative interior. This matters when $\mathcal{D}$ is lower-dimensional, e.g. the probability simplex $\{x \geq 0,\ \sum_i x_i = 1\}$; the equality constraints themselves never need to hold strictly.

In words: for convex problems with a strictly feasible point, the best lower bound is exact and is achieved by a finite multiplier. *Why each assumption matters:* convexity makes the perturbation function $p^\star(u)$ of the collapsible below convex, so its epigraph has a supporting hyperplane at $u = 0$ (theorem below); strict feasibility of every inequality keeps 0 in the interior of the domain of $p^\star$, which rules out a vertical supporting hyperplane, i.e. an infinite multiplier. The support is then the graph of an affine function $p^\star(0) - (\lambda^\star)^\top u$ that never exceeds $p^\star(u)$: its slope is minus the

optimal multiplier. The refinement that lets affine inequalities hold only non-strictly needs a separate (polyhedral) argument: the constraints $x \le 0, -x \le 0$ satisfy it at $\bar{x} = 0$, yet the domain of their $p^\star$ is $\{u : u_1 + u_2 \ge 0\}$, with 0 on its boundary. Slater is sufficient, not necessary: linear programs have zero gap whenever they are feasible and bounded.

THEOREM — SUPPORTING AND SEPARATING HYPERPLANES (BOYD & VANDENBERGHE 2004)

A hyperplane is $\{x : a^T x = c\}$ with $a \neq 0$. If nonempty convex sets $C, D \subset \mathbb{R}^n$ are disjoint and D is open, there are $a \neq 0, c \in \mathbb{R}$ such that $a^T x \ge c \ge a^T y$ for every $x \in C, y \in D$. Closedness of C is not required.

For a convex function $f : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$, finite somewhere and never $-\infty$, every u_0 in the interior of $\{u : f(u) < +\infty\}$ admits $b \in \mathbb{R}^m$ with $f(u) \ge f(u_0) + b^T(u - u_0)$ for all u. The relative-interior version measures directions inside the affine hull of that domain.

Intuition. An affine lower bound touches the graph at the point in question. Convexity keeps the graph above it; an interior point prevents a vertical support. Separation is the corresponding statement for two sets, and will be used again in the S-lemma proof.

Boyd & Vandenberghe 2004, Sections 2.5 and 5.3

Going deeper — The geometry of the duality gap (perturbation function) ▾

Drop the equality constraints for readability and define the **perturbation function** $p^\star(u) = \inf\{f_0(x) : f_i(x) \le u_i,\ i = 1..m\}$, so $p^\star(0) = p^\star$. It is nonincreasing in u (looser constraints, smaller minimum). The dual function is a Legendre-type transform (see Primer B) of it:

$$g(\lambda) = \inf_u \left[p^\star(u) + \lambda^\top u\right] \qquad (\lambda \ge 0).$$

Proof. ($\ge$) For any $x \in \mathcal{D}$ put $u = f(x)$; then x is feasible for the u-perturbed problem, so $f_0(x) \ge p^\star(f(x))$ and $L(x, \lambda) \ge p^\star(u) + \lambda^\top u \ge \inf_u[\cdot]$. Take the infimum over x. ($\le$) Fix u and any x with $f(x) \le u$; because $\lambda \ge 0$, $L(x, \lambda) \le f_0(x) + \lambda^\top u$; take the infimum over such x to get $g(\lambda) \le p^\star(u) + \lambda^\top u$, then over u. ■

So $g(\lambda)$ is the largest intercept c such that the affine function $u \mapsto c - \lambda^\top u$ lies below the graph of $p^\star$ everywhere; $d^\star$ is the best such intercept over all slopes $-\lambda \le 0$. Because $p^\star$ is nonincreasing (and the primal is feasible), affine minorants with other slopes add nothing, so $d^\star$ is the value at $u = 0$ of the *closed convex envelope* of $p^\star$, the supremum of all its affine minorants.

Zero duality gap means the graph of $p^\star$ meets this envelope at $u = 0$. A finite optimal multiplier is a further requirement: given zero gap, it exists iff a non-vertical supporting line touches the graph at $u = 0$; its slope is then $-\lambda^\star$, which gives the sensitivity interpretation $\lambda_i^\star = -\partial p^\star(0)/\partial u_i$ when $p^\star$ is differentiable. The two can come apart: for $\min x$ s.t. $x^2 \le 0$ we have $p^\star(u) = -\sqrt{u}$ for $u \ge 0$ and $p^\star = d^\star = 0$, but $g(\lambda) = -1/(4\lambda)$ only tends to 0 as $\lambda \to \infty$, because the tangent at $u = 0$ is vertical (exactly what Slater's condition rules out). For a non-convex primal the graph can have a dent at $u = 0$; the envelope then passes strictly below and the gap $p^\star - d^\star$ is the depth of the dent. A finite dual optimum can still exist there, but its line supports the envelope at $(0, d^\star)$, below the graph, which is why the touching-line test above presumes zero gap: for $\min x$ over $x \in \{-1, 1\}$ s.t. $-x \le 0$ we have $p^\star = 1$ and $g(\lambda) = -|1 - \lambda|$, so $d^\star = 0$ is attained at $\lambda^\star = 1$ despite a gap of 1. Dual ascent maximises g, which only sees the envelope, so it targets the multiplier of the convexified problem (and converges to it under the step-size conditions above, when it exists); but the primal minimisers $x(\lambda_k)$ need not converge and can oscillate between minimisers on either side of the dent (for one constraint, typically the two points whose mixture the envelope represents), none of which is both feasible and optimal. This "primal recovery" failure motivates the state-augmentation construction discussed in Module 8.

The $x^2 \le 0$ example in detail. For $\lambda > 0$ complete the square: $x + \lambda x^2 = \lambda\big(x + \frac{1}{2\lambda}\big)^2 - \frac{1}{4\lambda}$, so $g(\lambda) = -1/(4\lambda)$; at $\lambda = 0$, $g(0) = \inf_x x = -\infty$. The feasible set is $\{0\}$, so $p^\star = 0 = \sup_{\lambda \ge 0} g(\lambda) = d^\star$, but no finite λ attains the supremum.

> **THEOREM — FENCHEL–MOREAU, FINITE-DIMENSIONAL FORM**
>
> Let $f : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$ be *proper* (finite somewhere), convex and *lower semicontinuous*, i.e. its epigraph $\{(u, t) : t \ge f(u)\}$ is closed. Define the conjugate $f^*(b) = \sup_u(b^\top u - f(u))$ and the biconjugate $f^{**}(u) = \sup_b(b^\top u - f^*(b))$, the supremum of all affine lower bounds of f. Then $f = f^{**}$. For a proper f that has at least one affine lower bound, f^{**} is its *closed convex envelope*, the greatest lower-semicontinuous convex function below f. For a concave, upper-semicontinuous P apply this to $-P$: affine upper bounds recover P.
>
> *Intuition.* Each slope gives the best affine lower bound with that slope; their supremum fills non-convex dents from below, and convexity plus closedness mean there is no dent to fill. Here $g(\lambda) = \inf_u[p^\star(u) + \lambda^\top u] = -(p^\star)^*(-\lambda)$, so $d^\star$ is the closed convex envelope of $p^\star$ evaluated at $u = 0$ (Boyd & Vandenberghe 2004, Sec. 3.3.2 and Ch. 5).

THEOREM — PRIMAL RECOVERY FROM A SADDLE POINT

For the primal problem of this section, suppose $x^\star$ is feasible, $\lambda^\star \geq 0$, $x^\star$ *globally* minimises $L(\cdot, \lambda^\star, \nu^\star)$ over $\mathcal{D}$, and $\lambda_i^\star f_i(x^\star) = 0$ for every i. Then $x^\star$ is primal optimal, $(\lambda^\star, \nu^\star)$ is dual optimal, and the gap is zero. No convexity is needed: $g(\lambda^\star, \nu^\star) = L(x^\star, \lambda^\star, \nu^\star) = f_0(x^\star)$ because the constraint terms vanish, and weak duality $g(\lambda^\star, \nu^\star) \leq d^\star \leq p^\star \leq f_0(x^\star)$ forces equality throughout (Boyd & Vandenberghe 2004, Sec. 5.4–5.5).

Intuition. A multiplier alone is not a solution: a Lagrangian minimiser counts as a recovered primal solution only after feasibility and complementary slackness are checked. In the $\{-1, 1\}$ example both minimisers at $\lambda^\star = 1$ fail ($x = -1$ is infeasible, $x = 1$ violates $\lambda^\star f_1(x) = 0$), as they must when there is a gap. Letting a policy depend on extra state, such as the current multiplier, is a different construction with its own assumptions and guarantees; Module 8 develops it.

The one non-convex problem that matters most here has *no* dent: Paternain et al., NeurIPS 2019 prove that a CMDP with bounded rewards and a Slater point has zero duality gap, because its perturbation function $P(\xi) = \max_\pi V_0(\pi)$ s.t. $V_i(\pi) \geq c_i + \xi_i$ is concave: the set of occupancy measures is convex (see Primer E), so any two feasible policies can be mixed at the level of occupancy measures, and the value is linear in the occupancy measure. Fenchel–Moreau then gives $P^\star = D^\star$ (their Theorem 1). This holds over the class of *all* policies; for an ϵ-universal parameterised policy class their Theorem 2 only bounds how far the parameterised dual value $D_\theta^\star$ falls below the unrestricted optimum, $0 \leq P^\star - D_\theta^\star \leq (B_{r_0} + \|\lambda_\epsilon^\star\|_1 B_r)\, \Delta_\epsilon$, a term linear in ϵ. Here B_{r_i} bound the rewards, $B_r = \max_{i \geq 1} B_{r_i}$, $\lambda_\epsilon^\star$ is the optimal multiplier of the unrestricted problem with every constraint tightened by $B_r \Delta_\epsilon$, and $\Delta_\epsilon = \epsilon/(1-\gamma)^2$ for the paper's unnormalised values (the paper prints $\epsilon/(1-\gamma)$, the right scale for returns normalised by $1-\gamma$). The parameterised problem's own duality gap $D_\theta^\star - P_\theta^\star$ is not bounded directly. This is the regime of Module 8, which derives the corrected constant.

Why occupancy measures form a convex set. For a finite discounted MDP with initial distribution μ and transition probabilities $P(s \mid s', a')$, let $d(s,a) = (1-\gamma) \sum_{t \geq 0} \gamma^t \Pr(s_t = s, a_t = a)$. These nonnegative numbers satisfy the linear flow equations $\sum_a d(s,a) = (1-\gamma)\mu(s) + \gamma \sum_{s',a'} P(s \mid s', a')\, d(s', a')$, so a convex combination of two occupancy measures satisfies them too; conversely, every nonnegative solution is the occupancy measure of the stationary policy $\pi(a \mid s) = d(s,a) / \sum_b d(s,b)$ (any action where $\sum_b d(s,b) = 0$). Values are linear in d: $V(\pi) = (1-\gamma)^{-1} \sum_{s,a} d(s,a) r(s,a)$. Mixing two policies at the level of d therefore mixes their values linearly.

THEOREM — DISCOUNTED CMDP DUALITY AND APPROXIMATION (FINITE MDP)

Take finitely many states and actions, a fixed transition law and initial distribution, $0 \le \gamma < 1$, rewards $|r_i| \le B_{r_i}$ and unnormalised values $V_i(\pi) = \mathbb{E}_\pi \sum_{t\ge 0} \gamma^t r_i(s_t, a_t)$, over all stationary randomised policies. (i) If some policy has $V_i(\pi) > c_i$ for every constraint $i = 1, \dots, m$, then $P^\star = \max\{V_0(\pi) : V_i(\pi) \ge c_i \ \forall i\}$ equals $D^\star = \min_{\lambda \ge 0} \max_\pi \left[V_0(\pi) + \sum_i \lambda_i (V_i(\pi) - c_i)\right]$, with both optima attained. (ii) Suppose moreover that the class $\{\pi_\theta\}$ is *ϵ-universal*: for every stationary π some π_θ has $\sup_s \sum_a |\pi(a \mid s) - \pi_\theta(a \mid s)| \le \epsilon$. Put $\Delta_\epsilon = \epsilon/(1-\gamma)^2$ and $B_r = \max_{i \ge 1} B_{r_i}$, and assume the problem with tightened thresholds $c_i + B_r \Delta_\epsilon$ is feasible with optimal multiplier $\lambda_\epsilon^\star$. Then $D_\theta^\star = \inf_{\lambda \ge 0} \sup_\theta \left[V_0(\pi_\theta) + \sum_i \lambda_i (V_i(\pi_\theta) - c_i)\right]$ satisfies

$$0 \le P^\star - D_\theta^\star \le \big(B_{r_0} + \|\lambda_\epsilon^\star\|_1 B_r\big)\Delta_\epsilon.$$

In words: occupancy measures turn the unrestricted problem into a linear program, so there is no dent; restricting to the parameterised class costs at most a term linear in ϵ. Part (ii) does not bound the parameterised problem's own gap $D_\theta^\star - P_\theta^\star$, and the approximation must hold uniformly over all states (Paternain et al. 2019, Theorems 1–2, specialised to finite MDPs with the unnormalised constant).

Proof of (ii). Write $\delta = B_r \Delta_\epsilon$, $D(\lambda) = \max_\pi \left[V_0(\pi) + \lambda^\top (V(\pi) - c)\right]$ and P_δ for the tightened optimum. The tightened problem has dual function $D(\lambda) - \delta \sum_i \lambda_i$, and it is again a linear program in d, so its gap is zero: $P_\delta = D(\lambda_\epsilon^\star) - \delta \|\lambda_\epsilon^\star\|_1 \ge P^\star - \delta \|\lambda_\epsilon^\star\|_1$, using weak duality $D(\lambda) \ge P^\star$. Now approximate a tightened optimal policy by some π_θ. By the return-error bound below, each value moves by at most $B_{r_i}\Delta_\epsilon$, so π_θ still satisfies $V_i \ge c_i$ and has $V_0(\pi_\theta) \ge P_\delta - B_{r_0}\Delta_\epsilon$. Because this π_θ is feasible, every $\lambda \ge 0$ gives $\sup_\theta[\cdots] \ge V_0(\pi_\theta)$, hence $D_\theta^\star \ge P_\delta - B_{r_0}\Delta_\epsilon$; restricting the inner supremum to the class gives $D_\theta^\star \le D^\star = P^\star$. Combining the three inequalities gives the display. ■

Where $\epsilon/(1-\gamma)^2$ comes from. Run π and π_θ from the same initial distribution and let e_t be the ℓ_1 distance between their state distributions at time t, so $e_0 = 0$. Adding and subtracting the joint law that pairs the first state distribution with the second policy shows that the state-action distributions differ by at most $e_t + \epsilon$. A transition kernel cannot increase ℓ_1 distance: if P_{ji} is the probability of moving to state j from state-action pair i, then $\sum_j |\sum_i P_{ji} a_i| \le \sum_i |a_i| \sum_j P_{ji} = \sum_i |a_i|$. Hence $e_{t+1} \le e_t + \epsilon$, and by induction the time-t state-action error is at most $(t+1)\epsilon$. With $|r| \le B_r$ the value error is at most $B_r \epsilon \sum_{t\ge 0}(t+1)\gamma^t = B_r \epsilon/(1-\gamma)^2$, because $\left(\sum_{k\ge 0}\gamma^k\right)^2$

contains each power γ^t exactly $t+1$ times. For instance, $\gamma = 0.9$, $B_r = 1$, $\epsilon = 0.01$ already allow a value error of $0.01/0.1^2 = 1$.

KKT conditions

DEFINITION — KARUSH–KUHN–TUCKER CONDITIONS

First time? Primer B builds these conditions up from a picture, with worked examples, constraint qualifications and approximate KKT points.

For differentiable f_i, h_j (see Primer B) on an open domain $\mathcal{D}$ (e.g. $\mathcal{D} = \mathbb{R}^n$; Boyd & Vandenberghe Sec. 5.5.3 assume open domains), a triple $(x^\star, \lambda^\star, \nu^\star)$ is a KKT point if

$$\text{primal feasibility:} \quad x^\star \in \mathcal{D}, \;\; f_i(x^\star) \le 0, \;\; h_j(x^\star) = 0; \qquad \text{dual feasibility:} \quad \lambda^\star \geq$$
$$\text{complementary slackness:} \quad \lambda_i^\star f_i(x^\star) = 0 \;\; \forall i;$$
$$\text{stationarity:} \quad \nabla f_0(x^\star) + \sum_i \lambda_i^\star \nabla f_i(x^\star) + \sum_j \nu_j^\star \nabla h_j(x^\star) = 0.$$

Theorem. (a) If strong duality holds and $x^\star$, $(\lambda^\star, \nu^\star)$ are primal and dual optimal, they form a KKT point. (b) For a convex problem, every KKT point is primal-dual optimal with zero gap. *In words:* at a primal-dual optimal pair with zero gap, nonnegative prices on the active constraints balance the gradient of the objective; zero gap together with an attained dual optimum (both supplied by Slater's condition when $p^\star$ is finite) is what makes such prices exist (necessity), and convexity is what makes a local balance a global certificate (sufficiency). Zero gap alone is not enough: for $\min x$ s.t. $x^2 \le 0$ (the vertical-tangent example above) the gap is zero, yet stationarity $1 + 2\lambda x^\star = 1 \neq 0$ fails for every λ. If $\mathcal{D}$ is instead a closed convex set (say $x \geq 0$), stationarity must include the boundary of $\mathcal{D}$: $0 \in \nabla_x L(x^\star, \lambda^\star, \nu^\star) + N_{\mathcal{D}}(x^\star)$ with $N_{\mathcal{D}}$ the normal cone (see Primer B). Example: $\min x$ over $\mathcal{D} = [0, \infty)$ has zero gap and $x^\star = 0$, yet $f_0'(0) = 1 \neq 0$.
Here $N_{\mathcal{D}}(x) = \{v : v^\top (y - x) \le 0 \text{ for every } y \in \mathcal{D}\}$ is the set of outward normal vectors of $\mathcal{D}$ at x (only $v = 0$ at interior points, which recovers ordinary stationarity), and $0 \in a + N$ means $a + v = 0$ for some $v \in N$. In the example $N_{[0,\infty)}(0) = (-\infty, 0]$, so the gradient 1 is balanced by the normal $v = -1$.

Proof sketch — Why KKT is necessary under strong duality with attained optima and sufficient under convexity ▾

(a) Strong duality gives the chain

$$f_0(x^\star) = g(\lambda^\star, \nu^\star) = \inf_x L(x, \lambda^\star, \nu^\star) \le L(x^\star, \lambda^\star, \nu^\star) = f_0(x^\star) + \sum_i \lambda_i^\star f_i(x^\star) \le f_0(x^\star$$

where the last step uses $\lambda_i^\star \ge 0$, $f_i(x^\star) \le 0$, $h_j(x^\star) = 0$. The two ends are equal, so every inequality is an equality: $x^\star$ minimises $L(\cdot, \lambda^\star, \nu^\star)$ over the open set $\mathcal{D}$ (hence its gradient vanishes there: stationarity), and $\sum_i \lambda_i^\star f_i(x^\star) = 0$ with every term ≤ 0 (hence each term is zero: complementary slackness).

(b) If the problem is convex, $L(\cdot, \lambda^\star, \nu^\star)$ is convex in x and stationarity says $x^\star$ is its global minimiser, so $g(\lambda^\star, \nu^\star) = L(x^\star, \lambda^\star, \nu^\star) = f_0(x^\star)$ by complementary slackness. Weak duality closes the sandwich: $p^\star \ge d^\star \ge g(\lambda^\star, \nu^\star) = f_0(x^\star) \ge p^\star$. ■

Complementary slackness is the statement students most often misread: a multiplier can only be positive on an *active* constraint, and an inactive constraint has zero price. In the dual-ascent dynamics below this is exactly the "multiplier decays to zero while the constraint is slack" behaviour.

Dual ascent: what Lagrangian safe RL actually does

ALGORITHM IDEA — DUAL ASCENT IS PROJECTED GRADIENT ASCENT ON G

Let $x(\lambda) \in \arg\min_x L(x, \lambda)$ (inequality constraints only, for brevity). Then $f(x(\lambda)) = (f_1(x(\lambda)), \ldots, f_m(x(\lambda)))$ is a *supergradient* of g at λ:

$$g(\lambda') = \inf_x L(x, \lambda') \leq L(x(\lambda), \lambda') = f_0(x(\lambda)) + \lambda'^\top f(x(\lambda)) = g(\lambda) + (\lambda' - \lambda)^\top f(x(\lambda))$$

Geometrically, the affine function $\lambda' \mapsto g(\lambda) + (\lambda' - \lambda)^\top f(x(\lambda))$ lies above the concave g everywhere and touches it at λ, just as a tangent plane does when g is differentiable. If moreover x ranges over a compact set on which the f_i are continuous and the minimiser $x(\lambda)$ is unique, g is differentiable at λ with $\nabla g(\lambda) = f(x(\lambda))$ (Danskin's theorem). Uniqueness alone is not enough on an unbounded domain: for $f_0(x) = x^2 + x^4$, $f_1(x) = -x^4$ the minimiser at $\lambda = 1$ is unique ($x = 0$), yet $g(\lambda) = 0$ for $\lambda \leq 1$ and $g(\lambda) = -\infty$ for $\lambda > 1$. Projected supergradient ascent on the concave g is

$$\lambda_{k+1} = \left[\lambda_k + \eta\, f(x(\lambda_k))\right]_+.$$

With a constant step η and bounded supergradients, subgradient-method theory only guarantees that the best dual value found comes within $O(\eta)$ (see Primer 0) of $d^\star$; convergence of λ_k to a dual optimum (when one exists) needs diminishing steps, e.g. $\sum_k \eta_k = \infty$ and $\sum_k \eta_k^2 < \infty$.
For a vector v, $[v]_+$ means $([v]_+)_i = \max\{v_i, 0\}$, the Euclidean projection onto the nonnegative orthant. The supergradient formula assumes an attained, exact inner minimum. In reward-maximising RL, take gradient *ascent* steps on $J_r - \lambda J_c$, while keeping the update $\lambda \leftarrow [\lambda + \eta(J_c - d)]_+$. Approximate policy updates do not automatically inherit the exact dual-ascent convergence theorem.

Here $O(\eta)$ means an error bounded by $C\eta$ for sufficiently small η, with C independent of η. Later, $O(n^3)$ describes work bounded by a constant times n^3 as the matrix size grows. These statements suppress constants; they do not specify an exact error or runtime.

Algorithm 1: Dual ascent (the template behind every Lagrangian safe-RL method)

1. Initialise $\lambda_0 \geq 0$, step size $\eta > 0$.
2. for $k = 0, 1, 2, \ldots$:
3. primal step: $x_k \leftarrow \arg\min_x L(x, \lambda_k)$ *// in RL: a few policy-gradient steps (see Primer E) on $J_r - \lambda_k J_c$ (RCPO, PPO-Lagrangian)*

4. dual step: $\lambda_{k+1} \leftarrow [\lambda_k + \eta\, f(x_k)]_+$ *// in RL: $f(x_k) = J_c(\pi_k) - d$, projection keeps $\lambda \geq 0$*
5. return (x_k, λ_k).

Read the dual step as a controller: the multiplier *integrates* the constraint violation (see Primer D), growing while $J_c > d$ and shrinking (until it hits zero) while the constraint is slack. This is precisely the update of Tessler et al., ICLR 2019 (RCPO, with λ on the slowest timescale) and the reason Stooke et al., ICML 2020 could add proportional and derivative terms to it. Replacing the exact inner minimisation by one or two gradient steps turns dual ascent into gradient descent–ascent, whose oscillations and overshoot are the subject of Module 8. The same dualisation solves the trust-region subproblem of CPO through its low-dimensional dual (Module 9) and the CBF-QP safety filter in closed form via KKT (Module 10).

Worked example — A two-variable QP solved via KKT, its dual and the shadow price ▾

Problem (a quadratic program, see Primer B). $\min_{x \in \mathbb{R}^2} (x_1 - 2)^2 + (x_2 - 1)^2$ subject to $x_1 + x_2 \leq 1$. Convex, and $x = 0$ is strictly feasible, so strong duality holds.

KKT. With $f_1(x) = x_1 + x_2 - 1$, stationarity reads $2(x_1 - 2) + \lambda = 0$, $2(x_2 - 1) + \lambda = 0$, i.e. $x_1 = 2 - \lambda/2$, $x_2 = 1 - \lambda/2$. Case $\lambda = 0$: $x = (2, 1)$, but $f_1 = 2 > 0$, infeasible. Case $\lambda > 0$: complementary slackness forces $x_1 + x_2 = 1$, i.e. $3 - \lambda = 1$, so $\lambda^\star = 2$ and $x^\star = (1, 0)$ with $p^\star = 1 + 1 = 2$.

Dual function. Minimising the Lagrangian in closed form (same stationarity equations) gives

$$g(\lambda) = \tfrac{\lambda^2}{4} + \tfrac{\lambda^2}{4} + \lambda\big((2 - \tfrac{\lambda}{2}) + (1 - \tfrac{\lambda}{2}) - 1\big) = 2\lambda - \tfrac{\lambda^2}{2},$$

a concave parabola maximised at $\lambda = 2$ with $d^\star = 4 - 2 = 2 = p^\star$: zero gap, as promised.

Shadow price. Perturb the constraint to $x_1 + x_2 \leq 1 + u$. The optimum is the squared distance from $(2, 1)$ to the line $x_1 + x_2 = 1 + u$, $p^\star(u) = (2 - u)^2/2$ for $u \leq 2$, so $-\mathrm{d}p^\star/\mathrm{d}u\,|_{u=0} = 2 = \lambda^\star$. Relaxing the constraint by one unit is worth two units of objective, at the margin.

Worked example — computing a dual function by hand ▾

Take the problem from Primer B's KKT examples: minimise $f_0(x) = (x_1 - 2)^2 + (x_2 - 1)^2$ subject to $f_1(x) = x_1 + x_2 - 1 \leq 0$. Its optimum is $x^\star = (1, 0)$ with $p^\star = 2$.

Step 1: the Lagrangian. $L(x, \lambda) = (x_1 - 2)^2 + (x_2 - 1)^2 + \lambda(x_1 + x_2 - 1)$.

Step 2: minimise over x, ignoring the constraint. Setting the gradient to zero, $2(x_1 - 2) + \lambda = 0$ and $2(x_2 - 1) + \lambda = 0$, gives $x_1 = 2 - \lambda/2$ and $x_2 = 1 - \lambda/2$.

Step 3: substitute back.

$$g(\lambda) = \frac{\lambda^2}{4} + \frac{\lambda^2}{4} + \lambda\,(3 - \lambda - 1) = 2\lambda - \frac{\lambda^2}{2}.$$

Step 4: read off the bounds. Every price gives a lower bound on $p^\star = 2$: $g(0) = 0$, $g(1) = 1.5$, $g(3) = 1.5$. The best price maximises g: $g'(\lambda) = 2 - \lambda = 0$ gives $\lambda^\star = 2$ and $d^\star = g(2) = 2 = p^\star$. The gap is zero, as the strong duality theorem promises for this convex problem with a strictly feasible point (for example $x = (0, 0)$). The best price $\lambda^\star = 2$ is exactly the KKT multiplier found in Primer B.

Worked example — a duality gap from a yes/no decision ▾

Duality gaps appear when the problem is not convex. Let the decision be yes or no, $x \in \mathcal{D} = \{0, 1\}$, and minimise $f_0(x) = x$ subject to $f_1(x) = 0.5 - x \le 0$. Only $x = 1$ is feasible, so $p^\star = 1$.

The dual function minimises $L(x, \lambda) = x + \lambda(0.5 - x)$ over the two points of $\mathcal{D}$:

$$g(\lambda) = \min\big\{L(0, \lambda),\, L(1, \lambda)\big\} = \min\big\{0.5\lambda,\; 1 - 0.5\lambda\big\}.$$

The first term rises with λ and the second falls, so the best price is where they cross, $\lambda = 1$, with $d^\star = 0.5$. The gap is $p^\star - d^\star = 0.5$. The dual answers the question as if $x = 0.5$ were allowed: it only sees the convex relaxation $x \in [0, 1]$ of the yes/no choice. The same mechanism gives a gap when constrained RL optimises over a restricted, non-convex policy class (Module 8).

2. LMIs & Semidefinite Programs

In plain words. A symmetric matrix M is positive semidefinite, $M \succeq 0$, if the quadratic form $x^\top M x$ is never negative: picture a bowl that does not dip below zero in any direction. A linear matrix inequality (LMI) asks for unknown numbers that make a matrix, built linearly from those numbers, positive semidefinite. That sounds exotic, but stability, gain and Lipschitz certificates all take this form. And the set of solutions is always convex, so computers can search it reliably.

SYMBOLS IN THIS SECTION

- $\mathbb{S}^n$: symmetric $n \times n$ matrices. $M \succeq 0$: positive semidefinite (all eigenvalues ≥ 0); $M \succ 0$: positive definite (all > 0); $M \preceq 0$ means $-M \succeq 0$.
- $F(x) = F_0 + x_1 F_1 + \cdots + x_m F_m$: a symmetric matrix that depends affinely on the unknowns $x \in \mathbb{R}^m$; the constraint $F(x) \succeq 0$ is an LMI.
- SDP (semidefinite program): minimise a linear function of x subject to LMIs.
- P: a matrix unknown, as in a Lyapunov certificate $V(x) = x^\top P x$; $\rho(A)$: the spectral radius, the largest absolute value of an eigenvalue of A.

A linear matrix inequality is a constraint on a vector of unknowns that says "this symmetric matrix, which depends affinely on the unknowns, is positive semidefinite". Its power comes from a coincidence: the PSD cone is convex, yet it can express Lyapunov, gain, robustness and Lipschitz conditions that look hopelessly nonlinear when written out entrywise.

DEFINITION — THE PSD CONE AND THE LOEWNER ORDER

For $M \in \mathbb{S}^n$ the following are equivalent and define $M \succeq 0$: (i) $x^\top M x \geq 0$ for all $x \in \mathbb{R}^n$; (ii) all eigenvalues of M are ≥ 0; (iii) $M = R^\top R$ for some matrix R. Replacing $\geq$ by $>$ (for $x \neq 0$) defines $M \succ 0$; $M \preceq 0$ means $-M \succeq 0$; $A \succeq B$ means $A - B \succeq 0$. Two facts carry the whole page:

- **Cone.** $\mathbb{S}^n_+$ is a closed convex cone: $\theta A + \mu B \succeq 0$ for $A, B \succeq 0$ and $\theta, \mu \geq 0$, since $x^\top(\theta A + \mu B)x = \theta x^\top A x + \mu x^\top B x \geq 0$.
- **Congruence.** For any matrix R of compatible size, $M \succeq 0 \Rightarrow R^\top M R \succeq 0$, because $x^\top R^\top M R x = (Rx)^\top M (Rx)$. If R is square and invertible the implication is an equivalence (Rx ranges over all vectors).

DEFINITION — LINEAR MATRIX INEQUALITY AND SEMIDEFINITE PROGRAM

Given $F_0, F_1, \dots, F_m \in \mathbb{S}^n$, the constraint

$$F(x) = F_0 + \sum_{i=1}^{m} x_i F_i \;\succeq\; 0 \qquad (x \in \mathbb{R}^m)$$

is a **linear matrix inequality** (LMI) in x; $F(x) \succ 0$ is a strict LMI. A matrix unknown $P \in \mathbb{S}^n$ is covered by expanding $P = \sum_k p_k E_k$ in a basis of $\mathbb{S}^n$, so "$A^\top P A - P \prec 0$ is an LMI in P" is a statement of this form with $m = n(n+1)/2$. A **semidefinite program** minimises a linear objective over LMI constraints:

$$\min_x \; c^\top x \quad \text{s.t.} \quad F(x) \succeq 0.$$

Several LMIs are one LMI (block-diagonal matrices: Primer A): $F^{(1)}(x) \succeq 0, \dots, F^{(k)}(x) \succeq 0$ iff $\operatorname{blkdiag}\big(F^{(1)}(x), \dots, F^{(k)}(x)\big) \succeq 0$, and a linear inequality $a^\top x \le b$ is a 1×1 LMI, so LPs are SDPs with diagonal F.

For example, a symmetric 2×2 unknown $P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix} = p_1 E_1 + p_2 E_2 + p_3 E_3$ with $E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $E_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ has three scalar unknowns, and for fixed A, $P - A^\top P A = \sum_k p_k (E_k - A^\top E_k A)$. So $A^\top P A - P \prec 0$ is the strict LMI $F(p) \succ 0$ with $F_0 = 0$ and $F_k = E_k - A^\top E_k A$. In size n there are n diagonal and $n(n-1)/2$ off-diagonal unknowns, $n(n+1)/2$ in total.

THEOREM — THE FEASIBLE SET OF AN LMI IS CONVEX

If $F(x) \succeq 0$ and $F(y) \succeq 0$ then $F(\theta x + (1-\theta)y) \succeq 0$ for all $\theta \in [0,1]$. *Proof.* F is affine, so $F(\theta x + (1-\theta)y) = \theta F(x) + (1-\theta) F(y)$, a nonnegative combination of two PSD matrices, which is PSD by the cone property. ■ Hence an SDP is a convex problem: a linear objective over a convex set. Strong duality (Section 1) applies to SDPs under Slater's condition ($F(\bar{x}) \succ 0$ for some $\bar{x}$), and interior-point solvers return primal and dual certificates together.

Going deeper — The dual of an SDP, and what a dual certificate proves ▾

Dualise $\min_x c^\top x$ s.t. $F(x) \succeq 0$ exactly as in Section 1, but with a *matrix* multiplier $Z \in \mathbb{S}^n$ and the pairing $\operatorname{tr}(ZF)$: $L(x, Z) = c^\top x - \operatorname{tr}\big(Z F(x)\big) = \sum_i x_i \big(c_i - \operatorname{tr}(F_i Z)\big) - \operatorname{tr}(F_0 Z)$. This is affine in x, so its infimum is $-\infty$ unless every coefficient vanishes, and the dual problem is

$$d^\star = \max_{Z \succeq 0} \; -\operatorname{tr}(F_0 Z) \quad \text{s.t.} \quad \operatorname{tr}(F_i Z) = c_i, \;\; i = 1, \dots, m.$$

Weak duality in one line. For primal-feasible x and dual-feasible Z: $c^\top x + \operatorname{tr}(F_0 Z) = \sum_i x_i \operatorname{tr}(F_i Z) + \operatorname{tr}(F_0 Z) = \operatorname{tr}\left(F(x) Z\right) = \operatorname{tr}\left(Z^{1/2} F(x) Z^{1/2}\right) \geq 0$. The condition $Z \succeq 0$ plays the role of $\lambda \geq 0$: the PSD cone is self-dual ($\operatorname{tr}(FZ) \geq 0$ for every $F \succeq 0$ iff $Z \succeq 0$). If the primal is strictly feasible and $p^\star$ is finite, then $p^\star = d^\star$ and the dual optimum is attained (Slater for generalized inequalities, Boyd & Vandenberghe Sec. 5.9, Example 5.11), and an optimal pair satisfies $\operatorname{tr}\left(F(x^\star) Z^\star\right) = 0$, i.e. $F(x^\star) Z^\star = 0$: complementary slackness as a matrix product.

The two missing steps. Self-duality, converse direction: if $\operatorname{tr}(FZ) \geq 0$ for every $F \succeq 0$, test with $F = vv^\top$ to get $\operatorname{tr}(Zvv^\top) = v^\top Z v \geq 0$ for every v, i.e. $Z \succeq 0$. Complementary slackness: for $F, Z \succeq 0$, cyclicity of the trace gives $\operatorname{tr}(FZ) = \operatorname{tr}\left((F^{1/2} Z^{1/2})(F^{1/2} Z^{1/2})^\top\right) = \|F^{1/2} Z^{1/2}\|_F^2$, the sum of squared entries (Frobenius norm). If this trace is zero then $F^{1/2} Z^{1/2} = 0$, hence $FZ = F^{1/2}(F^{1/2} Z^{1/2}) Z^{1/2} = 0$.

Why it matters for certificates. Any dual-feasible Z proves the lower bound $c^\top x \geq -\operatorname{tr}(F_0 Z)$ without trusting the solver, and infeasibility has a certificate too: the strict LMI $F(x) \succ 0$ is infeasible iff there is $Z \succeq 0$, $Z \neq 0$, with $\operatorname{tr}(F_i Z) = 0$ for all i and $\operatorname{tr}(F_0 Z) \leq 0$ (Boyd & Vandenberghe, Example 5.14). The easy direction is the same trace argument: $F(x) \succ 0$ and $Z \neq 0$ would give $0 < \operatorname{tr}(F(x) Z) = \operatorname{tr}(F_0 Z) \leq 0$.

Feasibility versus optimisation, and the first LMI that matters

Half of the certificates in this section are pure *feasibility* questions ("does a Lyapunov matrix exist?", see Primer D), the other half are optimisation problems ("what is the smallest Lipschitz bound this method can certify?"). Solvers do not distinguish them: to test strict feasibility of $F(x) \succ 0$ one solves $\min t$ s.t. $F(x) + tI \succeq 0$ and checks whether $t^\star < 0$. The objectives we will meet are $\min \rho$ (LipSDP), $\min \gamma^2$ (ℓ_2 gain) and, for a certified ellipsoid (see Primer A) $\mathcal{E}(P) = \{x : x^\top P x \leq 1\}$, $\min \operatorname{tr} P$, the linear proxy for "large $\mathcal{E}(P)$" that Yin, Seiler & Arcak and Pauli et al. use for regions of attraction. The volume of $\mathcal{E}(P)$ is proportional to $(\det P)^{-1/2}$, so $\max \log\det P$ would find the *smallest* ellipsoid; the largest one needs $\min \log\det P$, which is not convex in P. When the constraints permit it, a remedy is the variable $S = P^{-1}$ (constraints rewritten as LMIs in S, usually by a congruence) and $\max \log\det S$ (see Primer B): a concave objective over LMIs, i.e. a determinant-maximisation problem rather than an SDP with a linear objective, still solved efficiently by interior-point methods (Boyd et al. 1994, Sec. 2.2.4).

Two details. First, a negative feasible margin is all that is needed: any feasible pair (x, t) with $t < 0$ gives $F(x) \succeq -tI \succ 0$, and conversely if $F(x) \succ 0$ then any $t < 0$ with $|t| \le \lambda_{\min}(F(x))$ is feasible; the auxiliary problem may be unbounded below, so no finite attained $t^\star$ is required. Second, the substitution $S = P^{-1}$ helps only if *every* constraint becomes an LMI in S: congruence of $A^\top P + PA \prec 0$ with S gives $SA^\top + AS \prec 0$, but other constraints on P may become nonlinear in S.

THEOREM — THE DISCRETE-TIME LYAPUNOV LMI

For $A \in \mathbb{R}^{n\times n}$ the following are equivalent: (i) A is Schur stable (see Primer D), i.e. its spectral radius satisfies $\rho(A) < 1$; (ii) the LMI $P \succ 0$, $A^\top PA - P \prec 0$ is feasible; (iii) for some (equivalently, every) $Q \succ 0$ the Stein equation $A^\top PA - P = -Q$ has a unique solution and it satisfies $P \succ 0$. The matrix P is the certificate: $V(x) = x^\top Px$ strictly decreases along $x_{t+1} = Ax_t$.

Proof sketch — LMI feasibility implies stability, and stability implies the Stein solution ▾

(ii) ⇒ (i). Since $A^\top PA - P \prec 0$ there is $\epsilon > 0$ with $A^\top PA - P \preceq -\epsilon I$ (take ϵ the smallest eigenvalue of $P - A^\top PA$). Along a trajectory, $V(x_{t+1}) - V(x_t) = x_t^\top (A^\top PA - P)x_t \le -\epsilon\|x_t\|^2 \le -\frac{\epsilon}{\lambda_{\max}(P)} V(x_t)$, so $V(x_t) \le (1 - \epsilon/\lambda_{\max}(P))^t V(x_0) \to 0$ geometrically; because $V(x) \ge \lambda_{\min}(P)\|x\|^2$ with $\lambda_{\min}(P) > 0$, every trajectory converges to 0, which for a linear map means $\rho(A) < 1$ (an eigenvector of an eigenvalue with $|\lambda| \ge 1$ would not decay).

(i) ⇒ (iii). If $\rho(A) < 1$ the series $P = \sum_{k\ge 0} (A^\top)^k Q A^k$ converges, is $\succeq Q \succ 0$, and satisfies $A^\top PA - P = \sum_{k\ge 1} (A^\top)^k Q A^k - \sum_{k\ge 0} (A^\top)^k Q A^k = -Q$. Uniqueness: the linear map $P \mapsto A^\top PA - P$ has eigenvalues $\lambda_i\lambda_j - 1$ (λ_i eigenvalues of A), all nonzero when $\rho(A) < 1$. Finally (iii) ⇒ (ii) is immediate with $Q = I$. ■ The second interactive explorer solves exactly this Stein equation for a 2×2 system and draws the certificate.

Convergence and uniqueness without operator eigenvalues. For Schur-stable A and any q with $\rho(A) < q < 1$ there is $c \ge 1$ with $\|A^k\|_2 \le c\, q^k$ (Primer D). Each summand therefore has norm $\|(A^\top)^k Q A^k\|_2 \le c^2 \|Q\|_2\, q^{2k}$, and the series converges by comparison with a geometric series. If P_1, P_2 both solve the Stein equation, $H = P_1 - P_2$ satisfies $H = A^\top HA$; iterating gives $H = (A^\top)^k H A^k \to 0$, so $H = 0$.

Why SDP solvers work, what they cost, and the tooling

The interior of the LMI set has a natural barrier (see Primer B), $\phi(x) = -\log\det F(x)$: it is smooth and convex on $\{F(x) \succ 0\}$ (strictly convex when $F_1, \dots, F_m$ are linearly independent,

which holds whenever the feasible set is nonempty and bounded) and blows up at the boundary. Nesterov and Nemirovski showed that it is *self-concordant* (made precise in the theorem below), which is what makes damped Newton steps on $t\,c^\top x + \phi(x)$ predictable. Following the central path while increasing t needs $O(\sqrt{n}\log(1/\varepsilon))$ Newton iterations for an $n \times n$ LMI and accuracy ε; in practice the iteration count is a few tens and nearly size independent (Boyd, El Ghaoui, Feron & Balakrishnan 1994, Ch. 2). The cost in each Newton step: forming the $m \times m$ Newton system costs on the order of $m\,n^3 + m^2 n^2$ and solving it m^3. With a handful of variables and a large matrix this is $O(n^3)$ per step; with a full matrix variable, $m \sim n^2$, it becomes $O(n^6)$. LipSDP-Neuron on a network with n hidden neurons sits in between: about n diagonal multipliers and a matrix of size about n (plus the input dimension), so a dense Newton step costs roughly $O(n^4)$. That is why the dense SDP becomes impractical for large networks and Module 12 turns to sparse and layer-wise variants.

THEOREM — LOG-DETERMINANT PATH FOLLOWING (BOYD ET AL. 1994)

Let $F(x) = F_0 + \sum_i x_i F_i \in \mathbb{S}^n$, let $\{x : F(x) \succ 0\}$ be nonempty, and remove redundant variable directions so that $\nabla^2\phi$ is positive definite, where $\phi = -\log\det F$. For a direction h, set $\psi(s) = \phi(x + sh)$. Then $|\psi'''(0)| \le 2\,\psi''(0)^{3/2}$ and $|\psi'(0)| \le \sqrt{n\,\psi''(0)}$: ϕ is a self-concordant barrier with parameter n. Suppose the linear-objective SDP has finite optimum, a central path exists, and a starting point in a fixed Newton neighbourhood of that path is available with certified gap bound $G_0 > \varepsilon > 0$. A standard short-step path-following method reaches duality gap at most ε in $O(\sqrt{n}\log(G_0/\varepsilon))$ Newton steps (up to an additive initial step count).

Intuition. The derivative inequalities limit how rapidly curvature changes along any line. That makes a Newton step and a small increase in the path parameter predictable. The count assumes a suitable start and exact arithmetic; finding the start, solving each linear system and controlling numerical error are additional work. For $F = x > 0$, $\phi = -\log x$ has $\phi^{(2)} = 1/x^2$ and $|\phi^{(3)}| = 2/x^3$, exactly the curvature bound.

Boyd et al. 1994, Chapter 2

Tooling. In Python, CVXPY parses LMIs written as `M << 0` and hands them to a solver; in MATLAB, YALMIP does the same. MOSEK and Clarabel are interior-point solvers (high accuracy, $O(n^3)$–$O(n^6)$ per iteration); SCS is a first-order ADMM solver that scales further at the price of 10^{-3}–10^{-5} accuracy. The stack varies by paper: Pauli et al.'s ADMM training (L-CSS 2022) and Zames–

Falb analysis (CDC 2021) solve their SDPs with YALMIP and MOSEK, whereas LipKernel's direct parameterisations need no solver at all and are implemented in PyTorch.

Background — ADMM and copies of the weights

ADMM is the alternating direction method of multipliers. It keeps two copies of the variable, writes the task as $\min_{W,Z} \ell(W) + I_C(Z)$ subject to $W = Z$, where I_C is zero on the certificate-feasible set C and $+\infty$ outside it, and alternates three steps: minimise the loss plus an agreement penalty over W, project onto C to update Z, and update a multiplier that accumulates the disagreement $W - Z$. This describes the training strategy; convergence guarantees for convex splitting do not automatically apply to a neural-network loss.

For example, with $C = [-1, 1]$, the constrained copy of a scalar weight 2 is pulled back to 1. With penalty $r > 0$ and scaled multiplier u, the agreement penalty is $(r/2)\|W - Z + u\|^2$; after the two minimisations, set $u^+ = u + W - Z$. This split makes the certificate step explicit.

COMMON MISTAKE — MATRIX INEQUALITY IS NOT ELEMENTWISE INEQUALITY

$M \succeq 0$ says $x^\top M x \geq 0$ for every x; it says nothing about the signs of the entries. $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has only positive entries but eigenvalues 3 and -1, so it is *not* PSD; $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has a negative entry and eigenvalues $1, 3$, so it *is* positive definite. Likewise $A \succeq B$ does not mean $A_{ij} \geq B_{ij}$, and $A \succeq B \succeq 0$ does not imply $A^2 \succeq B^2$. When a paper writes $M \preceq 0$ for a certificate, it means "all eigenvalues nonpositive"; checking a certificate returned by a solver means computing those eigenvalues.

CAVEAT — A SOLVER STATUS IS NOT A CERTIFICATE

Interior-point solvers return $M \preceq 0$ only up to a tolerance (typically 10^{-8}) and may return a matrix with a slightly positive eigenvalue. To certify, impose a margin ($M \preceq -\epsilon I$) or recompute the eigenvalues of the returned matrix afterwards. The certificate is the matrix, not the solver's "optimal" flag; this matters when the certified quantity is a safety bound that will be used downstream, as in Module 12.

A numerical eigenvalue check is evidence of feasibility. To turn it into a rigorous certificate, include an error bound: if $\widehat{M}$ is the computed matrix, $\|M - \widehat{M}\|_2 \leq \delta$, and a validated upper bound gives $\lambda_{\max}(\widehat{M}) \leq -\delta$, then $M \preceq 0$, because $\lambda_{\max}(M) \leq \lambda_{\max}(\widehat{M}) + \|M -$

$\widehat{M}\|_2$. A strict margin is useful when it exceeds the combined matrix-assembly and eigenvalue errors.

WHY IT MATTERS

Which of the problems in this section are SDPs? LipSDP with fixed weights (Fazlyab et al., NeurIPS 2019): $\min \rho$ s.t. $M(\rho, T) \preceq 0$, affine in (ρ, T). The stability LMI of Yin, Seiler & Arcak, TAC 2022: affine in (P, λ). Zames–Falb multipliers, dissipativity of RNNs, the ℓ_2-gain LMI of Section 6. What is *not* an SDP: any of these with the weights W_i as unknowns, because $W^\top W$ and $W^\top T W$ are quadratic. Section 3 and the walkthrough show how far the Schur complement can push the boundary, and Module 13 shows how to sidestep the solver entirely by parameterising solutions of the LMI.

Worked example — what the set described by an LMI looks like

Consider the LMI in two unknowns x and y:

$$\begin{bmatrix} x & 1 \\ 1 & y \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{F_0} + x \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{F_1} + y \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{F_2} \succeq 0.$$

A symmetric 2×2 matrix is positive semidefinite exactly when both diagonal entries and the determinant are nonnegative. Here that means $x \geq 0$, $y \geq 0$ and $xy - 1 \geq 0$. The solution set is the region on and above the hyperbola $y = 1/x$ in the first quadrant. It is convex, but its boundary is curved, so finitely many linear inequalities could not describe it.

- $(x, y) = (2, 0.5)$: $xy = 1$, on the boundary. The matrix has eigenvalues 0 and 2.5.
- $(x, y) = (2, 0.4)$: $xy = 0.8 < 1$, outside. The determinant is -0.2, so one eigenvalue is negative.
- $(x, y) = (0.5, 3)$: $xy = 1.5$, inside.

Averaging the two feasible points $(2, 0.5)$ and $(0.5, 3)$ gives $(1.25, 1.75)$, with $xy = 2.19 \geq 1$: feasible again, as convexity guarantees.

Worked example — a stability certificate that the identity matrix cannot give

The system $x_{t+1} = Ax_t$ with $A = \begin{bmatrix} 0.5 & 1 \\ 0 & 0.5 \end{bmatrix}$ is stable: both eigenvalues are 0.5, so $\rho(A) = 0.5 < 1$ and every trajectory goes to zero. The Lyapunov LMI asks for $P \succ 0$ with $A^\top PA - P \prec 0$, so that $V(x) = x^\top Px$ decreases at every step.

The obvious guess fails. With $P = I, V(x) = \|x\|^2$ is the squared length. But from $x_0 = (0, 1)$ the first step gives $x_1 = (1, 0.5)$, with $\|x_1\| = 1.118 > 1$: the length grows before it shrinks. Correspondingly, $A^\top A - I$ has a positive eigenvalue, so $P = I$ is not a certificate.

Solving for P. Solve the Stein equation $A^\top PA - P = -I$, which is linear in the three unknown entries of P:

$$P = \begin{bmatrix} 1.3333 & 0.8889 \\ 0.8889 & 4.2963 \end{bmatrix}, \qquad \text{eigenvalues } 1.087 \text{ and } 4.543.$$

So $P \succ 0$ and $A^\top PA - P = -I \prec 0$: a valid certificate. Along the same trajectory $V(x_0) = 4.2963$ and $V(x_1) = 3.2963$. V drops by exactly $\|x_0\|^2 = 1$, because $V(x_{t+1}) - V(x_t) = x_t^\top (A^\top PA - P)x_t = -\|x_t\|^2$. A tilted ellipse measures progress where the round circle could not.

3. The Schur Complement

In plain words. Checking whether a big symmetric matrix is positive definite can be split into two smaller checks: one diagonal block must be positive definite, and a "leftover" matrix, the Schur complement, must be too. Used in reverse, the same fact turns a nonlinear condition, such as "a squared length is at most t", into a linear matrix inequality in a larger matrix. That is how most LMIs on this page are built.

SYMBOLS IN THIS SECTION

- $M = \begin{bmatrix} A & B \\ B^\top & C \end{bmatrix}$: a symmetric block matrix with symmetric diagonal blocks A and C (here A is a block, not a system matrix).
- $C - B^\top A^{-1} B$ and $A - BC^{-1}B^\top$: the Schur complements of A and of C in M.
- I: the identity matrix; $\|x\|^2 = x^\top x$.

The Schur complement is the tool that turns "quadratic in the unknown" into "linear in a bigger matrix". Every LMI on this page that contains a term like $W^\top W$, $c^\top P^{-1} c$ or $PBR^{-1}B^\top P$ was

produced by it.

THEOREM — SCHUR COMPLEMENT (BOTH SIGNS)

Let $M = \begin{bmatrix} A & B \\ B^\top & C \end{bmatrix}$ with $A \in \mathbb{S}^p, C \in \mathbb{S}^q$.

- If $C \succ 0$: $M \succeq 0 \iff A - BC^{-1}B^\top \succeq 0$, and $M \succ 0 \iff A - BC^{-1}B^\top \succ 0$.
- If $A \succ 0$: $M \succeq 0 \iff C - B^\top A^{-1}B \succeq 0$, and $M \succ 0 \iff C - B^\top A^{-1}B \succ 0$.
- Negative form (apply the above to $-M$): if $C \prec 0$, $M \preceq 0 \iff A - BC^{-1}B^\top \preceq 0$.

The matrix $A - BC^{-1}B^\top$ is the Schur complement of C in M. *In words:* a block matrix with a definite corner block is (semi)definite iff what remains after completing the square on that corner is; the corner must be definite so that it can be inverted. (If C is only PSD, the nonstrict statement needs the pseudo-inverse (see Primer A) and the range condition $(I - CC^\dagger)B^\top = 0$; we will not need it.)

Derivation — Complete proof by congruence transformation

Assume $C \succ 0$ and define the invertible block-triangular matrix $R = \begin{bmatrix} I & -BC^{-1} \\ 0 & I \end{bmatrix}$. Multiply out, one factor at a time:

$$MR^\top = \begin{bmatrix} A & B \\ B^\top & C \end{bmatrix} \begin{bmatrix} I & 0 \\ -C^{-1}B^\top & I \end{bmatrix} = \begin{bmatrix} A - BC^{-1}B^\top & B \\ B^\top - CC^{-1}B^\top & C \end{bmatrix} = \begin{bmatrix} A - BC^{-1}B^\top & B \\ 0 & C \end{bmatrix}$$

$$R(MR^\top) = \begin{bmatrix} I & -BC^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} S & B \\ 0 & C \end{bmatrix} = \begin{bmatrix} S & B - BC^{-1}C \\ 0 & C \end{bmatrix} = \begin{bmatrix} S & 0 \\ 0 & C \end{bmatrix}, \qquad S := A -$$

So $RMR^\top = \mathrm{blkdiag}(S, C)$. By the congruence fact of Section 2 with the invertible $R^\top$, $M \succeq 0 \iff RMR^\top \succeq 0 \iff S \succeq 0$ and $C \succeq 0$; the second condition is given, so $M \succeq 0 \iff S \succeq 0$. The strict version is identical with $\succ$. Swapping the roles of the blocks (conjugate with $\begin{bmatrix} I & 0 \\ -B^\top A^{-1} & I \end{bmatrix}$) gives the second bullet, and applying the first bullet to $-M$ gives the negative form. ■

The scalar picture. For vectors x, y the identity behind the algebra is completing the square:

$$x^\top A x + 2x^\top B y + y^\top C y = (y + C^{-1}B^\top x)^\top C\,(y + C^{-1}B^\top x) + x^\top (A - BC^{-1}B^\top)x.$$

Minimising over y kills the first term (choose $y = -C^{-1}B^\top x$), so the quadratic form is nonnegative for all (x, y) iff its minimum over y, the Schur-complement form, is nonnegative in x.

The standard uses

(a) A quadratic constraint on the inverse becomes an LMI. For $P \succ 0$, the ellipsoid $\mathcal{E}(P) = \{x : x^\top P x \le 1\}$ lies in the slab $\{|c^\top x| \le 1\}$ iff $c^\top P^{-1} c \le 1$. Indeed, with $y = P^{1/2}x$, Cauchy–Schwarz (Primer A) gives $c^\top x = (P^{-1/2}c)^\top y \le \|P^{-1/2}c\|\,\|y\|$ with equality attainable, so $\max_{\mathcal{E}(P)} c^\top x = \sqrt{c^\top P^{-1} c}$. The condition $1 - c^\top P^{-1} c \ge 0$ is nonlinear in P, but it is the Schur complement of P in

$$\begin{bmatrix} 1 & c^\top \\ c & P \end{bmatrix} \succeq 0,$$

which is linear in P. Read with a point x in place of c, the same LMI says $x^\top P^{-1} x \le 1$, i.e. "x lies in the ellipsoid with shape matrix P^{-1}", and it is jointly linear in (x, P): the standard way to put a point, or the vertices of a polytope, inside an ellipsoid whose shape is a decision variable. Back to the slab: with $c = W_i^\top / (\bar{v}_i - v_{*,i})$ (W_i the i-th row of the first weight matrix, the ellipsoid centred at the equilibrium) and a congruence with $\mathrm{diag}(\bar{v}_i - v_{*,i}, I)$, it becomes the constraint $\begin{bmatrix} (\bar{v}_i - v_{*,i})^2 & W_i \\ W_i^\top & P \end{bmatrix} \succeq 0$ of Yin, Seiler & Arcak, TAC 2022 and of Pauli et al., CDC 2021: it keeps the certified ellipsoid inside the region where the local sector bounds of the activations are valid (Module 14).

Derivation — Keeping the ellipsoid inside both ends of the activation interval

Write $\delta x = x - x_*$ and $v_i - v_{*,i} = W_i \delta x$. To guarantee $\underline{v}_i \le v_i \le \bar{v}_i$, define $m_i = \min\{\bar{v}_i - v_{*,i}, v_{*,i} - \underline{v}_i\} > 0$. It suffices to require $|W_i \delta x| \le m_i$ throughout $\delta x^T P \delta x \le 1$, equivalently $\begin{bmatrix} m_i^2 & W_i \\ W_i^T & P \end{bmatrix} \succeq 0$. Using only $\bar{v}_i - v_{*,i}$ assumes this is the smaller available margin.

(b) Riccati inequality. With $R \succ 0$, the continuous-time Riccati inequality $A^\top P + PA + Q + PBR^{-1}B^\top P \preceq 0$ is quadratic in P. Apply the negative form with the $(2,2)$ block $-R \prec 0$:

$$\begin{bmatrix} A^\top P + PA + Q & PB \\ B^\top P & -R \end{bmatrix} \preceq 0 \iff A^\top P + PA + Q - (PB)(-R)^{-1}(B^\top P) \preceq 0 \iff$$

and the left-hand side is an LMI in P. Exercise 2.3 does the same for the ℓ_2-gain version.

(c) Spectral-norm bounds (Primer A). For $\gamma > 0$, $\|W\|_2 \le \gamma \iff W^\top W \preceq \gamma^2 I \iff \begin{bmatrix} \gamma I & W^\top \\ W & \gamma I \end{bmatrix} \succeq 0$ (Schur complement of the lower-right block: $\gamma I - W^\top (\gamma I)^{-1} W \succeq 0$). The right-hand side is linear in (W, γ), which is why norm-bounded layers are "LMI-representable", and why layer conditions such as the SLL condition $W^\top W \preceq T$ (T diagonal) and the Sandwich layer's LMI in Module 13 can be satisfied by construction.

(d) From analysis to training. In LipSDP the last layer enters through $W_l^\top W_l$. With the weights fixed this is a constant, but for training Pauli et al., L-CSS 2022 need the certificate to be linear in W_l; the Schur complement with a $-I$ block achieves exactly that. Step 6 of the walkthrough does the computation.

Worked example — two uses of the Schur complement ▾

Testing a matrix. Is $M = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ positive definite? The top-left block $A = 2$ is positive, and its Schur complement is $C - B^\top A^{-1} B = 1 - 1 \cdot \frac{1}{2} \cdot 1 = 0.5 > 0$. So $M \succ 0$. Check: the eigenvalues are $(3 \pm \sqrt{5})/2$, about 0.382 and 2.618, both positive.

Turning a nonlinear condition into an LMI. The condition $\|x\|^2 \le t$ is quadratic in x. With $C = I \succ 0$, the Schur complement theorem gives

$$\begin{bmatrix} t & x^\top \\ x & I \end{bmatrix} \succeq 0 \quad \iff \quad t - x^\top I^{-1} x = t - \|x\|^2 \ \ge \ 0,$$

and the matrix on the left is affine in the unknowns (t, x): an LMI. For $x = (3, 4)$ we have $\|x\|^2 = 25$, and the block matrix is positive semidefinite for $t = 25$ (on the boundary) and $t = 26$, but not for $t = 24$.

4. The S-Procedure / S-Lemma

In plain words. Many certificates need a statement of the form "whenever this quadratic inequality holds, that one holds too". Checking it point by point is impossible, because there are infinitely many points. The S-procedure instead looks for one number $\tau \geq 0$ such that "the second quadratic minus τ times the first" is nonnegative *everywhere*. If such a τ exists, the implication is proved: wherever the first quadratic is nonnegative, the second is at least τ times it, hence nonnegative too. With a single constraint that holds strictly somewhere, the S-lemma adds the converse: if the implication is true, such a τ exists.

SYMBOLS IN THIS SECTION

- $\sigma_0, \sigma_1, \ldots, \sigma_m$: quadratic functions of a vector ξ. The question is whether $\sigma_1(\xi) \geq 0, \ldots, \sigma_m(\xi) \geq 0$ imply $\sigma_0(\xi) \geq 0$.
- $\tau_i \geq 0$: the S-procedure multipliers, one per constraint (Lagrange multipliers in disguise).
- Strict feasibility: some $\bar{\xi}$ with $\sigma_1(\bar{\xi}) > 0$.

A robustness question almost always has the shape "for all signals that satisfy *this* quadratic constraint, is *that* quadratic form nonpositive?" The S-procedure replaces the universal quantifier by one nonnegative multiplier per constraint and one matrix inequality. It is a Lagrangian relaxation in disguise, and for a single strictly feasible constraint it is exact.

DEFINITION — THE S-PROCEDURE (SUFFICIENT CONDITION)

Let $\sigma_0, \ldots, \sigma_m : \mathbb{R}^n \to \mathbb{R}$ be quadratic functions, $\sigma_i(\xi) = \xi^\top A_i \xi + 2b_i^\top \xi + c_i$ with $A_i \in \mathbb{S}^n$. If there exist $\tau_1, \ldots, \tau_m \geq 0$ such that

$$\sigma_0(\xi) - \sum_{i=1}^{m} \tau_i \, \sigma_i(\xi) \;\geq\; 0 \qquad \text{for all } \xi \in \mathbb{R}^n,$$

then $\sigma_0(\xi) \geq 0$ for every ξ with $\sigma_1(\xi) \geq 0, \ldots, \sigma_m(\xi) \geq 0$. Because a quadratic function is nonnegative on all of $\mathbb{R}^n$ iff its homogenised matrix is PSD, the premise is the LMI

$$\begin{bmatrix} A_0 & b_0 \\ b_0^\top & c_0 \end{bmatrix} - \sum_{i=1}^{m} \tau_i \begin{bmatrix} A_i & b_i \\ b_i^\top & c_i \end{bmatrix} \succeq 0 \qquad (\text{for homogeneous forms: } A_0 - \textstyle\sum_i \tau_i A_i \succeq 0),$$

which is linear in the multipliers τ.

Proof sketch — Sufficiency (two lines) and the homogenisation claim ▾

Sufficiency. Take ξ with $\sigma_i(\xi) \geq 0$ for all i. Since $\tau_i \geq 0$, $\sum_i \tau_i \sigma_i(\xi) \geq 0$, and the premise gives $\sigma_0(\xi) \geq \sum_i \tau_i \sigma_i(\xi) \geq 0$. ■ Compare with weak duality: the multipliers price the constraints and the premise says the priced objective is nonnegative everywhere; the conclusion follows on the feasible set because the prices are nonnegative.

Homogenisation. Write $\sigma(\xi) = \begin{bmatrix} \xi \\ 1 \end{bmatrix}^\top N \begin{bmatrix} \xi \\ 1 \end{bmatrix}$ with $N = \begin{bmatrix} A & b \\ b^\top & c \end{bmatrix}$. If $N \succeq 0$ then clearly $\sigma \geq 0$. Conversely, if $\sigma(\xi) \geq 0$ for all ξ, then for $t \neq 0$, $\begin{bmatrix} \xi \\ t \end{bmatrix}^\top N \begin{bmatrix} \xi \\ t \end{bmatrix} = t^2 \sigma(\xi/t) \geq 0$, and the case $t = 0$ follows by continuity; so $N \succeq 0$. ■

THEOREM — S-LEMMA (YAKUBOVICH; ONE CONSTRAINT IS LOSSLESS)

Let $A, B \in \mathbb{S}^n$ and suppose there is a point $\bar{\xi}$ with $\bar{\xi}^\top A \bar{\xi} > 0$ (*strict feasibility*). Then

$$\Big[\ \xi^\top A \xi \geq 0 \ \Rightarrow\ \xi^\top B \xi \geq 0\ \Big] \quad \Longleftrightarrow \quad \exists\, \tau \geq 0 : \ \ B - \tau A \succeq 0.$$

The same equivalence holds for inhomogeneous quadratic functions σ_1, σ_0 (with $\sigma_1(\bar{\xi}) > 0$) and the LMI of the definition above with $m = 1$. For $m \geq 2$ constraints the S-procedure is in general only sufficient. See the survey by Pólik & Terlaky, SIAM Review 2007, which attributes the result to Yakubovich (1971) and collects the exceptions.

In words: if a quadratic inequality follows from another one, then it follows for the most naive reason, namely because it is a nonnegative multiple of it plus something globally nonnegative. *Why strict feasibility matters:* take $\sigma_1(\xi) = -\xi^2$ and $\sigma_0(\xi) = \xi$ on $\mathbb{R}$. The feasible set is $\{0\}$, so the implication holds trivially, but $\xi + \tau\xi^2 < 0$ for small negative ξ whatever $\tau \geq 0$ is: no multiplier exists. The direction "$\Leftarrow$" is the S-procedure; the content of the lemma is "$\Rightarrow$".

Proof sketch — Losslessness via Dines' theorem and a separating line ▾

The proof rests on a classical fact about pairs of real quadratic forms: by Dines, Bull. AMS 1941, the joint range $K = \{(\xi^\top A \xi,\ \xi^\top B \xi) : \xi \in \mathbb{R}^n\} \subset \mathbb{R}^2$ is a *convex cone*. (This is false for three forms in general, which is one reason the S-lemma stops at one constraint.)

Assume the implication holds. Then no ξ has $\xi^\top A \xi \geq 0$ and $\xi^\top B \xi < 0$, so K does not meet the open convex quadrant $\mathcal{M} = \{(u, v) : u > 0,\ v < 0\}$. By the separating-hyperplane theorem of Section 1 (which needs no closedness of K, because $\mathcal{M}$ is open) there are $(p, q) \neq 0$ and c with $pu + qv \geq c$ on K and $pu + qv \leq c$ on $\mathcal{M}$; and $c = 0$, since $0 \in K$ gives $c \leq 0$ while points of $\mathcal{M}$ arbitrarily close to the origin give $c \geq 0$. So $pu + qv \geq 0$ on K and $pu + qv \leq 0$ on $\mathcal{M}$. On

$\mathcal{M}$, letting $u \to \infty$ forces $p \le 0$ and letting $v \to -\infty$ forces $q \ge 0$. On K the inequality reads $p\,\xi^\top A\xi + q\,\xi^\top B\xi \ge 0$ for all ξ.

If $q > 0$, divide by q and set $\tau = -p/q \ge 0$: $\xi^\top(B - \tau A)\xi \ge 0$ for all ξ, i.e. $B - \tau A \succeq 0$. If $q = 0$ then $p < 0$ and the inequality says $\xi^\top A\xi \le 0$ for all ξ, contradicting the strictly feasible $\bar{\xi}$. So $q > 0$ and the multiplier exists. ■ Strict feasibility is used exactly once, to exclude the degenerate separating line $u = 0$.

The multiplier reading

Consider the non-convex problem $p^\star = \min_\xi q_0(\xi)$ s.t. $q_1(\xi) \le 0$ with one quadratic constraint (a trust-region subproblem when $q_1(\xi) = \|\xi\|^2 - 1$). Its dual is $d^\star = \max_{\tau \ge 0} \inf_\xi [q_0(\xi) + \tau q_1(\xi)]$. For any level t,

$$t \le p^\star \iff \big[q_1(\xi) \le 0 \Rightarrow q_0(\xi) - t \ge 0\big] \iff \exists \tau \ge 0 :\ q_0(\xi) - t + \tau q_1(\xi) \ge 0\ \forall \xi$$

where the middle equivalence is the S-lemma (applied to $-q_1$ and $q_0 - t$, assuming a strictly feasible point $q_1(\bar{\xi}) < 0$). Taking $t = p^\star$ (assumed finite) gives $p^\star \le d^\star$, weak duality gives the reverse, and the τ produced by the S-lemma attains the dual optimum. Hence $p^\star = d^\star$: **a non-convex QCQP (quadratically constrained quadratic program) with a single quadratic constraint has no duality gap**, and the S-procedure multiplier τ is its Lagrange multiplier. With two or more quadratic constraints the gap can be positive, and the S-procedure certifies only the dual value $d^\star \le p^\star$: a valid but possibly conservative bound.

KEY EQUATION — FROM QUADRATIC CONSTRAINTS TO AN LMI (THE ENGINE OF LIPSDP)

Suppose a stacked vector ξ (for a network: all increments of inputs and activations) is known to satisfy n quadratic constraints $\xi^\top Q_i \xi \ge 0$, $i = 1, \dots, n$, and we want to certify $\xi^\top M \xi \le 0$ for all such ξ. The S-procedure says it suffices that

$$M + \sum_{i=1}^{n} \lambda_i Q_i \ \preceq\ 0 \qquad \text{for some } \lambda_1, \dots, \lambda_n \ge 0,$$

because then $\xi^\top M\xi \le \xi^\top M\xi + \sum_i \lambda_i \xi^\top Q_i \xi = \xi^\top \big(M + \sum_i \lambda_i Q_i\big)\xi \le 0$. When Q_i is the slope-restriction constraint of neuron i (Section 5), $\sum_i \lambda_i Q_i$ assembles into the block matrix $\begin{bmatrix} -2\alpha\beta T & (\alpha+\beta)T \\ (\alpha+\beta)T & -2T \end{bmatrix}$ with $T = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$: **the diagonal multiplier of LipSDP is the vector of S-procedure multipliers, one per neuron.**

KEY INSIGHT — TWO SOURCES OF CONSERVATISM, AND ONE EXACT CASE

A certificate of this kind is loose for two independent reasons. (1) *Abstraction:* the set $\{\xi : \xi^\top Q_i \xi \geq 0 \ \forall i\}$ is an outer approximation of the pairs $(\Delta x, \Delta\varphi)$ that the activation can actually produce; any property proved on the larger set holds on the smaller one, but not conversely. (2) *Several constraints:* with $n \geq 2$ neurons the S-procedure is only sufficient. In the scalar network of the walkthrough there is one neuron, so (2) disappears, and for ReLU (1) disappears too, because every slope in $[0, 1]$ is a chord slope of ReLU; the SDP then returns the true Lipschitz constant. For tanh the abstraction is not exact (no chord of tanh has slope exactly 0 or 1), yet the bound is still tight because chord slopes come arbitrarily close to 1 near the origin; for an activation whose slopes stay well below 1, such as $\varphi(v) = v/2$ (also slope-restricted in $[0, 1]$), the $[0, 1]$ certificate is loose by that factor. In general LipSDP returns an upper bound. Module 12 quantifies the gap between this bound, the naive product bound and empirical estimates (interactive).

LEMMA — FINSLER (THE S-PROCEDURE FOR AN EQUALITY CONSTRAINT) AND THE ELIMINATION LEMMA

Let $Q \in \mathbb{S}^n$, $B \in \mathbb{R}^{m\times n}$, and let the columns of N form a basis of $\ker B$. The following are equivalent: (i) $\xi^\top Q \xi < 0$ for all $\xi \neq 0$ with $B\xi = 0$; (ii) $N^\top Q N \prec 0$; (iii) $Q - \mu B^\top B \prec 0$ for some $\mu \in \mathbb{R}$; (iv) $Q + XB + B^\top X^\top \prec 0$ for some matrix $X \in \mathbb{R}^{n\times m}$. (i) $\iff$ (ii) is a change of coordinates on $\ker B$: every ξ with $B\xi = 0$ is $\xi = N\eta$, and $\xi^\top Q \xi = \eta^\top N^\top Q N \eta$, with $\xi \neq 0$ iff $\eta \neq 0$ because the columns of N are independent. (iii)$\Rightarrow$(iv): take $X = -\frac{\mu}{2} B^\top$. (iv)$\Rightarrow$(i): for $B\xi = 0$ the added term is $2\xi^\top X B \xi = 0$. The only nontrivial step, (i)$\Rightarrow$(iii), is a compactness argument: Q is negative definite on $\ker B$, and a large enough μ makes $-\mu B^\top B$ dominate off it. Read (iii) as an S-procedure for the single constraint $\|B\xi\|^2 \leq 0$ (equivalently $B\xi = 0$) with multiplier μ, and (iv) as the same idea with a free matrix multiplier X attached to the linear constraint $B\xi = 0$. Its generalisation, the **elimination lemma**, states that, for $G \in \mathbb{S}^n$, $U \in \mathbb{R}^{n\times p}$, $V \in \mathbb{R}^{n\times q}$, the inequality $G + UXV^\top + VX^\top U^\top \succ 0$ is feasible in $X \in \mathbb{R}^{p\times q}$ iff $\tilde{U}^\top G \tilde{U} \succ 0$ and $\tilde{V}^\top G \tilde{V} \succ 0$, where the columns of $\tilde{U}, \tilde{V}$ form bases (independent columns, not merely spanning sets) of $\ker U^\top$ and $\ker V^\top$. It is the standard way to remove controller variables from synthesis LMIs, and slack variables such as X in (iv) are the standard way to decouple a Lyapunov matrix from system matrices (Boyd et al. 1994, Sec. 2.6.2 and its Notes, which also give Finsler's lemma).

Worked example — finding the multiplier that proves an implication ▾

Claim: if $x^2 \le 1$, then $2 - x - x^2 \ge 0$. In S-procedure form, $\sigma_1(x) = 1 - x^2 \ge 0$ should imply $\sigma_0(x) = 2 - x - x^2 \ge 0$. (The claim is true, because $2 - x - x^2 = (1 - x)(2 + x)$ is nonnegative on $[-1, 1]$.)

Step 1: the certificate condition. We need $\tau \ge 0$ with $\sigma_0(x) - \tau\sigma_1(x) \ge 0$ for *all* real x:

$$(2 - x - x^2) - \tau(1 - x^2) = (\tau - 1)x^2 - x + (2 - \tau) \;\ge\; 0 \quad \text{for all } x.$$

Step 2: when is a quadratic nonnegative everywhere? Its leading coefficient must be positive, $\tau > 1$, and its discriminant must be at most zero: $1 - 4(\tau - 1)(2 - \tau) \le 0$. Expanding gives $4\tau^2 - 12\tau + 9 \le 0$, that is, $(2\tau - 3)^2 \le 0$, so $\tau = 1.5$ exactly.

Step 3: the certificate. With $\tau = 1.5$ the difference is $0.5x^2 - x + 0.5 = 0.5(x - 1)^2 \ge 0$, a perfect square. So for every x with $1 - x^2 \ge 0$ we get $2 - x - x^2 \ge 1.5\,(1 - x^2) \ge 0$. Any other τ, for example 1.4 or 1.6, makes the difference negative somewhere. The S-lemma guaranteed that this search would succeed, because the constraint is strictly feasible ($\sigma_1(0) = 1 > 0$) and the implication is true.

5. Quadratic Constraints

In plain words. A neural network's activation function, or an uncertain part of a plant, is hard to handle exactly. A quadratic constraint replaces it by one simple fact it is known to satisfy, written as a quadratic inequality between its input and its output. For ReLU, the fact is "the output lies between 0 and the input": a sector. Certificates then hold for *every* function obeying that fact, so they are safe but can be loose. The S-procedure of Section 4 combines such facts, each with a nonnegative weight, into one LMI.

SYMBOLS IN THIS SECTION

- φ: a scalar nonlinearity, for example an activation, with input v and output $\varphi(v)$.
- Sector $[\alpha, \beta]$: $\alpha v^2 \le v\,\varphi(v) \le \beta v^2$; the graph lies between the lines of slopes α and β. Slope restriction $[\alpha, \beta]$: the same for every chord, $\alpha \le \frac{\varphi(v) - \varphi(\bar{v})}{v - \bar{v}} \le \beta$.
- $\Delta v = v - \bar{v}$, $\Delta\varphi = \varphi(v) - \varphi(\bar{v})$: increments between two inputs. Incremental QCs compare two arbitrary inputs; non-incremental ones compare with a fixed point.
- $T = \mathrm{diag}(\lambda_i)$ with $\lambda_i \ge 0$: one multiplier per neuron (not the Lagrange multiplier of Section 1).

A quadratic constraint (QC) is the control-theoretic way to say "I do not know this nonlinearity exactly, but I know a quadratic inequality its input–output pairs satisfy." Activations, uncertain plants and delays are all treated the same way, which is what lets the same LMI machinery certify a Lipschitz constant and a closed-loop region of attraction.

DEFINITION — SECTOR-BOUNDED AND SLOPE-RESTRICTED NONLINEARITIES

A function $\varphi : \mathbb{R} \to \mathbb{R}$ with $\varphi(0) = 0$ is **sector-bounded** in $[\alpha, \beta]$ if $\alpha v^2 \le v\varphi(v) \le \beta v^2$ for all v, equivalently $(\varphi(v) - \alpha v)(\beta v - \varphi(v)) \ge 0$, equivalently

$$\begin{bmatrix} v \\ \varphi(v) \end{bmatrix}^\top \begin{bmatrix} -2\alpha\beta & \alpha+\beta \\ \alpha+\beta & -2 \end{bmatrix} \begin{bmatrix} v \\ \varphi(v) \end{bmatrix} \ge 0.$$

It is **slope-restricted** in $[\alpha, \beta]$ ($0 \le \alpha < \beta < \infty$ in LipSDP's Definition 1) if every chord has slope in $[\alpha, \beta]$:

$$\alpha \le \frac{\varphi(a) - \varphi(b)}{a - b} \le \beta \qquad \forall a \ne b,$$

i.e. the *increment* $(a - b,\ \varphi(a) - \varphi(b))$ lies in the sector $[\alpha, \beta]$. Multiplying by $(a-b)^2$ and rearranging gives the same 2×2 matrix applied to the increment. Slope restriction with $\varphi(0) = 0$ implies the sector bound (take $b = 0$); the converse fails (a sector-bounded function need not be monotone).

The matrix identity is worth checking once: $(\varphi - \alpha v)(\beta v - \varphi) = -\alpha\beta v^2 + (\alpha+\beta)v\varphi - \varphi^2$, and doubling it gives $-2\alpha\beta v^2 + 2(\alpha+\beta)v\varphi - 2\varphi^2$, which is exactly the quadratic form. A **loop transformation** normalises any sector: if φ is in $[\alpha, \beta]$, then $\tilde{\varphi}(v) = (\varphi(v) - \alpha v)/(\beta - \alpha)$ is in $[0, 1]$ (and slope restriction transforms the same way), with the linear part αv absorbed into the rest of the loop. This is why many results are stated for $[0, 1]$ only, and why the absolute-stability criteria for such Lur'e loops (see Primer D) (circle, Popov, Zames–Falb) are usually derived in the normalised setting; LipSDP instead keeps α, β explicit in the multiplier. Tight intervals for common activations: ReLU and tanh are slope-restricted in $[0, 1]$ (tanh has slope $1 - \tanh^2(v) \in (0, 1]$, so $\alpha = 0$ is tight globally); the sigmoid in $[0, \frac{1}{4}]$ (hence also in $[0, 1]$; these are *slope* intervals, and since $\sigma(0) = \frac{1}{2}$ the origin-centred sector definition applies to the shifted $\sigma(v) - \frac{1}{2}$); leaky ReLU with slope $a \in [0, 1)$ on negative inputs in $[a, 1]$. Softmax is not elementwise (see Primer E) but is the gradient of the convex log-sum-exp, and gradients of β-smooth convex functions satisfy the incremental QC with $T = \lambda I$ (theorem below). GroupSort and MaxMin are 1-Lipschitz but *not* slope-restricted, which is why Module 12 needs new QCs for them.

THEOREM — COCOERCIVITY OF CONVEX GRADIENTS (FAZLYAB ET AL. 2019)

Let $F : \mathbb{R}^n \to \mathbb{R}$ be differentiable and convex, and suppose $\|\nabla F(x) - \nabla F(y)\|_2 \leq \beta\|x - y\|_2$ for all x, y, with $\beta > 0$. Then $\Delta g = \nabla F(x) - \nabla F(y)$ satisfies $\beta(x - y)^\top \Delta g \geq \|\Delta g\|_2^2$ (*cocoercivity*). Multiplying by $2\lambda \geq 0$ gives exactly the incremental sector QC of ∇F with $\alpha = 0$ and $T = \lambda I$.

Intuition. Convexity aligns the gradient change with the displacement; smoothness prevents that gradient change from being too large relative to its alignment. Lipschitz continuity alone does not imply this inequality. For softmax, $F(x) = \log \sum_i e^{x_i}$ and $p = \nabla F(x)$ has entries $p_i = e^{x_i} / \sum_j e^{x_j}$; the Hessian is $\mathrm{diag}(p) - pp^\top$. For any z, its quadratic form is $\sum_i p_i z_i^2 - (\sum_i p_i z_i)^2$, between 0 and $\|z\|_2^2$ by Cauchy–Schwarz and $0 \leq p_i \leq 1$. Hence the conservative choice $\beta = 1$ is valid.

Fazlyab et al. 2019, Sec. 2.2 (softmax example)

DEFINITION — INCREMENTAL VERSUS NON-INCREMENTAL QUADRATIC CONSTRAINTS

A map $\phi : \mathbb{R}^n \to \mathbb{R}^n$ satisfies the **incremental QC** defined by a set $\mathcal{Q} \subset \mathbb{S}^{2n}$ if for every $Q \in \mathcal{Q}$ and all x, y:

$$\begin{bmatrix} x - y \\ \phi(x) - \phi(y) \end{bmatrix}^\top Q \begin{bmatrix} x - y \\ \phi(x) - \phi(y) \end{bmatrix} \geq 0$$

(LipSDP Definition 2). A **non-incremental** QC relates $(x - x_*,\ \phi(x) - \phi(x_*))$ to a single fixed point x_* (an equilibrium), as in the local sector constraints of Yin, Seiler & Arcak. Incremental constraints are the right abstraction for global Lipschitz bounds and contraction (two arbitrary trajectories); non-incremental ones for stability of one equilibrium, where they can be made *local* and much tighter. Restricted to $[-\bar{v}, \bar{v}]$ with $\bar{v} > 0$, tanh satisfies the local sector $[\tanh(\bar{v})/\bar{v},\ 1]$ instead of the global $[0, 1]$. Around an equilibrium input $v_* \in (\underline{v}, \bar{v})$ the *offset* local sector, i.e. the sector condition for $(v - v_*,\ \varphi(v) - \varphi(v_*))$ with $v \in [\underline{v}, \bar{v}]$, has $\beta = 1$ and $\alpha = \min\left\{\frac{\tanh\bar{v} - \tanh v_*}{\bar{v} - v_*}, \frac{\tanh v_* - \tanh\underline{v}}{v_* - \underline{v}}\right\}$, the smaller of the two endpoint chord slopes (Yin, Seiler & Arcak, Sec. II-C). The interval itself is certified by the Schur-complement LMI of Section 3(a).

Background — Contraction and incremental stability ▾

A discrete-time map F is a *contraction* in a chosen norm if $\|F(x) - F(y)\| \leq q\|x - y\|$ for all x, y with a fixed $q < 1$. Iteration then gives $\|x_t - y_t\| \leq q^t \|x_0 - y_0\|$ for any two trajectories

$x_{t+1} = F(x_t), y_{t+1} = F(y_t)$. An incremental QC can help prove such a bound, but a finite Lipschitz constant greater than or equal to one does not establish contraction.

For $F(x) = x/2$, two initial points a distance 8 apart are distances $4, 2, 1$ apart after one, two and three steps. A weighted norm $\|x\|_P = \sqrt{x^T P x}$ with $P \succ 0$ is a constant contraction metric when the same shrinking inequality holds in it. This is stronger than merely bounding a network output difference.

LEMMA — DIAGONAL MULTIPLIERS FOR ELEMENTWISE SLOPE-RESTRICTED ACTIVATIONS (FAZLYAB ET AL. 2019, LEMMA 1)

Let $\varphi : \mathbb{R} \to \mathbb{R}$ be slope-restricted in $[\alpha, \beta]$ and $\phi(x) = [\varphi(x_1) \ \cdots \ \varphi(x_n)]^\top$. Then for every $T \in \mathcal{T}_n = \{T = \sum_{i=1}^n \lambda_i e_i e_i^\top : \ \lambda_i \geq 0\}$ (diagonal, PSD) and all $x, y \in \mathbb{R}^n$,

$$\begin{bmatrix} x - y \\ \phi(x) - \phi(y) \end{bmatrix}^\top \begin{bmatrix} -2\alpha\beta T & (\alpha+\beta)T \\ (\alpha+\beta)T & -2T \end{bmatrix} \begin{bmatrix} x - y \\ \phi(x) - \phi(y) \end{bmatrix} \geq 0.$$

Proof. Expanding the form with T diagonal gives $\sum_i \lambda_i \big[-2\alpha\beta(x_i - y_i)^2 + 2(\alpha + \beta)(x_i - y_i)(\varphi(x_i) - \varphi(y_i)) - 2(\varphi(x_i) - \varphi(y_i))^2 \big]$, and each bracket is the scalar slope-restriction QC of neuron i, hence ≥ 0; a nonnegative combination of nonnegative terms is nonnegative. ■ (Fazlyab et al., NeurIPS 2019; note their $\rho = L^2$.)

PITFALL — WHY THE MULTIPLIER MUST BE DIAGONAL FOR INCREMENTAL CONSTRAINTS (THE LIPSDP-NETWORK ERROR)

The NeurIPS 2019 version of LipSDP also allowed coupling terms $\lambda_{ij}(e_i - e_j)(e_i - e_j)^\top$ in T ("LipSDP-Network"). Expanding such a term gives λ_{ij} times the scalar QC evaluated at the *difference pair* $\big(\Delta x_i - \Delta x_j, \ \Delta\varphi_i - \Delta\varphi_j\big)$, which is nonnegative only if that pair has chord slope in $[\alpha, \beta]$. But $\Delta\varphi_i = s_i \Delta x_i$ and $\Delta\varphi_j = s_j \Delta x_j$ with two *independent* chord slopes $s_i, s_j \in [\alpha, \beta]$ taken at different places on the graph of φ, and $(s_i \Delta x_i - s_j \Delta x_j)/(\Delta x_i - \Delta x_j)$ can be anything. Pauli, Koch, Berberich, Kohler & Allgöwer, L-CSS 2022 give the counterexample: ReLU, $T = (e_1 - e_2)(e_1 - e_2)^\top$, $x = (0, 1), y = (-1.5, 0)$, so $\Delta x = (1.5, 1)$, $\Delta\varphi = (0, 1)$ and, with $\alpha = 0, \beta = 1$, the form equals $2\Delta x^\top T \Delta\varphi - 2\Delta\varphi^\top T \Delta\varphi = 2(-0.5) - 2(1) = -3 < 0$ (the difference pair has slope -2). They also exhibit a two-neuron tanh network fitting $\cos$ on $[-\pi/2, \pi/2]$, true Lipschitz constant about 1, for which the coupled LMI is feasible for arbitrarily small L. The arXiv text of the paper prints the value of the form as -2; evaluating the stated numbers gives -3, negative either way. The current arXiv version of LipSDP keeps only diagonal T (LipSDP-Neuron and LipSDP-Layer). Coupled "repeated nonlinearity" multipliers remain valid for *non-incremental* QCs at a single input, where $\varphi(z_i)$ and $\varphi(z_j)$ are the same function at two arguments and the chord

between them genuinely has slope in $[\alpha, \beta]$; Revay, Wang & Manchester, TAC 2024 (Remark 5) state the same restriction for incremental IQCs and contraction. Full story in Module 12.

Integral quadratic constraints, briefly

When the "nonlinearity" is a dynamic operator $w = \Delta(v)$ (an uncertain plant, a delay, or an activation seen in feedback over time), pointwise QCs are replaced by **integral quadratic constraints** (Megretski & Rantzer, TAC 1997): Δ satisfies the IQC defined by the multiplier Π (a bounded, Hermitian-matrix-valued function of ω) if, for every square-integrable input v (see Primer D),

$$\int_{-\infty}^{\infty} \begin{bmatrix} \hat{v}(j\omega) \\ \hat{w}(j\omega) \end{bmatrix}^{*} \Pi(j\omega) \begin{bmatrix} \hat{v}(j\omega) \\ \hat{w}(j\omega) \end{bmatrix} \mathrm{d}\omega \;\geq\; 0 \qquad \text{whenever } w = \Delta(v).$$

Background — Complex vectors and Hermitian matrices

Here $j^2 = -1$, $\hat{v}(j\omega)$ is the Fourier transform of the time signal v, and $M^* = \overline{M}^T$ is conjugate transpose. A Hermitian matrix satisfies $M = M^*$; its quadratic form $z^* M z$ is real, and $M \succeq 0$ means this is nonnegative for every complex vector z. Thus complex congruence $R^* M R$ plays the same role as real congruence $R^T M R$.

For example, $z = (1, j)^T$ gives $z^* z = 2$, whereas $z^T z = 0$: conjugation is essential to energy. Also $e^{j\theta} = \cos\theta + j \sin\theta$, and $\overline{a + jb} = a - jb$. The largest singular value of G is $\sqrt{\lambda_{\max}(G^* G)}$. These are the complex versions of the real matrix operations used above.

Their theorem is the basic robustness result for such loops.

THEOREM — IQC ROBUST STABILITY (MEGRETSKI & RANTZER 1997, THEOREM 1)

Signals live in $\mathcal{L}_{2e}$, i.e. have finite energy $\|v\|_{[0,T]}^2 = \int_0^T \|v(t)\|^2 dt$ on every finite interval; $\mathcal{L}_2$ are those with finite energy on $[0,\infty)$. Let G be a stable, proper, real-rational transfer matrix ($G \in \mathcal{RH}_\infty$: entries are ratios of real polynomials, numerator degree at most denominator degree, no poles in the closed right half-plane), and let $\Delta : \mathcal{L}_{2e} \to \mathcal{L}_{2e}$ be *causal* (its output up to time T depends only on its input up to T) and *bounded* ($\Delta(0) = 0$ and $\|\Delta(v)\|_{[0,T]} \le K\|v\|_{[0,T]}$ for all v, T; see Primer D). Let Π be measurable, Hermitian-valued and bounded ($\|\Pi(j\omega)\|_2 \le c$ for all ω). Consider the loop $v = Gw + d_v$, $w = \tau\Delta(v) + d_w$ with external signals d_v, d_w. Suppose

- (i) for every $\tau \in [0,1]$ the loop is *well posed*: every (d_v, d_w) determines a unique causal solution (v, w);
- (ii) for every $\tau \in [0,1]$, $\tau\Delta$ satisfies the IQC defined by Π for every $v \in \mathcal{L}_2$;
- (iii) there is $\epsilon > 0$ with $\begin{bmatrix} G(j\omega) \\ I \end{bmatrix}^* \Pi(j\omega) \begin{bmatrix} G(j\omega) \\ I \end{bmatrix} \preceq -\epsilon I$ for all $\omega \in \mathbb{R}$.

Then the loop with $\tau = 1$ is *stable*: the map from (d_v, d_w) to (v, w) has a finite induced energy gain. *In words:* the plant strictly violates the quadratic inequality that the uncertainty satisfies, so no nonzero signal can circulate with growing energy. Assumption (iii) must hold with a uniform margin ϵ; (i) and (ii) must hold along the whole path $\tau \in [0,1]$ because the proof starts from the trivially stable open loop $\tau = 0$ and increases τ continuously.

The proof is a homotopy in τ; the dissipativity route of Scherer's tutorial replaces the homotopy by *hard* IQCs (valid on every finite horizon) and a storage function, so check which hypotheses a paper uses. A pointwise-in-time QC, such as the sector or slope constraint of a static nonlinearity, integrates to an IQC with a constant multiplier that holds on every finite horizon, i.e. a hard IQC. The converse fails: a constant multiplier alone does not make an IQC hard for a dynamic operator. The delay $w(t) = v(t-h)$, $h > 0$, with zero initial history ($v(t) = 0$ for $t < 0$) satisfies the IQC with $\Pi = \mathrm{diag}(-I, I)$ over the infinite horizon (input and output energies are equal), but on a finite horizon $[0,T]$ the integral is $\int_0^T (\|w\|^2 - \|v\|^2)\, dt = -\int_{\max(0,T-h)}^T \|v(t)\|^2 dt$, minus the input energy in the last delay interval, which is negative whenever the input has energy there. With the constant sector multiplier, the theorem reproduces the circle criterion; the Popov criterion (Primer D), for time-invariant sector nonlinearities, uses the frequency-dependent multiplier $1 + q\, j\omega$ with a scalar $q \ge 0$ chosen as part of the certificate; because this factor grows without bound in ω when $q > 0$, Popov is a separate theorem rather than a direct instance of the bounded-multiplier statement above. A *dynamic* multiplier (Π depends on ω) captures memory: for slope-restricted

nonlinearities the Zames–Falb class is the classical example, and Pauli et al., CDC 2021 bring its acausal FIR version (see Primer D) to neural-network loops. The computational recipe is always the same: factorise $\Pi = \Psi^* P \Psi$ (here P is the multiplier's middle matrix, not a storage matrix), append the state of the filter Ψ to the plant state, and the frequency-domain test becomes an LMI in a storage matrix X and the multiplier parameters P, via the KYP lemma of Section 6 or, in the time domain, via dissipativity and the S-procedure; the tutorial of Scherer, IEEE CSM 2022 is the reference for this route, which Module 14 follows.

THEOREM — A SCALAR FIR ZAMES–FALB IQC

Let $\varphi : \mathbb{R} \to \mathbb{R}$ be nondecreasing, globally Lipschitz, and $\varphi(0) = 0$. For any real two-sided square-summable sequence v, set $w_t = \varphi(v_t)$. Choose finitely many coefficients $h_k \geq 0$ ($k \in \mathbb{Z}$, $h_0 = 0$) with $\sum_k h_k < 1$ and define the FIR filter $(Hv)_t = \sum_k h_k v_{t-k}$; it is *acausal* if some h_k with $k < 0$ is nonzero, since it then uses future samples. Then

$$\sum_{t \in \mathbb{Z}} w_t \big(v_t - (Hv)_t\big) \geq 0.$$

With $M(e^{j\omega}) = 1 - \sum_k h_k e^{-jk\omega}$ this is the IQC with $\Pi = \begin{bmatrix} 0 & M^* \\ M & 0 \end{bmatrix}$, up to a positive factor. Zero-extending a one-sided sequence gives this infinite-horizon statement. For example, $(Hv)_t = \frac{1}{4} v_{t-1} + \frac{1}{4} v_{t+1}$ is admissible and acausal. Lipschitz continuity ensures $w \in \ell_2$; nonnegativity and the coefficient sum are essential to this sufficient class.

Intuition. Pairing ordered samples of a monotone graph gives at least as much inner product as shifting one sequence relative to the other. A nonnegative weighted sum of these shift inequalities gives the result. Future samples are allowed in an analysis filter; this theorem alone is neither an online implementation nor a hard finite-horizon IQC.

Scalar special case of the acausal FIR class of Pauli et al., CDC 2021.

THEOREM — A FACTORISED FREQUENCY TEST AS AN AUGMENTED LMI

Assume the chosen multiplier has a stable proper finite-dimensional real-rational factorisation $\Pi = \Psi^* P \Psi$ with a constant middle matrix $P = P^\top$, and G is stable proper real-rational. Realise the cascade $r = \Psi[G; I]w$ as $\dot{\chi} = A_a\chi + B_a w$, $r = C_a\chi + D_a w$, with A_a Hurwitz. Then the strict frequency inequality $[G; I]^*\Pi[G; I] \prec 0$, including its limit at infinity, is equivalent to the existence of a symmetric X satisfying

$$\begin{bmatrix} A_a^T X + X A_a & X B_a \\ B_a^T X & 0 \end{bmatrix} + \begin{bmatrix} C_a & D_a \end{bmatrix}^T P \begin{bmatrix} C_a & D_a \end{bmatrix} \prec 0.$$

No controllability is needed for this strict KYP form. It is affine in (X, P) only when the filter and plant realisations are fixed. A frequency factorisation alone does not imply a hard IQC. A direct finite-horizon dissipation proof additionally needs a valid finite-horizon IQC for this filter (including any terminal correction), the stated initial-state convention and a storage with the required nonnegative terminal contribution.

Intuition. The filter turns a frequency weight into an ordinary quadratic output supply. Adding its state makes that supply available to KYP. Stability and finite dimension are assumptions on the chosen factorisation, not automatic properties of an arbitrary bounded function of frequency.

Strict KYP lemma of Section 6 applied to the filtered plant; see Scherer, IEEE CSM 2022.

Derivation — The augmented plant–filter realisation ▾

With plant $\dot{x} = Ax + Bw$, $v = Cx + Dw$, split the filter input blocks as $B_\psi = [B_v\ B_w]$, $D_\psi = [D_v\ D_w]$, where the filter is $\dot{\eta} = A_\psi\eta + B_\psi[v; w]$, $r = C_\psi\eta + D_\psi[v; w]$, and put $\chi = (x, \eta)$. Substituting $v = Cx + Dw$ into the filter equations gives

$$\dot{\chi} = \underbrace{\begin{bmatrix} A & 0 \\ B_v C & A_\psi \end{bmatrix}}_{A_a}\chi + \underbrace{\begin{bmatrix} B \\ B_v D + B_w \end{bmatrix}}_{B_a} w, \qquad r = \underbrace{\begin{bmatrix} D_v C & C_\psi \end{bmatrix}}_{C_a}\chi + \underbrace{(D_v D + D_w)}_{D_a} w.$$

Use zero initial plant and filter states for the transfer-function identity. This triangular A_a is Hurwitz when A and A_ψ are Hurwitz. In the strict frequency test X is only required to be symmetric; a direct storage proof must separately justify its terminal sign.

Worked example — a Lipschitz bound for one neuron from a quadratic constraint ▾

Take the one-neuron network $f(x) = w_2\,\mathrm{ReLU}(w_1 x)$ with $w_1 = 2$ and $w_2 = 3$. Its true Lipschitz constant is $|w_1 w_2| = 6$, the slope where the neuron is on. Can the quadratic-constraint machinery find it without knowing where the neuron is on or off?

Step 1: the fact. ReLU is slope-restricted in $[0, 1]$. Let $\Delta v = w_1 \Delta x$ be the pre-activation increment and Δz the neuron's output increment. The incremental QC of the lemma above, with $\alpha = 0$ and $\beta = 1$, reads $2\lambda\,\Delta z\,(w_1 \Delta x - \Delta z) \geq 0$ for any $\lambda \geq 0$.

Step 2: what we want. $|\Delta f| \leq L\,|\Delta x|$, that is, $L^2 \Delta x^2 - w_2^2 \Delta z^2 \geq 0$ for all increments the neuron can produce.

Step 3: the S-procedure. It suffices that

$$L^2 \Delta x^2 - w_2^2 \Delta z^2 - 2\lambda\,\Delta z\,(w_1 \Delta x - \Delta z) \;\geq\; 0 \;\text{ for all } (\Delta x, \Delta z) \quad\Longleftrightarrow\quad \begin{bmatrix} L^2 & - \\ -\lambda w_1 & 2\lambda \end{bmatrix}$$

Step 4: solve. The 2×2 test needs $2\lambda - 9 \geq 0$ and a nonnegative determinant, $L^2(2\lambda - 9) - 4\lambda^2 \geq 0$. At $\lambda = 9/2$ the determinant is $-4\lambda^2 < 0$ whatever L is, so $\lambda > 9/2$, and then $L^2 \geq 4\lambda^2/(2\lambda - 9)$. Minimising over λ gives $\lambda = 9$ and $L^2 = 36$: $L = 6$**, the exact constant**. The matrix is then $\begin{bmatrix} 36 & -18 \\ -18 & 9 \end{bmatrix}$, with eigenvalues 0 and 45.

Any other admissible multiplier, $\lambda > 9/2$, still gives a *valid* bound, just a looser one: $\lambda = 12$ gives $L \approx 6.20$. Choosing the multipliers well is exactly what the SDP in LipSDP does, for thousands of neurons at once (Module 12).

6. Dissipativity & the KYP Lemma

In plain words. Think of a system as a tank of energy. The input pours energy in, the output takes energy out, and the tank's content, the storage V, can never be negative. A system is dissipative if the stored energy grows by at most what the supply rate s allows. With the supply rate $s = \gamma^2 \|w\|^2 - \|z\|^2$, dissipativity says "the output energy is at most γ^2 times the input energy": a gain

bound. For linear systems with quadratic storage, finding V is an LMI, and the KYP lemma links that LMI to the frequency response.

SYMBOLS IN THIS SECTION

- $x_{t+1} = f(x_t, w_t)$, $z_t = h(x_t, w_t)$: a system with state x, input w and output z.
- $V(x) \geq 0$: the storage function; $s(w, z)$: the supply rate; the dissipation inequality is $V(x_{t+1}) - V(x_t) \leq s(w_t, z_t)$.
- γ: an ℓ_2-gain bound, $\sum_t \|z_t\|^2 \leq \gamma^2 \sum_t \|w_t\|^2$ from zero initial state; (Q, S, R): the matrices of a general quadratic supply rate.
- $G(e^{j\omega})$: the frequency response. For a stable linear system the best γ is its peak magnitude.

Dissipativity is the energy bookkeeping that turns "the system cannot amplify its input by more than γ" into a matrix inequality, and it is the reason a stack of neural-network layers can be certified layer by layer.

DEFINITION — DISSIPATIVITY, STORAGE FUNCTION, SUPPLY RATE (WILLEMS 1972)

A system $x_{t+1} = f(x_t, w_t)$, $z_t = h(x_t, w_t)$ is **dissipative** with respect to the supply rate $s(w, z)$ if there is a **storage function** $V \geq 0$ with

$$V(x_{t+1}) - V(x_t) \;\leq\; s(w_t, z_t) \qquad \text{for all } t \text{ and all trajectories}$$

(continuous time: $\dot{V} \leq s(w, z)$). Summing the dissipation inequality over $t = 0, \ldots, N-1$ and using $V(x_N) \geq 0$ gives $\sum_{t<N} s(w_t, z_t) \geq -V(x_0)$: the net "energy" that can be extracted from the system, $-\sum_t s(w_t, z_t)$, never exceeds its initial storage $V(x_0)$. A dissipative system can store or dissipate the supplied energy, but it cannot generate energy (Willems, Arch. Rational Mech. Anal. 1972). A **QSR supply rate** is a quadratic form in the output and input,

$$s(w, z) = \begin{bmatrix} z \\ w \end{bmatrix}^\top \begin{bmatrix} Q & S \\ S^\top & R \end{bmatrix} \begin{bmatrix} z \\ w \end{bmatrix}.$$

Special cases: Lyapunov stability ($w \equiv 0$, $s = 0$: V is nonincreasing along trajectories, which proves Lyapunov stability of the equilibrium $x = 0$ when V is continuous and positive definite, $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$; a storage that is merely ≥ 0 proves nothing, since $V \equiv 0$ always qualifies); ℓ_2 gain at most γ ($Q = -I$, $S = 0$, $R = \gamma^2 I$, $s = \gamma^2 \|w\|^2 - \|z\|^2$); passivity ($Q = R = 0$, $S = I$, $s = 2w^\top z$). Papers differ on whether the vector is ordered (z, w) or (w, z); check before copying a (Q, S, R) triple.

For the gain supply the telescoped inequality reads $\sum_{t<N}\|z_t\|^2 \le \gamma^2 \sum_{t<N}\|w_t\|^2 + V(x_0)$; with $V(x_0)=0$ (in particular $x_0=0$ when $V(0)=0$, as for every quadratic storage) and $N\to\infty$ this is $\|z\|_2 \le \gamma\|w\|_2$, an ℓ_2-gain bound; otherwise the initial storage enters as the bias term $V(x_0)$. Dissipativity is thus a Lyapunov argument with an input, and, like Lyapunov's theorem, it becomes an LMI as soon as the system is linear and the storage is quadratic.

KEY EQUATION — THE DISSIPATIVITY LMI OF A DISCRETE-TIME LINEAR SYSTEM

For $x_{t+1}=Ax_t+Bw_t$, $z_t=Cx_t+Dw_t$ with storage $V(x)=x^\top Px$, $P\succeq 0$, the dissipation inequality holds for all (x_t,w_t) iff

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^\top \begin{bmatrix} P & 0 \\ 0 & -P \end{bmatrix} \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} - \begin{bmatrix} C & D \\ 0 & I \end{bmatrix}^\top \begin{bmatrix} Q & S \\ S^\top & R \end{bmatrix} \begin{bmatrix} C & D \\ 0 & I \end{bmatrix} \preceq 0,$$

i.e. $\begin{bmatrix} A^\top PA - P & A^\top PB \\ B^\top PA & B^\top PB \end{bmatrix} - \begin{bmatrix} C & D \\ 0 & I \end{bmatrix}^\top \Pi \begin{bmatrix} C & D \\ 0 & I \end{bmatrix} \preceq 0$ with Π the QSR matrix.

Derivation. $V(x^+)-V(x)-s(w,z)$ is the quadratic form of the displayed matrix in the vector (x,w), because $x^+=Ax+Bw$ and $(z,w)=\begin{bmatrix} C & D \\ 0 & I \end{bmatrix}(x,w)$; a quadratic form is nonpositive for all vectors iff its matrix is $\preceq 0$. The matrix is affine in (P,Π), so it is an LMI in P (for fixed supply) and also in the supply parameters, e.g. in γ^2. This is the structure of the certificate $\begin{bmatrix} I & 0 \\ A_{\mathrm{tot}} & B_{\mathrm{tot}} \\ C_{\mathrm{tot}} & D_{\mathrm{tot}} \end{bmatrix}^\top \operatorname{diag}(-X,X,P)\,[\cdot] \prec 0$ in the Zames–Falb paper and of the layer conditions in Pauli, Gramlich & Allgöwer, L4DC 2023, where a 1-D convolution is a finite-impulse-response system with nilpotent A.

Derivation — Reading the augmented Zames–Falb certificate

Let χ be the combined plant/filter state with $\chi^+=A_{\mathrm{tot}}\chi+B_{\mathrm{tot}}w$, let $r=C_{\mathrm{tot}}\chi+D_{\mathrm{tot}}w$ be the filter output, and let $V(\chi)=\chi^\top X\chi$. The dot in the certificate repeats the left factor $H=\begin{bmatrix} I & 0 \\ A_{\mathrm{tot}} & B_{\mathrm{tot}} \\ C_{\mathrm{tot}} & D_{\mathrm{tot}} \end{bmatrix}$, and $H(\chi,w)=(\chi,\chi^+,r)$, so

$$(\chi,w)^\top H^\top \operatorname{diag}(-X,X,P) H(\chi,w) = V(\chi^+)-V(\chi)+r^\top Pr.$$

Hence $H^\top\operatorname{diag}(-X,X,P)H\prec 0$ says $V(\chi^+)-V(\chi)<-r^\top Pr$ for every nonzero (χ,w): a dissipation inequality with supply $-r^\top Pr$. Summed over time, the IQC makes $\sum_t r_t^\top Pr_t\ge 0$, so

the storage must decrease; this is how the multiplier enters the stability proof.

DERIVATION: FROM THE DISSIPATION INEQUALITY TO THE ℓ2-GAIN LMI (CONTINUOUS TIME)

1. Supply | 2. Expand | 3. LMI | 4. Schur | 5. KYP

Step 1: The gain supply rate

For $\dot{x} = Ax + Bw$, $z = Cx + Dw$, a *sufficient* certificate for "the gain from w to z is at most γ" is dissipativity with respect to $s(w, z) = \gamma^2 \|w\|^2 - \|z\|^2$ with a quadratic storage $V(x) = x^\top P x$, $P \succeq 0$: it proves the energy-gain bound from zero initial state (see the insight below). The converse, that every such gain bound admits a quadratic certificate, needs the stability and KYP assumptions of step 5.

$$\dot{V}(x) \;\leq\; \gamma^2 \|w\|^2 - \|z\|^2 \qquad \text{for all } (x, w).$$

KEY INSIGHT

Integrating from 0 to T with $x(0) = 0$ gives $\int_0^T \|z\|^2 \, \mathrm{d}t \leq \gamma^2 \int_0^T \|w\|^2 \, \mathrm{d}t - V(x(T)) \leq \gamma^2 \int_0^T \|w\|^2 \, \mathrm{d}t$, the $\mathcal{L}_2$-gain bound. Nonnegativity of V is used exactly here.

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THEOREM — KALMAN–YAKUBOVICH–POPOV LEMMA (RANTZER 1996 FORM)

Let $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, $M \in \mathbb{S}^{n+m}$, with $\det(j\omega I - A) \neq 0$ for all $\omega \in \mathbb{R}$ and (A, B) controllable (see Primer D). Then the following are equivalent:

- (i) $\begin{bmatrix}(j\omega I - A)^{-1}B \\ I\end{bmatrix}^* M \begin{bmatrix}(j\omega I - A)^{-1}B \\ I\end{bmatrix} \preceq 0$ for all $\omega \in \mathbb{R} \cup \{\infty\}$;
- (ii) there exists $P \in \mathbb{S}^n$ with $M + \begin{bmatrix}A^\top P + PA & PB \\ B^\top P & 0\end{bmatrix} \preceq 0$.

The equivalence for strict inequalities holds even without controllability (Rantzer, Systems & Control Letters 1996). *In words:* a frequency-domain inequality that must hold at infinitely many frequencies is equivalent to one finite-dimensional LMI. With $M = \begin{bmatrix}C^\top C & C^\top D \\ D^\top C & D^\top D - \gamma^2 I\end{bmatrix}$, (i) reads $G(j\omega)^* G(j\omega) \preceq \gamma^2 I$ for all ω, with $G(s) = C(sI - A)^{-1}B + D$, and (ii) is the gain LMI of the walkthrough above without its side constraint $P \succeq 0$. If A is Hurwitz, (i) says exactly $\|G\|_\infty \leq \gamma$, i.e. the $\mathcal{L}_2$ gain is at most γ, and every P satisfying (ii) is automatically $\succeq 0$: the $(1,1)$ block gives $A^\top P + PA \preceq -C^\top C \preceq 0$, and for Hurwitz A this Lyapunov inequality forces $P \succeq 0$. This is the **bounded-real lemma**.

Stability cannot be dropped: $A = B = C = 1, D = 0$ satisfies (i) with $\gamma = 1$, since $|1/(j\omega - 1)| \leq 1$, and (ii) with $P = -1$, yet the system is unstable and its gain from w to z is unbounded. With the passivity supply $s = 2w^\top z$ instead, $\dot{V} - s \leq 0$ is (ii) with $M = \begin{bmatrix}0 & -C^\top \\ -C & -(D + D^\top)\end{bmatrix}$, and (i) reads $G(j\omega) + G(j\omega)^* \succeq 0$: the **positive-real lemma**.

The value at $\omega = \infty$ means the limiting inequality as $|\omega| \to \infty$. Since $(j\omega I - A)^{-1}B \to 0$, the stacked matrix tends to $[0; I]$, so this tests the lower-right block of M. It includes the direct input-output term even when the dynamic part disappears.

Proof sketch — The easy direction (ii) ⇒ (i), and what the hard direction buys ▾

Fix ω and write $\Phi(\omega) = \begin{bmatrix}(j\omega I - A)^{-1}B \\ I\end{bmatrix}$. Multiply (ii) from the left by Φ^* and from the right by Φ (congruence, Section 2, so $\preceq 0$ is preserved). The M-term becomes the left-hand side of (i). For the P-term, put $X = (j\omega I - A)^{-1}B$, so that $j\omega X = AX + B$, and compute

$$\Phi^* \begin{bmatrix}A^\top P + PA & PB \\ B^\top P & 0\end{bmatrix} \Phi = X^*(A^\top P + PA)X + X^* PB + B^\top PX = (AX + B)^* PX$$

So the P-term contributes nothing and (i) follows. ■ The same cancellation, in the time domain, is the statement that a quadratic storage function integrates to zero along a periodic (sinusoidal) steady state. The hard direction (i) $\Rightarrow$ (ii) is what makes SDP-based robust control complete rather than merely sufficient; Rantzer's proof is a separation argument on a convex set, in the same spirit as the S-lemma proof of Section 4. For the discrete-time version (no eigenvalues of A on the unit circle) replace $(j\omega I - A)^{-1}$ by $(e^{j\omega} I - A)^{-1}$, $\omega \in [0, 2\pi)$, and the *whole* P-term by $\begin{bmatrix} A^\top P A - P & A^\top P B \\ B^\top P A & B^\top P B \end{bmatrix}$, the storage block of the discrete-time dissipativity LMI above (replacing only the $(1,1)$ block would be wrong). The easy direction is the same computation: with $z = e^{j\omega}$ and $X = (zI - A)^{-1} B$ we have $zX = AX + B$, so the P-term contributes $(AX + B)^* P (AX + B) - X^* P X = (|z|^2 - 1) X^* P X = 0$.

Derivation — Why a Hurwitz A forces $P \succeq 0$ ▾

Suppose $A^\top P + P A \preceq 0$ (the $(1,1)$ block of (ii) when the $(1,1)$ block of M is $\succeq 0$, as for the gain and passivity supplies) and set $Q = -(A^\top P + P A) \succeq 0$. Along the unforced system $x(t) = e^{At} x_0$, $\frac{d}{dt}\, x^\top P x = x^\top (A^\top P + P A) x = -x^\top Q x$. Because A is Hurwitz, $x(t) \to 0$, and integrating from 0 to ∞ gives $x_0^\top P x_0 = \int_0^\infty x(t)^\top Q x(t)\, dt \geq 0$ for every x_0, so $P \succeq 0$.

THEOREM — POSITIVE-REAL LEMMA, STABLE CONTROLLABLE FORM (RANTZER 1996)

Let $A \in \mathbb{R}^{n\times n}$ be Hurwitz, $B \in \mathbb{R}^{n\times m}$, $C \in \mathbb{R}^{m\times n}$, $D \in \mathbb{R}^{m\times m}$, and assume (A, B) controllable. For the square transfer matrix $G(s) = C(sI - A)^{-1}B + D$, the condition $G(j\omega) + G(j\omega)^* \succeq 0$ at all real frequencies and at infinity is equivalent to the existence of $P = P^T \succeq 0$ with

$$\begin{bmatrix} A^T P + PA & PB - C^T \\ B^T P - C & -(D + D^T) \end{bmatrix} \preceq 0.$$

Equivalently, $V = x^T Px$ satisfies $\dot{V} \leq 2w^T z$ for every state and input. The implication from this storage inequality to the frequency inequality needs no controllability; that assumption supplies the nonstrict converse here.

Intuition. Input and output have matching dimensions so their inner product represents supplied power. The matrix inequality says that the stored energy cannot increase faster than this power supply. Hurwitz stability forces the KYP matrix to be a nonnegative storage, by the unforced-trajectory calculation above.

The KYP lemma above with the passivity supply (Rantzer 1996).

Incremental dissipativity and layer-wise composition

Replace the signals by *differences* of two trajectories, $\Delta x = x^a - x^b$, Δw, Δz, and the storage by a function of Δx: a system is **incrementally dissipative** if $V(\Delta x_{t+1}) - V(\Delta x_t) \leq s(\Delta w_t, \Delta z_t)$. For a static layer $y = f(u)$ there is no state and the inequality reduces to $s(\Delta u, \Delta y) \geq 0$; with the weighted gain supply $s = \|\Delta u\|^2_{X_{k-1}} - \|\Delta y\|^2_{X_k}$ (where $\|v\|^2_X = v^\top X v, X \succ 0$) this says $\|\Delta y\|_{X_k} \leq \|\Delta u\|_{X_{k-1}}$: a Lipschitz bound in weighted norms.

THEOREM — CASCADES OF INCREMENTALLY DISSIPATIVE LAYERS ARE LIPSCHITZ (LIPKERNEL, THEOREM 7)

Let layers $k = 1, \ldots, l$ map $u_k \mapsto y_k$ with $u_{k+1} = y_k$, and let each satisfy $\|\Delta u_k\|^2_{X_{k-1}} - \|\Delta y_k\|^2_{X_k} \geq 0$ for matrices $X_0, \ldots, X_l \succ 0$. Then $\|\Delta y_l\|^2_{X_l} \leq \|\Delta u_1\|^2_{X_0}$. With $X_0 = \rho^2 I$ and $X_l = I$ the network is ρ-Lipschitz (in LipKernel's convention ρ is the constant itself, not its square). *Proof.* Sum the layer inequalities. Because $\Delta u_{k+1} = \Delta y_k$, the term $-\|\Delta y_k\|^2_{X_k}$ of layer k cancels the term $+\|\Delta u_{k+1}\|^2_{X_k}$ of layer $k+1$, and the sum telescopes to $\|\Delta u_1\|^2_{X_0} - \|\Delta y_l\|^2_{X_l} \geq 0$. ■ (Pauli, Wang, Manchester & Allgöwer, Automatica 2026.)

KEY INSIGHT — THE INTERFACE MATRICES ARE WHAT BEATS THE PRODUCT BOUND

If every X_k is a scalar multiple of the identity, $X_k = c_k I$, layer k's condition says its Lipschitz constant is at most $\sqrt{c_{k-1}/c_k}$ and the telescoped bound is the product $\sqrt{c_0/c_l} = \rho$ of the layer-wise constants: the naive bound of Module 12. General matrices X_k let a layer be "loose" in directions the next layer attenuates, which is why layer-wise dissipativity with matrix interfaces (GLipSDP, Pauli, Gramlich & Allgöwer 2024, Theorem 1) is lossless with respect to full LipSDP for fully connected networks (made precise in the theorem below), and why LipKernel can enforce each layer's LMI by construction and still obtain a tighter end-to-end bound than the product of layer-wise constants (in LipKernel's two-layer illustration, bound 1 with ellipsoidal interfaces versus almost 2 with spherical ones; Module 13). Each layer LMI is exactly the dissipativity LMI above, with the activation handled by the slope-restriction QC and a diagonal multiplier: dissipativity supplies the storage, the QC supplies the abstraction, the S-procedure glues them, and the Schur complement makes the result linear.

THEOREM — SPLITTING A FULLY CONNECTED LIPSDP CERTIFICATE

Consider fixed affine weights in a feedforward chain $z_k = \varphi_k(W_k z_{k-1} + b_k)$, $k = 1, \dots, l$, and final affine output $W_{l+1} z_l + b_{l+1}$. Each φ_k is elementwise slope-restricted in $[0, 1]$. Use exactly the scalar sector QCs with nonnegative diagonal multipliers T_k and no cross-neuron couplings. For arbitrary increment variables $v_0, \dots, v_l$, define

$$q_k = 2 v_k^T T_k W_k v_{k-1} - 2 v_k^T T_k v_k.$$

The global LipSDP inequality $\rho \|v_0\|^2 - \|W_{l+1} v_l\|^2 - \sum_k q_k \geq 0$ for all increments is equivalent to existence of symmetric interface matrices $X_1, \dots, X_{l-1}$, with $X_0 = \rho I$, $X_l = W_{l+1}^\top W_{l+1}$, such that

$$\begin{bmatrix} X_{k-1} & -W_k^T T_k \\ -T_k W_k & 2T_k - X_k \end{bmatrix} \succeq 0, \qquad k = 1, \dots, l.$$

The interface matrices have no imposed diagonal or scalar structure; semidefinite boundary cases are allowed. Each X_{k-1} is automatically PSD as a principal block. Optimising over the same multiplier family therefore gives equal infimal values of ρ.

Intuition. Summing the local inequalities proves one direction. The converse decomposes the positive semidefinite block-tridiagonal global matrix along its chain, as in the cited splitting theorem. Thus splitting adds no loss to this SDP; the activation abstraction and S-procedure can still overestimate the true Lipschitz constant. Singular interface matrices give quadratic forms, not norms, but the telescoping argument still holds.

Pauli, Gramlich & Allgöwer 2024, Theorem 1 and Sec. IV.

Worked example — the gain of a first-order system from a storage function

System: $x_{t+1} = 0.5\, x_t + w_t$ with output $z_t = x_t$. Try storage $V(x) = p\, x^2$ with supply rate $s(w, z) = \gamma^2 w^2 - z^2$. The dissipation inequality $p(0.5x + w)^2 - p x^2 \leq \gamma^2 w^2 - x^2$ must hold for all x and w. Collecting terms gives

$$\begin{bmatrix} x \\ w \end{bmatrix}^\top \begin{bmatrix} 1 - 0.75p & p/2 \\ p/2 & p - \gamma^2 \end{bmatrix} \begin{bmatrix} x \\ w \end{bmatrix} \leq 0 \ \text{ for all } (x, w) \quad \Longleftrightarrow \quad \begin{bmatrix} 1 - 0.75p & p/2 \\ p/2 & p - \gamma^2 \end{bmatrix} \preceq$$

For a 2×2 matrix to be $\preceq 0$, both diagonal entries must be ≤ 0 and the determinant ≥ 0. The diagonal gives $p \geq 4/3$, but at $p = 4/3$ the determinant is $-4/9 < 0$, so $p > 4/3$. The determinant condition then becomes $\gamma^2 \geq p + p^2/(3p - 4)$. The right side is smallest at $p = 2$, where it equals 4. So $\gamma = 2$, certified by $V(x) = 2x^2$. The matrix is then $\begin{bmatrix} -0.5 & 1 \\ 1 & -2 \end{bmatrix}$, with eigenvalues 0 and -2.5.

Sanity check. A constant input $w_t = 1$ settles at $x = 0.5x + 1$, that is, $x = 2$: the output is twice the input. The frequency response $G(e^{j\omega}) = 1/(e^{j\omega} - 0.5)$ has its largest magnitude at $\omega = 0$, also $1/(1 - 0.5) = 2$. The storage-function LMI found the exact gain, as the KYP lemma promises for linear systems.

Walkthrough: From a Quadratic Constraint to an LMI

LipSDP in miniature. Take the scalar one-hidden-neuron network $f(x) = w_1\, \varphi(w_0 x + b_0) + b_1$ with φ slope-restricted in $[0, 1]$ (ReLU, tanh), and find the smallest ρ such that $|f(x) - f(y)|^2 \leq \rho\, |x - y|^2$ for all x, y. Every step is a section of this page; Module 12 repeats the same six steps with matrices.

DERIVATION: LIPSDP FOR ONE SCALAR NEURON

1. Setup | 2. QC | 3. Target | 4. S-procedure | 5. Schur | 6. Training

Step 1: Increments, and the bias disappears

Take two inputs $x, y \in \mathbb{R}$ and write $\Delta x = x - y$. The pre-activations differ by $\Delta v = (w_0 x + b_0) - (w_0 y + b_0) = w_0 \Delta x$: the bias cancels, which is why this QC-based certificate never depends on biases. (The true Lipschitz constant of a larger network can: $\mathrm{ReLU}(b - \mathrm{ReLU}(x))$ is constant for $b \leq 0$ but has slope -1 on $(0, b)$ for $b > 0$. For a single neuron it cannot, because shifting the argument of φ does not change its slopes.) Write $\Delta\varphi = \varphi(w_0 x + b_0) - \varphi(w_0 y + b_0)$ for the activation increment. Then $f(x) - f(y) = w_1 \Delta\varphi$, and the goal is the smallest ρ such that $w_1^2 \Delta\varphi^2 \leq \rho\, \Delta x^2$ for every pair of inputs.

$$f(x) - f(y) = w_1 \Delta\varphi, \qquad \text{find } \min \rho \text{ s.t. } w_1^2 \Delta\varphi^2 - \rho\, \Delta x^2 \leq 0 \ \ \forall x, y.$$

KEY INSIGHT

The unknown is the pair $(\Delta x, \Delta\varphi)$. The difficulty is that $\Delta\varphi$ depends on x and y through a nonlinear function we do not want to enumerate; the next step replaces it by an inequality.

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Interactive: S-Lemma in 2-D

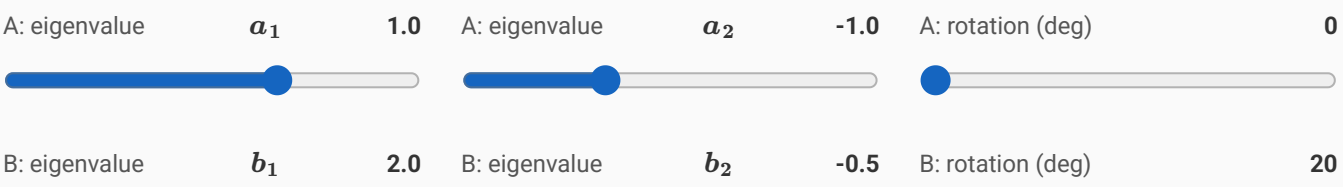
Two quadratic forms on $\mathbb{R}^2$: the constraint $\xi^\top A\xi \geq 0$ and the target $\xi^\top B\xi \geq 0$. Set each matrix through its eigenvalues and rotation, then move the multiplier τ and watch whether $B - \tau A \succeq 0$ (its two eigenvalues use a closed formula evaluated in floating point). Writing a nonzero $\xi = ru$ with $\|u\| = 1$ gives $\xi^\top A\xi = r^2\, u^\top Au$, so the sign of a homogeneous quadratic form depends only on the direction of ξ (and both forms vanish at $\xi = 0$); that is why everything is drawn on the unit circle.

Derivation — How the explorer searches for the best multiplier ▾

The button approximately maximises $h(\tau) = \lambda_{\min}(B - \tau A) = \min_{\|u\|=1}(u^T Bu - \tau u^T Au)$. This is concave because it is an infimum of affine functions of τ. Ternary search (see Primer B) compares two interior points and discards a third of the interval that cannot contain a better value. For these sliders (eigenvalues in $[-3, 3]$ in steps of 0.1), strict feasibility gives $\lambda_{\max}(A) \geq 0.1$ and $\|B\|_2 \leq 3$, so $h(\tau) \leq 3 - 0.1\tau$, while $h(0) \geq -3$; hence $h(\tau) < h(0)$ for $\tau > 60$, and no maximiser lies above 60.

Automatic maximisation is disabled when $A \preceq 0$: then every $u^\top Au \leq 0$, so $h(\tau)$ is nondecreasing and it may grow without bound, approach an unattained limit, or attain a maximum. For example, when $A = 0$, it is constant and every $\tau \geq 0$ is a maximiser. The current multiplier can still be checked directly.

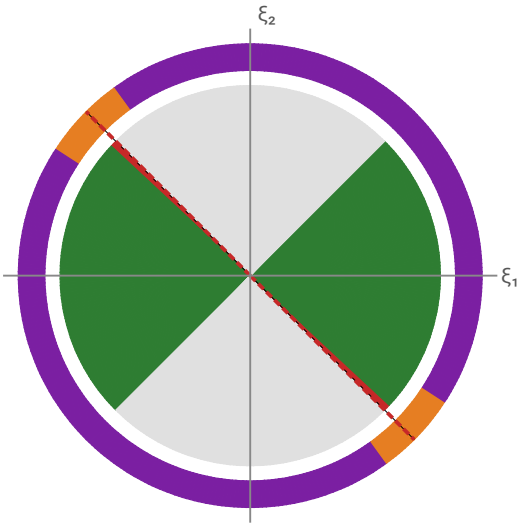
S-LEMMA IN 2-D



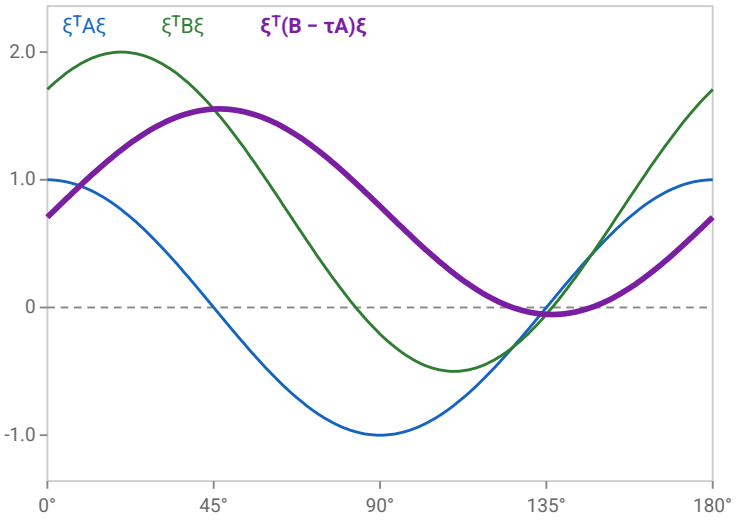
Multiplier τ **1.00**

Set τ to the best multiplier $\tau^\star$

A = [[1.00, 0.00], [0.00, -1.00]], eig = (-1.00, 1.00); **B** = [[1.71, 0.80], [0.80, -0.21]], eig = (-0.50, 2.00)
B − τA = [[0.71, 0.80], [0.80, 0.79]], eigenvalues (-0.05, 1.55) → **not PSD: this τ does not certify the implication**
Best multiplier: τ* = 0.96 with $\lambda_{min}(B - \tau^*A)$ = -0.05 (λ_{min} is concave in τ; approximately maximised by ternary search on [0, 60], which contains every maximiser for these slider ranges)
Implication (analytic minimum, evaluated with tolerance 1e-9: minimum of $\xi^T B\xi$ over the feasible unit directions, attained at an arc endpoint or at the minimiser of $\xi^T B\xi$): **fails**: min = -0.0535 at θ ≈ 135.00° (red dashed). Strict feasibility: $\max_\theta \xi^T A\xi = \lambda_{max}(A) = 1.00 > 0$, so the S-lemma applies.
S-lemma verdict: a multiplier exists ($\lambda_{min}(B - \tau^*A) \geq 0$) is **false**; the implication is **false** → **consistent, as the lemma guarantees**.



inner: $\xi^T A\xi \geq 0 \Rightarrow \xi^T B\xi \geq 0$? (green ok, red fails)
ring: $\xi^T(B - \tau A)\xi$ (purple ≥ 0, orange < 0)



direction angle θ of ξ = (cos θ, sin θ)

Left: directions where the constraint holds are coloured green if the target also holds and red if it fails; grey directions are outside the constraint set. The wedges are sampled per degree, so a very thin violation may not show; the verdict below uses analytic formulas evaluated in floating point, and a red dashed ray marks the worst feasible direction when the implication fails. The outer ring shows the sign of $\xi^\top(B - \tau A)\xi$: the certificate is valid iff the ring is entirely purple. Right: the three forms as functions of the angle; $B - \tau A \succeq 0$ iff the thick purple curve stays above zero.

Derivation — How the verdict tests every feasible direction

The implication test minimises analytically over direction rather than checking only the drawn wedges, but evaluates the formulas in floating point with tolerance 10^{-9}. For $M = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$, $u(\theta)^T M u(\theta) = (a + d)/2 + (a - d)\cos(2\theta)/2 + b\sin(2\theta)$. On each feasible angular

arc, its minimum occurs at an endpoint or an included minimum of this sinusoid. Values within the tolerance of zero are treated as zero; they are not exact proofs of boundary feasibility.

Interactive: Lyapunov LMI Feasibility

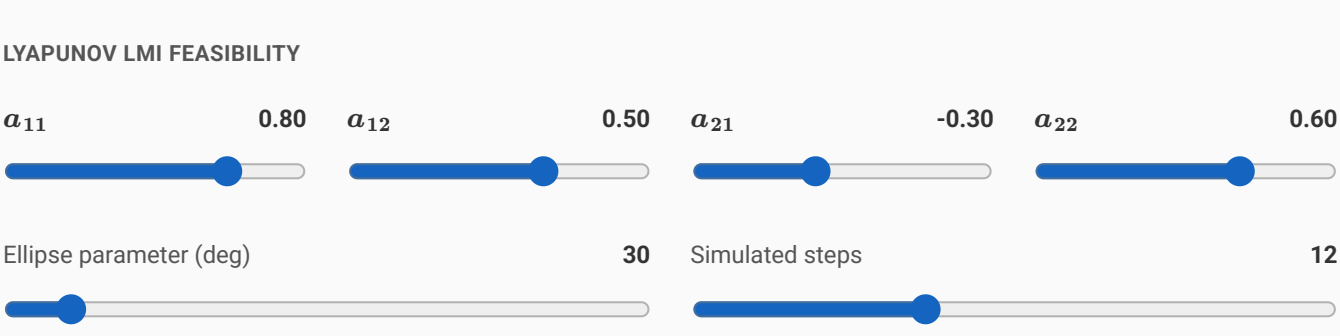
A 2×2 discrete-time system $x_{t+1} = Ax_t$. The explorer solves the Stein equation $A^\top PA - P = -I$ in closed form (three unknowns, one linear system), tests $P \succ 0$, and draws the eigenvalues of A against the unit circle, the certified level set $x^\top Px = 1$ and one trajectory. By the theorem of Section 2, $P \succ 0$ iff the LMI $A^\top PA - P \prec 0$, $P \succ 0$ is feasible iff A is Schur stable.

Derivation — The three linear equations the explorer solves ▾

Write $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $P = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$. Matching the three independent entries of $A^T PA - P = -I$ gives $(a^2 - 1)p + 2acq + c^2 r = -1$, $abp + (ad + bc - 1)q + cdr = 0$, and $b^2 p + 2bdq + (d^2 - 1)r = -1$. Gaussian elimination solves for (p, q, r). Positive definiteness then requires $p > 0$ and $pr - q^2 > 0$. If elimination meets a pivot below its numerical threshold, the explorer reports that no reliable unique solution was found. In exact arithmetic the Stein map $P \mapsto A^\top PA - P$ is singular iff $\lambda_i \lambda_j = 1$ for some pair of eigenvalues of A, which rules out Schur stability; a tiny pivot is only numerical evidence of (near-)singularity, not a proof.

Derivation — How the initial state is placed on the ellipse ▾

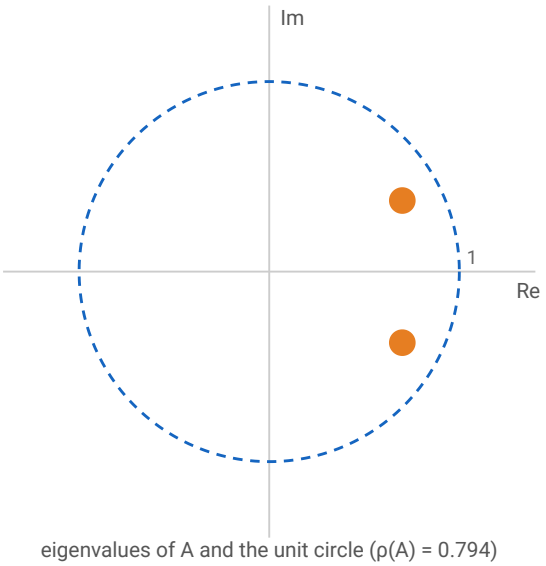
The slider controls the ellipse parameter θ: if P has eigenvalues $\mu_1, \mu_2 > 0$ with orthonormal eigenvectors u_1, u_2, then $x_0 = \frac{\cos\theta}{\sqrt{\mu_1}} u_1 + \frac{\sin\theta}{\sqrt{\mu_2}} u_2$. This guarantees $x_0^T Px_0 = 1$ but generally differs from the polar angle of x_0. Without a positive definite P, the initial point is instead chosen on a circle of radius 0.6. Hollow eigenvalue markers indicate values beyond the plotting range.



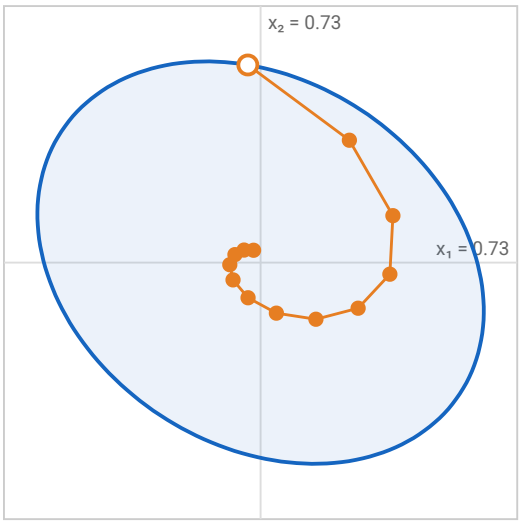
A = [[0.80, 0.50], [-0.30, 0.60]]: $\lambda_{1,2} = 0.700 \pm 0.374j$ (complex pair, $|\lambda|^2 = \det A$), spectral radius $\rho(A) = 0.794$ → **Schur stable**

Stein solution **P** = [[2.651, 0.707], [0.707, 3.260]], eigenvalues (2.186, 3.725) → **$P \succ 0$: LMI feasible, P is the certificate**. Decrease: $V(x_{t+1}) - V(x_t) = -\|x_t\|^2 \le -V(x_t)/\lambda_{max}(P)$, so $V(x_{t+1}) \le 0.732 \cdot V(x_t)$.

Along the trajectory: $V(x_0) = 1.000$, $V(x_1) = 0.684$, $V(x_2) = 0.500$, $V(x_3) = 0.342$, $V(x_4) = 0.207$, $V(x_5) = 0.114$, $V(x_6) = 0.063$, $V(x_7) = 0.041$, ….



eigenvalues of A and the unit circle (ρ(A) = 0.794)



level set $x^T P x = 1$ (blue) and the trajectory $x_{t+1} = A x_t$ (orange)

Left: eigenvalues of A (orange) and the unit circle. Right: the ellipse $x^\top P x = 1$ (blue) and the trajectory (orange) started on it; along a valid certificate $V(x_{t+1}) - V(x_t) = -\|x_t\|^2 < 0$, so the trajectory can never leave the ellipse. Try $a_{11} = a_{22} = 1.05, a_{12} = a_{21} = 0$: the Stein solution has negative entries and no certificate exists.

From the mathematics to a real decision

Learning objectives

- Use a quadratic storage function to certify a coupled thermal model over a whole operating set.
- Interpret an LMI through eigenvalues and distinguish its sufficient region from the physical constraint set.
- Carry a disturbance bound into an invariant radius instead of claiming nominal convergence.

A commissioning decision

Two temperature sensors monitor adjacent zones of the robot enclosure. Define the dimensionless deviations $x_1 = (T_1 - 40)/5$ and $x_2 = (T_2 - 40)/5$, with temperatures in degrees Celsius. The

permitted operating box is $|x_1| \le 1$, $|x_2| \le 1$, corresponding to 35–45 degrees in each zone. A candidate fixed controller has already been substituted into a linear minute-sampled model.

$$x_{t+1} = Ax_t, \qquad A = \begin{bmatrix} 0.6 & 0.1 \\ 0.1 & 0.7 \end{bmatrix}.$$

Assume this map is exact inside the operating box, there is no disturbance for the initial calculation, and the equilibrium really is $(40, 40)$ degrees. These assumptions make the example an analysis of a proposed closed loop, not a procedure for identifying its coefficients. The practical question is whether a specified family of initial temperatures stays inside the box while approaching equilibrium.

Use the quadratic energy $V(x) = x^\top P x$ with the proposed storage matrix $P = I$. Here I is the two-dimensional identity. A positive definite storage weights directions in state space; it is not physical heat energy unless a separate physical derivation supplies that interpretation. We need its decrease, rather than that extra interpretation, for this decision.

Worked decision, with its limits

Form the residual. The Lyapunov inequality is $A^\top P A - P \prec 0$. With $P = I$, multiplication gives

$$Q = I - A^\top A = \begin{bmatrix} 0.63 & -0.13 \\ -0.13 & 0.50 \end{bmatrix}, \qquad V(Ax) - V(x) = -x^\top Q x.$$

The two eigenvalues of Q are approximately 0.419656 and 0.710344, both positive. Therefore the energy decreases for every nonzero vector, rather than only for a plotted trajectory. Equivalently, the largest eigenvalue of the symmetric positive definite A is $q = 0.761803$, so $\|Ax\|_2 \le q\|x\|_2$. This establishes a contraction in the chosen Euclidean norm.

Connect energy to temperatures. If $V(x) \le 1$, then each coordinate satisfies $|x_i| \le \|x\|_2 \le 1$. Thus the unit disk lies inside the physical operating box. Its invariance follows because V decreases. The model was assumed only inside the box, but the induction remains valid: the initial state lies in the disk, the model maps it back into the disk, and the same assumption applies again.

Check one starting condition. At $x_0 = (0.6, -0.8)$, the temperatures are $(43, 36)$ degrees and $V(x_0) = 1$. The next state is $(0.28, -0.50)$, giving temperatures $(41.4, 37.5)$ and energy

$0.28^2 + 0.50^2 = 0.3284$. The specific decrease exceeds the uniform guarantee because this vector is not aligned with the slowest contracting direction.

Read the limitation. The disk is a sufficient certified region. A corner such as $(1, -1)$ lies in the permitted box but outside this disk because its energy is 2. That does not mean it is unsafe; the certificate simply does not cover that starting point. To enlarge the region one could seek a different P and prove its ellipsoid remains inside the box. Stability alone never supplies that containment argument.

A tempting wrong approach

> COMMON MISTAKE — IGNORING THE COUPLING
>
> Checking only $0.6 < 1$ and $0.7 < 1$ treats the sensors as independent scalar loops. Off-diagonal terms can defeat that reasoning: replacing both off-diagonal entries by 0.6 creates a symmetric matrix with a largest eigenvalue above 1. The original residual tests the complete map, including heat transferred between zones. Similarly, requiring every entry of Q to be positive would reject a valid certificate because its off-diagonal entries are negative.

Transfer the argument

▼ **Exercise 2.B1 — Medium: Certify a rectangular startup specification**

The startup procedure guarantees $|x_i| \le 0.7$. Show that every allowed startup is covered by the disk certificate. Give a uniform coordinate bound at startup and after one update.

Review: Lyapunov matrix inequalities.

▼ **Show hint**

Bound V over the box, then apply the induced-norm contraction to its largest possible radius.

▼ **Show worked solution**

Cover every permitted startup. Here $P = I$, so the disk certificate is $V(x) = x_1^2 + x_2^2 \le 1$. For any startup satisfying $|x_1|, |x_2| \le 0.7$, square the coordinate bounds and add:

$$V(x_0) \le (0.7)^2 + (0.7)^2 = 0.49 + 0.49 = 0.98 \le 1.$$

The maximum 0.98 is attained at each startup-box corner. Thus the whole startup box, including its boundary, lies in the certified invariant disk. The nominal model is assumed exact inside the operating box; disk containment and invariance keep every subsequent state in that domain.

Obtain a uniform bound from the storage. Each coordinate magnitude is at most the vector's Euclidean norm, since $x_i^2 \le x_1^2 + x_2^2$. Therefore the energy calculation gives

$$|x_{i,0}| \le \|x_0\|_2 \le \sqrt{0.98} \approx 0.989949.$$

The original startup specification supplies the sharper initial coordinate bound 0.7. The radius bound is useful because it does not depend on the direction of the state.

Propagate the radius one step. The symmetric matrix A has characteristic equation $\lambda^2 - 1.3\lambda + 0.41 = 0$, with eigenvalues $(13 \pm \sqrt{5})/20$. Both are positive, so its Euclidean induced norm is the larger one, $q = (13 + \sqrt{5})/20 \approx 0.761803$. Consequently,

$$|x_{i,1}| \le \|Ax_0\|_2 \le q\|x_0\|_2 \le q\sqrt{0.98} \approx 0.754147.$$

This is a valid common coordinate bound after one update. If we also use the rectangular startup information directly, the triangle inequality gives $|x_{1,1}| \le 0.6(0.7) + 0.1(0.7) = 0.49$ and $|x_{2,1}| \le 0.1(0.7) + 0.7(0.7) = 0.56$. Thus 0.56 is a sharper common one-step bound for this particular startup box; the storage bound remains convenient for subsequent arbitrary directions.

Read the temperature units. Since $T_i = 40 + 5x_i$, a dimensionless coordinate bound b means $40 - 5b \le T_i \le 40 + 5b$ degrees Celsius. The startup bound 0.7 gives 36.5–43.5 degrees, and the direct common one-step bound 0.56 gives 37.2–42.8 degrees. All are inside the permitted 35–45 range.

▼ Exercise 2.B2 — Hard: Replace convergence by a disturbance tube

Now assume $x_{t+1} = Ax_t + w_t$ with $\|w_t\|_2 \le 0.05$ every minute. Derive an invariant Euclidean radius and decide whether the unit disk remains invariant. Can every trajectory still converge to zero?

Review: Storage and gain bounds.

▼ Show hint

Use $\|Ax + w\|_2 \le q\|x\|_2 + 0.05$. Consider a constant disturbance along a unit eigenvector for q.

▼ Show worked solution

Include the disturbance at every step. Let $r_t = \|x_t\|_2$. The induced-norm bound for the stated A, followed by the triangle inequality, gives

$$r_{t+1} = \|Ax_t + w_t\|_2 \le \|Ax_t\|_2 + \|w_t\|_2 \le qr_t + 0.05, \qquad q = \frac{13+\sqrt{5}}{20} < 1.$$

To make a disk of radius $R \ge 0$ invariant, we need the right-hand side to be at most R whenever $r_t \le R$. Since $q \ge 0$, its largest value on that disk is $qR + 0.05$. Solve

$$qR + 0.05 \le R \iff 0.05 \le (1-q)R \iff R \ge R_* := \frac{0.05}{1-q} = \frac{1}{7-\sqrt{5}} = \frac{7+\sqrt{5}}{44} \approx$$

Division preserves the inequality because $1 - q > 0$. Every radius $R \ge R_*$ passes this algebraic norm test. To apply it using the lab's locally valid model, the disk must also lie inside the operating box; $R \le 1$ is sufficient. Thus $R_* \le R \le 1$ gives invariant disks fully covered by the stated model assumption. A larger disk may contain states outside that domain, so the norm calculation alone would not certify their updates. In particular, for $R = 1$, $qR + 0.05 \approx 0.811803 \le 1$, so the unit disk remains invariant for every disturbance in the given ball. As that disk lies in the operating box, induction also keeps the local model assumption applicable.

Show the transient and ultimate bounds. Iterating the scalar inequality yields $r_1 \le qr_0 + 0.05$, $r_2 \le q^2 r_0 + 0.05(q+1)$, and in general

$$r_t \le q^t r_0 + 0.05 \sum_{j=0}^{t-1} q^j = q^t r_0 + \frac{0.05(1-q^t)}{1-q} = R_* + q^t(r_0 - R_*).$$

The sum follows by multiplying the finite geometric sum by q and subtracting it from the original sum. Because $q^t \to 0$, this proves $\limsup_{t\to\infty} r_t \le R_*$. If a state starts outside the smallest disk, the bound approaches its radius; it does not assert entry into that exact disk in a finite number of steps.

Why convergence to zero is no longer guaranteed. Take a unit eigenvector v of A with $Av = qv$, and the allowed constant disturbance $w_t = 0.05v$. At $x_* = R_* v$ we have

$$Ax_* + w_t = (qR_* + 0.05)v = R_* v = x_*.$$

This nonzero equilibrium lies inside the unit disk and satisfies the model assumptions. Starting there produces a trajectory that never converges to zero. It also shows that no smaller centered Euclidean disk is robustly invariant: at the point Rv with $R < R_*$ the same disturbance makes the next radius $qR + 0.05 > R$. The justified conclusion is an invariant disturbance tube and an ultimate bound. Convergence to zero would need further conditions, for example disturbances tending to zero.

Synthesis and bridge

An LMI is useful when its matrix inequality closes the whole decision chain: decrease of storage, an invariant sublevel set, containment in physical limits, and applicability of the model on that set. Numerical feasibility is evidence for only one link. The calculation also shows how the same storage can answer a revised commissioning request after disturbances are introduced.

Next, imagine the heat-transfer residual is learned rather than exactly known. A small posterior variance does not automatically replace the disturbance assumption. The GP toolkit explains which extra assumptions turn a statistical model into an error envelope usable in this kind of set argument.

Exercises

Graded practice — Build the argument yourself

There are 12 new problems: four Easy, four Medium, and four Hard. Start with the level that lets you make progress without opening the solution. Easy problems rebuild individual operations; Medium problems combine them; Hard problems ask for proofs, boundary cases, and the limits of a guarantee. The required concepts are explained on this page or in the linked earlier material.

Easy — Warm up one skill at a time.

▼ **Exercise 2.P1 — Easy: KKT with one active constraint**

Minimize $(x-2)^2$ over $x \le 1$. Write the Lagrangian with constraint $x - 1 \le 0$, and find the optimum and multiplier using stationarity, feasibility and complementary slackness.

Review if needed: Primer B: constrained optimization. **Apply here:** this module's explanation.

▼ **Show hint**

The unconstrained minimizer is excluded, so check the endpoint $x = 1$.

▼ **Show worked solution**

> The Lagrangian is $L(x, \lambda) = (x-2)^2 + \lambda(x-1)$ with $\lambda \ge 0$. Stationarity is $2(x-2) + \lambda = 0$. The feasible point closest to 2 is $x^\star = 1$, with value 1; substituting gives $\lambda^\star = 2$.
>
> Primal feasibility holds at equality, dual feasibility holds because $2 \ge 0$, and complementary slackness is $2(1-1) = 0$. The objective is convex and the constraint affine; a strictly feasible point is $x = 0$. Thus the KKT conditions certify a global optimum. The nonzero price reflects that relaxing the bound lets the objective improve.

▼ **Exercise 2.P2 — Easy: Positive entries do not mean PSD**

Compare $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Compute their eigenvalues and identify which is PSD. Find one vector exposing the negative direction of the other.

Review if needed: Primer A: positive semidefiniteness. **Apply here:** this module's explanation.

▼ **Show hint**

For equal diagonal entries, $(1, 1)$ and $(1, -1)$ are eigenvectors.

▼ **Show worked solution**

Choose the directions explicitly. For both matrices, use $q_+ = (1, 1)^\top$ and $q_- = (1, -1)^\top$. They are linearly independent, so two eigenvalues found along these directions account for the complete spectrum of each 2×2 matrix.

Multiply by A. Direct matrix-vector products give

$$Aq_+ = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3q_+, \qquad Aq_- = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = q_-.$$

Thus A has eigenvalues 3 and 1. Both are strictly positive, so $A \succ 0$ and in particular $A \succeq 0$.

Multiply by B. In the same directions,

$$Bq_+ = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3q_+, \qquad Bq_- = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -q_-.$$

The eigenvalues are 3 and -1, so B is indefinite and is not PSD. The requested negative witness is $v = q_-$:

$$v^\top B v = (1, -1)(-1, 1)^\top = -1 - 1 = -2 < 0.$$

PSD requires a nonnegative quadratic form in every direction. Positive matrix entries do not establish that condition: B has positive entries but a negative direction.

▼ Exercise 2.P3 — Easy: First Schur-complement calculation

Find all real t for which $M(t) = \begin{bmatrix} t & 2 \\ 2 & 1 \end{bmatrix} \succeq 0$. When is it positive definite?

Review if needed: Primer A: block matrices. **Apply here:** this module's explanation.

▼ Show hint

Use the bottom-right block 1, which is strictly positive.

▼ Show worked solution

The Schur complement of the bottom-right entry is $t - 2(1^{-1})2 = t - 4$. Since that corner is positive, $M(t) \succeq 0$ exactly when $t - 4 \geq 0$, or $t \geq 4$. Strict positive definiteness requires $t > 4$.

At $t = 4$, the matrix is $vv^\top$ for $v = (2, 1)^\top$: it is PSD but singular. The vector $(1, -2)^\top$ lies in its nullspace. Checking this boundary case helps keep semidefinite and definite conclusions separate.

▼ Exercise 2.P4 — Easy: Read a scalar sector constraint

Let $y = \max\{0, x\}$. Verify $y(x - y) \geq 0$ separately for $x \geq 0$ and $x < 0$. Write the inequality as a quadratic form in $z = (x, y)^\top$.

Review if needed: Primer E: ReLU activation. **Apply here:** this module's explanation.

▼ Show hint

For ReLU the product is zero in both cases; the cross term of $z^\top Q z$ is twice the off-diagonal entry.

▼ Show worked solution

If $x \geq 0$, then $y = x$, so $y(x - y) = x \cdot 0 = 0$. If $x < 0$, then $y = 0$ and the product is again zero. Hence ReLU obeys the sector inequality.

$$y(x - y) = xy - y^2 = \begin{bmatrix} x \\ y \end{bmatrix}^\top \begin{bmatrix} 0 & 1/2 \\ 1/2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \geq 0.$$

The quadratic constraint also admits many pairs that are not on the ReLU graph, such as $(x, y) = (2, 1)$. Certificates built only from it deliberately cover a larger set of input-output pairs.

Medium — Combine definitions and compute a certificate.

▼ **Exercise 2.P5 — Medium: A scalar Lyapunov LMI**

For $x_{t+1} = 0.8x_t$, find $P > 0$ satisfying $0.8^2 P - P = -1$. Show how the result proves decrease of $V(x) = Px^2$. Would any $P > 0$ work for the strict decrease inequality?

Review if needed: Primer D: Lyapunov decrease. **Apply here:** this module's explanation.

▼ **Show hint**

Solve $-0.36P = -1$, then separate the normalization from the sign condition.

▼ **Show worked solution**

> The normalized equation gives $P = 1/0.36 = 25/9 \approx 2.77778$. Along the system, $V(x_{t+1}) - V(x_t) = (0.64 - 1)Px_t^2 = -x_t^2$, strictly negative for $x_t \neq 0$. Since $P > 0$, V is positive definite and certifies asymptotic stability.
>
> Every $P > 0$ satisfies the strict inequality because $-0.36P < 0$. The equation with right-hand side -1 chooses one scale for the certificate; stability depends on the sign, not that normalization. For an unstable scalar factor 1.1, the coefficient becomes $+0.21$, and no positive P can give decrease.

▼ Exercise 2.P6 — Medium: An S-procedure certificate on an interval

Prove that $1 - x^2 \geq 0$ implies $2 - x^2 \geq 0$ using a nonnegative multiplier τ and a globally nonnegative residual. Then find every $\tau \geq 0$ for which the residual works.

Review if needed: Primer 0: implications. **Apply here:** this module's explanation.

▼ Show hint

Expand $(2 - x^2) - \tau(1 - x^2)$ and require its constant and quadratic coefficients to be nonnegative.

▼ Show worked solution

Look for a globally nonnegative residual. The S-procedure subtracts a nonnegative multiple of the premise from the desired conclusion. Expanding gives

$$r_\tau(x) = (2 - x^2) - \tau(1 - x^2) = (2 - \tau) + (\tau - 1)x^2.$$

For this polynomial to be nonnegative for every real x, first set $x = 0$: then $2 - \tau \geq 0$, so $\tau \leq 2$. Next, if $\tau - 1 < 0$, the negative quadratic term dominates the constant as $|x| \to \infty$, making the residual negative. Thus $\tau \geq 1$ is also necessary. Conversely, when $1 \leq \tau \leq 2$, both coefficients are nonnegative and $x^2 \geq 0$, so the residual is nonnegative everywhere. Exactly the multipliers in $[1, 2]$ work.

Use the certificate on the interval. At any x with $1 - x^2 \geq 0$,

$$2 - x^2 = r_\tau(x) + \tau(1 - x^2) \geq 0 + 0 = 0.$$

The first term is nonnegative globally, and the second is nonnegative because both τ and the premise are nonnegative. For example, $\tau = 1$ gives residual 1; the boundary choice $\tau = 2$ gives residual x^2. Either proves the implication, including $x = \pm 1$.

This sufficiency argument needs no losslessness theorem. The one-constraint S-lemma would also supply the converse because the premise is strictly satisfied at $x = 0$, where $1 - x^2 = 1 > 0$.

▼ Exercise 2.P7 — Medium: Lift a quadratic epigraph into an LMI

For $x = (x_1, x_2)^\top$ and scalar t, express $x_1^2 + x_2^2 \le t$ as an LMI with an identity block. Verify the least feasible t when $x = (3, 4)^\top$.

Review if needed: Primer B: convex sets and epigraphs. **Apply here:** this module's explanation.

▼ Show hint

Put t in the top-left corner and x in the off-diagonal blocks.

▼ Show worked solution

$$\begin{bmatrix} t & x^\top \\ x & I_2 \end{bmatrix} = \begin{bmatrix} t & x_1 & x_2 \\ x_1 & 1 & 0 \\ x_2 & 0 & 1 \end{bmatrix} \succeq 0.$$

The bottom-right block is $I_2 \succ 0$, so the Schur complement gives $t - x^\top I_2^{-1} x = t - x_1^2 - x_2^2 \ge 0$. The larger matrix is affine in all three unknowns, even though the original inequality contains squares.

For $(3, 4)$, $x^\top x = 9 + 16 = 25$, so the least feasible value is $t = 25$. At equality the block matrix is singular; requiring it to be positive definite would impose the stricter condition $t > 25$.

▼ Exercise 2.P8 — Medium: Turn energy bookkeeping into a gain bound

A discrete-time system obeys $V_{t+1} - V_t \le 4w_t^2 - z_t^2$, with $V_t \ge 0$ and $V_0 = 0$. Derive a finite-horizon bound between input and output energies. If $\sum_{t=0}^{T-1} w_t^2 = 3$, what output-energy bound follows?

Review if needed: Primer D: signal energy and gain. **Apply here:** this module's explanation.

▼ Show hint

Sum the storage differences; the intermediate terms telescope.

▼ Show worked solution

Summing from $t = 0$ to $T - 1$ gives $V_T - V_0 \le 4 \sum w_t^2 - \sum z_t^2$. Rearranging and using $V_0 = 0$, $V_T \ge 0$ yields $\sum z_t^2 \le 4 \sum w_t^2 - V_T \le 4 \sum w_t^2$.

The output energy is therefore at most $4(3) = 12$. Taking square roots gives a signal-norm gain bound of 2, not 4. With nonzero initial storage the bound would instead contain $+V_0$, representing energy initially present in the system. An infinite-horizon statement follows by taking limits when the input has finite total energy.

Hard — Explain why the argument works and where it stops.

▼ **Exercise 2.P9 — Hard: Compute the whole dual function**

For the first practice problem, compute $g(\lambda) = \inf_{x \in \mathbb{R}} [(x-2)^2 + \lambda(x-1)]$ for $\lambda \geq 0$. Maximize it and compare with the constrained primal value.

Review if needed: Primer B: derivatives and minimizers. **Apply here:** this module's explanation.

▼ **Show hint**

Complete the square in x, or substitute the stationary point $x = 2 - \lambda/2$.

▼ **Show worked solution**

$$L(x, \lambda) = \left(x - 2 + \frac{\lambda}{2}\right)^2 + \lambda - \frac{\lambda^2}{4}, \qquad g(\lambda) = \lambda - \frac{\lambda^2}{4}.$$

The squared term has infimum zero, attained at $x = 2 - \lambda/2$. The dual function is a concave parabola. Its derivative is $1 - \lambda/2$, so its maximum on $\lambda \geq 0$ occurs at $\lambda = 2$, with $g(2) = 1$.

The primal optimum is also 1 at $x = 1$, giving zero gap and primal recovery. For any other nonnegative λ, $1 - g(\lambda) = (\lambda - 2)^2/4 \geq 0$: every price supplies a valid lower bound, while the best price makes it exact.

▼ Exercise 2.P10 — Hard: A singular corner needs a range check

For real numbers a, b, characterize when $\begin{bmatrix} a & b \\ b & 0 \end{bmatrix} \succeq 0$. Explain why replacing the inverse of the zero corner by zero would give an incomplete test.

Review if needed: Primer A: PSD quadratic forms. **Apply here:** this module's explanation.

▼ Show hint

Test the quadratic form on $(1, t)^\top$ for arbitrary t.

▼ Show worked solution

The quadratic form at $(1, t)^\top$ is $a + 2bt$. If $b \neq 0$, choosing t with the opposite sign and sufficiently large magnitude makes it negative. Therefore PSD forces $b = 0$. Once $b = 0$, the matrix is PSD exactly when $a \geq 0$.

Using a pseudoinverse of the zero corner gives only the condition $a \geq 0$ and misses the necessary $b = 0$. The general semidefinite Schur-complement rule needs a range condition in addition to the pseudoinverse. The definite-corner version avoids this issue because an invertible corner has the entire space as its range.

▼ Exercise 2.P11 — Hard: Compose a slope bound with weights

Let $F(x) = 3\varphi(2x + 1) - 4$, where φ is slope-restricted in $[0, 1]$, meaning $0 \leq [\varphi(a) - \varphi(b)]/(a - b) \leq 1$ for $a \neq b$. Prove that F is 6-Lipschitz and its squared-gain bound is $\rho = 36$. Is the bound attained for ReLU?

Review if needed: Primer B: Lipschitz continuity. **Apply here:** this module's explanation.

▼ Show hint

Biases cancel in differences. Apply the slope bound to the two hidden inputs, then multiply by the output weight.

▼ Show worked solution

Slope restriction implies $|\varphi(a) - \varphi(b)| \leq |a - b|$. Hence $|F(x) - F(y)| = 3|\varphi(2x + 1) - \varphi(2y + 1)| \leq 3|2x - 2y| = 6|x - y|$. Squaring gives $|F(x) - F(y)|^2 \leq 36|x - y|^2$, so the page's squared convention is $\rho = 36$.

For ReLU, choose two distinct inputs $x, y > -1/2$. Both hidden inputs are positive, and $F(x) - F(y) = 6(x - y)$, so the ratio is exactly 6 and no smaller global Lipschitz constant works. A certificate that reports $\rho = 36$ must be square-rooted before being read as a norm gain.

▼ **Exercise 2.P12 — Hard: A feasible certificate can be conservative**

Consider $A = \begin{bmatrix} 0.5 & 2 \\ 0 & 0.5 \end{bmatrix}$. Show that the system is Schur stable but $P = I$ fails the Lyapunov decrease test. Solve $A^\top P A - P = -I$ for symmetric $P = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$ and check $P \succ 0$.

Review if needed: Primer D: eigenvalues and discrete-time stability. **Apply here:** this module's explanation.

▼ **Show hint**

The diagonal entries give the eigenvalues. Match the three independent entries of the Stein equation in order p, q, r.

▼ **Show worked solution**

Separate stability from this particular storage choice. The matrix is upper triangular, so its characteristic polynomial is $(0.5-\lambda)^2$. Both eigenvalues are 0.5, and their moduli are below one. Hence A is Schur stable, even though its off-diagonal coupling can cause an initial increase in Euclidean length.

With $P = I$, calculate

$$A^\top A - I = \begin{bmatrix} 0.5 & 0 \\ 2 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 2 \\ 0 & 0.5 \end{bmatrix} - I = \begin{bmatrix} -3/4 & 1 \\ 1 & 13/4 \end{bmatrix}.$$

At $x = (0,1)^\top$, the quadratic form is $13/4 > 0$. Equivalently, $Ax = (2, 0.5)^\top$ has squared norm $4 + 0.25 = 4.25$, compared with the initial squared norm 1. Therefore $P = I$ fails strict Lyapunov decrease.

Solve for a weighted storage. Write $P = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$. Multiply the matrices before matching entries:

$$PA = \begin{bmatrix} p/2 & 2p + q/2 \\ q/2 & 2q + r/2 \end{bmatrix}, \qquad A^\top PA = \begin{bmatrix} p/4 & p + q/4 \\ p + q/4 & 4p + 2q + r/4 \end{bmatrix}.$$

Thus the three independent entries of $A^\top PA - P = -I$ are

$$-\frac{3}{4}p = -1, \qquad p - \frac{3}{4}q = 0, \qquad 4p + 2q - \frac{3}{4}r = -1.$$

From the first equation, $p = 4/3$. From the second, $q = (4/3)p = 16/9$. In the third, move the terms to obtain

$$\frac{3}{4}r = 1 + 4p + 2q = 1 + \frac{16}{3} + \frac{32}{9} = \frac{9 + 48 + 32}{9} = \frac{89}{9}, \qquad r = \frac{4}{3}\frac{89}{9} = \frac{356}{27}.$$

Check positive definiteness. For a symmetric 2×2 matrix, a positive first diagonal entry and positive determinant prove positive definiteness. Here

$$p = \frac{4}{3} > 0, \qquad pr - q^2 = \frac{4}{3}\frac{356}{27} - \left(\frac{16}{9}\right)^2 = \frac{1424 - 256}{81} = \frac{1168}{81} > 0.$$

Therefore $P \succ 0$, and by construction $V(Ax) - V(x) = x^\top (A^\top P A - P)x = -\|x\|_2^2$, strictly negative for nonzero x. Stability is certified by this weighted energy. Failure of the identity matrix was failure of one candidate; it was not infeasibility of the complete Lyapunov LMI.

Further practice — Original problems and research connections

The original exercises below retain their numbering. Some compare later methods or ask for longer research derivations; use the graded set above first, and return to a research-connection problem after reading the relevant linked module.

▼ **Exercise 2.1 — Weak duality for a maximisation problem (derivation)**

Consider the RL-style problem $p^\star = \sup_{\pi \in \Pi} J_r(\pi)$ s.t. $J_c(\pi) \le d$, with an arbitrary nonempty (possibly non-convex) set Π and bounded J_r, J_c. Supremum notation is needed because this policy class need not attain its best value. Define $L(\pi, \lambda) = J_r(\pi) - \lambda(J_c(\pi) - d)$ and $D(\lambda) = \sup_{\pi \in \Pi} L(\pi, \lambda)$. (a) Prove $D(\lambda) \ge p^\star$ for every $\lambda \ge 0$. (b) Prove that D is convex. (c) Explain which step would fail if $\lambda < 0$ were allowed.

▼ **Show answer**

(a) Let $\tilde{\pi}$ be feasible, $J_c(\tilde{\pi}) \le d$. Then $-\lambda(J_c(\tilde{\pi}) - d) \ge 0$ because $\lambda \ge 0$ and $J_c(\tilde{\pi}) - d \le 0$, so $L(\tilde{\pi}, \lambda) \ge J_r(\tilde{\pi})$. Hence $D(\lambda) = \sup_\pi L(\pi, \lambda) \ge L(\tilde{\pi}, \lambda) \ge J_r(\tilde{\pi})$. Taking the supremum over feasible $\tilde{\pi}$ gives $D(\lambda) \ge p^\star$. No property of Π was used, which is why the bound holds for neural-network policy classes.

(b) For fixed π, $\lambda \mapsto L(\pi, \lambda) = J_r(\pi) + \lambda d - \lambda J_c(\pi)$ is affine. A pointwise supremum of affine functions is convex: for $\theta \in [0, 1]$, $L(\pi, \theta\lambda + (1-\theta)\lambda') = \theta L(\pi, \lambda) + (1-\theta) L(\pi, \lambda') \le \theta D(\lambda) + (1-\theta) D(\lambda')$, and taking the supremum of the left side over π gives $D(\theta\lambda + (1-\theta)\lambda') \le \theta D(\lambda) + (1-\theta) D(\lambda')$.

(c) Only the sign step in (a): with $\lambda < 0$ and a strictly feasible $\tilde{\pi}$ ($J_c < d$) the term $-\lambda(J_c - d)$ is negative, so $L(\tilde{\pi}, \lambda) < J_r(\tilde{\pi})$ and the chain breaks. Convexity in (b) survives, but $D(\lambda)$ with $\lambda < 0$ is no longer guaranteed to upper-bound $p^\star$, so minimising over all real λ can undershoot $p^\star$. Example: one step, two actions with (reward, cost) $= (1, 0)$ and $(0, 2)$, threshold $d = 1$. The first action is feasible, so $p^\star = 1$, and $D(\lambda) = \max\{1 + \lambda,\ -\lambda\}$ (a mixed policy never beats the better action, because L is linear in the mixing weights). Over $\lambda \ge 0$ the minimum is $D(0) = 1 = p^\star$; over all real λ it is $D(-\tfrac{1}{2}) = \tfrac{1}{2} < p^\star$, a false "bound". (Whether the unrestricted problem is bounded below at all depends on the instance.) This is why the projection $[\cdot]_+$ in the dual update is not optional.

▼ **Exercise 2.2 — The dual of a small LP**

Derive the Lagrange dual of $\min 2x_1 + 3x_2$ s.t. $x_1 + x_2 \geq 4, x_1 + 3x_2 \geq 6, x \geq 0$ (keep $x \geq 0$ as the domain $\mathcal{D}$). Solve both problems and verify zero gap and complementary slackness.

▼ **Show answer**

Form the dual with the stipulated domain. Keep $x_1, x_2 \geq 0$ as the domain of the infimum. Write the two other inequalities as $4 - x_1 - x_2 \leq 0$ and $6 - x_1 - 3x_2 \leq 0$. Their multipliers y_1, y_2 are nonnegative, giving

$$\begin{aligned} L(x, y) &= 2x_1 + 3x_2 + y_1(4 - x_1 - x_2) + y_2(6 - x_1 - 3x_2) \\ &= 4y_1 + 6y_2 + (2 - y_1 - y_2)x_1 + (3 - y_1 - 3y_2)x_2. \end{aligned}$$

For a scalar coefficient c, $\inf_{u \geq 0} cu = 0$ if $c \geq 0$, attained at $u = 0$; if $c < 0$, taking $u \to +\infty$ sends cu to $-\infty$. Apply this independently to the two coordinates:

$$g(y) = \begin{cases} 4y_1 + 6y_2, & y_1 + y_2 \leq 2, \quad y_1 + 3y_2 \leq 3, \\ -\infty, & \text{otherwise.} \end{cases}$$

The dual maximizes $4y_1 + 6y_2$ subject to these two inequalities and $y_1, y_2 \geq 0$. Keeping the nonnegative domain is why the coefficient conditions are inequalities; with an unrestricted real x, each coefficient would have to equal zero.

Solve and check the primal. At the intersection of the two requirement boundaries, subtract $x_1 + x_2 = 4$ from $x_1 + 3x_2 = 6$. This gives $2x_2 = 2$, so $x_2 = 1$ and $x_1 = 4 - 1 = 3$. Both coordinates are nonnegative, both requirements hold at equality, and the cost is $2(3) + 3(1) = 9$. The other boundary vertices are $(6, 0)$ and $(0, 4)$, each with cost 12; the certificate below proves global optimality without relying on a vertex argument for the unbounded feasible region.

Solve and check the dual. Making both dual inequalities tight gives $y_1 + y_2 = 2$ and $y_1 + 3y_2 = 3$. Subtraction yields $2y_2 = 1$, so $y_2 = 1/2$ and $y_1 = 3/2$. They are nonnegative, and the objective is $4(3/2) + 6(1/2) = 6 + 3 = 9$. The remaining vertices $(0, 0), (2, 0), (0, 1)$ have values $0, 8, 6$.

Prove the matching points optimal. For every primal-feasible x and dual-feasible y, the difference between their objectives is

$$\begin{aligned} 2x_1 + 3x_2 - (4y_1 + 6y_2) = (2 - y_1 - y_2)x_1 + (3 - y_1 - 3y_2)x_2 \\ + y_1(x_1 + x_2 - 4) + y_2(x_1 + 3x_2 - 6) \geq 0. \end{aligned}$$

Every displayed term is nonnegative by its corresponding feasibility condition. The feasible points $(3, 1)$ and $(3/2, 1/2)$ have equal value 9, so weak duality proves $p^\star = d^\star = 9$ and both are optimal.

Check complementary slackness. At this pair the products are $(3/2)(3+1-4)=0$ and $(1/2)(3+3-6)=0$ for the primal requirements, and $3(2-3/2-1/2)=0$ and $1(3-3/2-3/2)=0$ for the nonnegative-domain coordinates. Both Lagrangian coefficients vanish, so $(3,1)$ minimizes $L(\cdot, y^\star)$ over the domain.

Interpret the prices within their range. With requirements r_1, r_2, the same active intersection gives $x_2=(r_2-r_1)/2$ and $x_1=(3r_1-r_2)/2$. While $r_1 \le r_2 \le 3r_1$, these coordinates stay nonnegative and the cost becomes $(3/2)r_1+(1/2)r_2$. Thus a unit increase of the first requirement from 4 to 5 adds 1.5 to the cost, and a unit increase of the second from 6 to 7 adds 0.5, with the other requirement fixed in each calculation. Larger changes need a new active-set check.

▼ **Exercise 2.3 — Schur-complement the bounded-real Riccati inequality (derivation)**

Let $\gamma^2 I - D^\top D \succ 0$. Show that the Riccati inequality

$$A^\top P + PA + C^\top C + (PB + C^\top D)(\gamma^2 I - D^\top D)^{-1}(B^\top P + D^\top C) \preceq 0$$

is equivalent to the 2×2 block LMI of the ℓ_2-gain walkthrough, and that this is in turn equivalent to

$$\begin{bmatrix} A^\top P + PA & PB & C^\top \\ B^\top P & -\gamma^2 I & D^\top \\ C & D & -I \end{bmatrix} \preceq 0.$$

▼ **Show answer**

First equivalence. Put $X = A^\top P + PA + C^\top C, Y = PB + C^\top D, Z = D^\top D - \gamma^2 I \prec 0$. The negative-form Schur complement (Section 3, $(2,2)$ block negative definite) gives

$$\begin{bmatrix} X & Y \\ Y^\top & Z \end{bmatrix} \preceq 0 \iff X - YZ^{-1}Y^\top \preceq 0 \iff X + Y(\gamma^2 I - D^\top D)^{-1}Y^\top \preceq 0,$$

using $Z^{-1} = -(\gamma^2 I - D^\top D)^{-1}$. The last expression is the Riccati inequality.

Second equivalence. In the 3×3 block matrix the $(3,3)$ block is $-I \prec 0$. Its negative-form Schur complement is

$$\begin{bmatrix} A^\top P + PA & PB \\ B^\top P & -\gamma^2 I \end{bmatrix} - \begin{bmatrix} C^\top \\ D^\top \end{bmatrix} (-I)^{-1} \begin{bmatrix} C & D \end{bmatrix} = \begin{bmatrix} A^\top P + PA + C^\top C & PB + C^\top D \\ B^\top P + D^\top C & D^\top D - \gamma^2 I \end{bmatrix},$$

exactly the 2×2 block LMI. Note the direction of the trade: the Riccati form is quadratic in P, the 2×2 form is linear in P but contains the constant products $C^\top C$, $D^\top D$, and the 3×3 form is jointly affine in (P, C, D, γ^2) with A, B fixed, which is what one types into a solver and what allows C and D (or W_l in LipSDP) to become variables.

▼ **Exercise 2.4 — The S-lemma bounds a quadratic over an ellipsoid**

Let $P \succ 0$ and $B \in \mathbb{S}^n$. (a) Use the S-lemma to show that $\max\{x^\top B x : \ x^\top P x \le 1\} = \min\{\tau \ge 0 : \ \tau P - B \succeq 0\} = \max\{0, \ \lambda_{\max}(P^{-1/2} B P^{-1/2})\}$. (b) Evaluate for $P = \operatorname{diag}(1, 4)$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and confirm with Cauchy–Schwarz.

▼ **Show answer**

(a) Turn an upper bound into a quadratic implication. Since $P \succ 0$, the ellipsoid $x^\top P x \le 1$ is nonempty, closed and bounded: $x^\top P x \ge \lambda_{\min}(P)\|x\|_2^2$ with $\lambda_{\min}(P) > 0$. The continuous objective therefore attains its maximum. A real number t is an upper bound on that maximum exactly when

$$1 - x^\top P x \ge 0 \quad \Longrightarrow \quad t - x^\top B x \ge 0.$$

The premise is strictly satisfied at $x = 0$. The one-constraint, inhomogeneous S-lemma consequently applies in both directions: the implication holds if and only if some $\tau \ge 0$ makes

$$t - x^\top B x - \tau(1 - x^\top P x) = x^\top(\tau P - B)x + (t - \tau) \ge 0 \quad \text{for every } x.$$

This residual is globally nonnegative exactly when $\tau P - B \succeq 0$ and $t - \tau \ge 0$. Necessity of the constant condition follows by setting $x = 0$. If the matrix had a negative direction v, taking $x = sv$ with $|s|$ large would make the residual negative, proving necessity of its PSD condition. The two conditions are also sufficient by direct addition.

For each admissible τ, the smallest allowed t is τ. Hence

$$\max_{x^\top P x \le 1} x^\top B x = \min_{\tau \ge 0,\, \tau P - B \succeq 0} \tau.$$

Compute that multiplier. Set $S = P^{-1/2} B P^{-1/2}$, where $P^{-1/2}$ is the symmetric inverse square root of P. Congruence by this invertible matrix preserves PSD and gives

$$\tau P - B \succeq 0 \iff \tau I - S \succeq 0 \iff \tau \ge \lambda_{\max}(S).$$

The last equivalence follows because the eigenvalues of $\tau I - S$ are $\tau - \lambda_i(S)$. Together with $\tau \ge 0$, the smallest admissible value is $\tau^\star = \max\{0, \lambda_{\max}(S)\}$, and it is attained.

Check the boundary cases and attainment directly. If $\mu = \lambda_{\max}(S) > 0$, choose a unit eigenvector v with $Sv = \mu v$ and $x = P^{-1/2} v$. Then $x^\top P x = 1$ and $x^\top B x = \mu$. If $\mu \le 0$, every objective is nonpositive and $x = 0$ attains 0. In particular, when $B \prec 0$, omitting the maximum with zero would incorrectly report a negative value even though the origin is feasible. When $\mu = 0$, nonzero null directions may also attain zero. The S-lemma multiplier is the Lagrange multiplier and these matching bounds establish zero duality gap.

(b) Substitute the given matrices. Here $P^{-1/2} = \operatorname{diag}(1, 1/2)$, so

$$S = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{bmatrix}, \qquad \operatorname{tr} S = \frac{5}{4}, \qquad \det S = \frac{1}{4} - \frac{1}{4} = 0.$$

The characteristic polynomial is $\lambda(\lambda - 5/4)$, so the eigenvalues are 0 and $5/4$. The maximum is $5/4 = 1.25$. The Cauchy–Schwarz check uses $u = (x_1, 2x_2)$ and $v = (1, 1/2)$:

$$x^\top B x = (x_1 + x_2)^2 = \big(x_1 + (2x_2)/2\big)^2 \le (x_1^2 + 4x_2^2)\left(1 + \frac{1}{4}\right) \le \frac{5}{4}.$$

Equality requires u proportional to v and the ellipsoid constraint active. Taking $u = (2/\sqrt{5})v$ gives $x = (2/\sqrt{5}, 1/(2\sqrt{5}))^\top$; its constraint is $4/5 + 1/5 = 1$ and its objective is $(5/(2\sqrt{5}))^2 = 5/4$. Its negative is another maximizer. As a final certificate check, $\tau^\star P - B = \begin{bmatrix} 1/4 & -1 \\ -1 & 4 \end{bmatrix} = (1/2, -2)^\top (1/2, -2) \succeq 0$.

▼ **Exercise 2.5 — The slope-restriction QC for tanh**

(a) Show that $\tanh$ is slope-restricted in $[0, 1]$ and sector-bounded in $[0, 1]$, and that $\alpha = 0$ cannot be improved globally. (b) Write the incremental QC with $\alpha = 0$, $\beta = 1$ and verify it numerically for $a = 1, b = -1$. (c) On the interval $|v| \leq \bar{v}$, what local slope bound $\alpha_{\text{loc}} > 0$ holds, and why does it help in Module 14?

▼ Show answer

(a) Use the derivative to control every chord. For $\varphi(v) = \tanh v$, $\varphi'(v) = 1 - \tanh^2 v$. Since $|\tanh v| < 1$ at every finite v, $0 < \varphi'(v) \le 1$. For $a \neq b$, the mean value theorem gives a point c between a and b such that

$$\frac{\tanh a - \tanh b}{a - b} = \varphi'(c) \in (0, 1] \subset [0, 1].$$

Thus tanh is slope-restricted in $[0, 1]$. Set $b = 0$ and use $\tanh 0 = 0$. For $v \neq 0$, multiply $0 \le \tanh(v)/v \le 1$ by the nonnegative quantity v^2, obtaining $0 \le v \tanh(v) \le v^2$. At $v = 0$ both sides vanish. This proves the origin-centered sector bound without multiplying an inequality by a possibly negative v.

No positive global lower slope works. For the chord from N to $N + 1$, the mean value theorem evaluates the derivative at $c \ge N$; it is at most $1 - \tanh^2 N$, which tends to zero as $N \to \infty$. Hence some chord violates any proposed $\alpha > 0$. A positive global lower sector bound also fails, because $0 \le \tanh v / v \le 1/v \to 0$ as $v \to +\infty$.

(b) Expand the incremental QC. Write $\Delta v = a - b$ and $\Delta\varphi = \tanh a - \tanh b$. For $\Delta v \neq 0$, let $s = \Delta\varphi / \Delta v \in [0, 1]$. Then

$$2\Delta v\, \Delta\varphi - 2\Delta\varphi^2 = 2(\Delta v)^2 s(1 - s) \ge 0.$$

When $a = b$, both increments vanish, so the same inequality holds. Its quadratic-form representation is

$$\begin{bmatrix} \Delta v \\ \Delta\varphi \end{bmatrix}^{\top} \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} \Delta v \\ \Delta\varphi \end{bmatrix} = 2\Delta v\, \Delta\varphi - 2\Delta\varphi^2 \ge 0.$$

At $a = 1, b = -1$, oddness gives $\Delta v = 2$ and $\Delta\varphi = 2\tanh 1 \approx 1.523188312$. Substitution gives $2(2)(1.523188312) - 2(1.523188312)^2 \approx 1.45255 > 0$. In exact terms the expression is $8 \tanh 1 (1 - \tanh 1)$, which is positive because $0 < \tanh 1 < 1$.

(c) Distinguish a local slope bound from a local sector. For a finite $\bar{v} \ge 0$, the derivative on $[-\bar{v}, \bar{v}]$ is at least $1 - \tanh^2 \bar{v}$. Thus every chord whose endpoints lie in that interval satisfies

$$\alpha_{\text{loc}} \le \frac{\tanh a - \tanh b}{a - b} \le 1, \qquad \alpha_{\text{loc}} = 1 - \tanh^2 \bar{v} > 0.$$

For $\bar{v} > 0$, a stronger lower bound is available for chords specifically from the origin, which is the non-incremental sector used in the stability argument. To see it, define $r(v) = \tanh(v)/v$ for $v > 0$. Its derivative is $r'(v) = [v\varphi'(v) - \varphi(v)]/v^2$. Since φ' decreases on $[0, \infty)$, $\varphi(v) = \int_0^v \varphi'(u)\, du \geq v\varphi'(v)$; therefore $r'(v) \leq 0$. Oddness makes the ratio the same at v and $-v$. Hence on the interval the origin-centered sector is

$$\frac{\tanh \bar{v}}{\bar{v}}\, v^2 \leq v \tanh v \leq v^2.$$

The mean value theorem on $[0, \bar{v}]$ also gives $\tanh \bar{v}/\bar{v} \geq 1 - \tanh^2 \bar{v}$. For $\bar{v} = 1$, these bounds are approximately 0.761594 and 0.419974, respectively. At $\bar{v} = 0$ the interval contains only zero; there are no distinct endpoint chords, and the ratio has limiting value 1.

Explain the role in Module 14. A positive lower slope or sector restricts the allowable pairs more than the global $[0, 1]$ information does. Its QC matrix is $\begin{bmatrix} -2\alpha\beta & \alpha + \beta \\ \alpha + \beta & -2 \end{bmatrix}$ with $\beta = 1$; it can yield a less conservative certificate for the fixed interval. The certificate must also prove that all relevant activation inputs stay in that interval, using the containment LMI described on this page. Around a nonzero equilibrium, the offset local sector must be used. A larger certified region is not automatic: a smaller interval tightens the sector while making interval containment harder.

▼ **Exercise 2.6 — The ℓ_2-gain LMI of a scalar discrete-time system by hand (derivation)**

Take $x_{t+1} = ax_t + bw_t$, $z_t = cx_t$ with $a = 0.5, b = 1, c = 1$. (a) Derive the dissipativity LMI for $V = px^2$ and the supply $\gamma^2 w^2 - z^2$. (b) Show that $\gamma = 2$ is certified with $p = 2$. (c) Show that no $\gamma < 2$ can be certified, and reconcile this with the frequency response.

▼ **Show answer**

(a) Expand the storage balance before forming the matrix. With $V(x) = px^2$, nonnegative storage requires $p \geq 0$. The gain supply is $\gamma^2 w^2 - z^2$, with $\gamma \geq 0$. Substitute $x^+ = ax + bw$ and $z = cx$:

$$\begin{aligned} V(x^+) - V(x) - (\gamma^2 w^2 - z^2) &= p(ax + bw)^2 - px^2 - \gamma^2 w^2 + c^2 x^2 \\ &= (pa^2 - p + c^2)x^2 + 2pab\, xw + (pb^2 - \gamma^2)w^2. \end{aligned}$$

A symmetric quadratic form has twice its off-diagonal entry as the mixed-term coefficient. Therefore the balance is nonpositive for every real x, w exactly when

$$M(p, \gamma) = \begin{bmatrix} pa^2 - p + c^2 & pab \\ pab & pb^2 - \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 - \frac{3}{4}p & \frac{1}{2}p \\ \frac{1}{2}p & p - \gamma^2 \end{bmatrix} \preceq 0, \qquad p \geq 0.$$

With the system coefficients fixed this is affine in p and γ^2, so it is an LMI in those two unknowns.

(b) Check the proposed certificate. At $p = 2, \gamma = 2$,

$$M = \begin{bmatrix} -1/2 & 1 \\ 1 & -2 \end{bmatrix}, \qquad (x, w)M(x, w)^\top = -\frac{1}{2}(x - 2w)^2 \leq 0.$$

This perfect-square identity proves negative semidefiniteness directly. Alternatively, the trace is $-5/2$ and the determinant is $(-1/2)(-2) - 1 = 0$, giving eigenvalues $-5/2$ and 0. Sum the resulting dissipation inequality from $t = 0$ to $T - 1$:

$$2x_T^2 - 2x_0^2 \leq 4\sum_{t=0}^{T-1} w_t^2 - \sum_{t=0}^{T-1} z_t^2, \qquad \sum_{t=0}^{T-1} z_t^2 \leq 4\sum_{t=0}^{T-1} w_t^2 + 2x_0^2.$$

The last step drops the nonnegative terminal storage. From $x_0 = 0$, taking square roots gives norm gain at most 2; nonzero initial state contributes the separate storage term. For square-summable input, monotone finite-horizon energy sums give the same infinite-horizon bound.

(c) Exclude every smaller certificate. A symmetric 2×2 matrix is negative semidefinite exactly when both diagonal entries are nonpositive and its determinant is nonnegative. The first diagonal condition is $1 - 3p/4 \leq 0$, hence $p \geq 4/3$. At $p = 4/3$ the determinant is $0 \cdot (p - \gamma^2) - (p/2)^2 = -4/9 < 0$, so this boundary is impossible. Thus $p > 4/3$.

The determinant condition now reads

$$\left(1-\frac{3}{4}p\right)(p-\gamma^2)-\frac{1}{4}p^2\geq 0 \iff \frac{3p-4}{4}(\gamma^2-p)\geq\frac{1}{4}p^2 \iff \gamma^2\geq p+\frac{p^2}{3p-4}.$$

Here $3p-4>0$, so the last division preserves the inequality. This bound also implies the second diagonal condition $\gamma^2\geq p$. To minimize it without skipping a calculus step, subtract 4 and put terms over a common denominator:

$$p+\frac{p^2}{3p-4}-4=\frac{(p-4)(3p-4)+p^2}{3p-4}=\frac{4p^2-16p+16}{3p-4}=\frac{4(p-2)^2}{3p-4}\geq 0.$$

Equality occurs exactly at $p=2$, which lies in the allowed interval. Consequently every feasible certificate has $\gamma^2\geq 4$, and the certificate in part (b) attains the smallest gain bound $\gamma^\star=2$.

Reconcile with the frequency response. The stable system has $H(z)=1/(z-0.5)$. On the unit circle $z=e^{j\omega}$,

$$|e^{j\omega}-0.5|^2=(\cos\omega-0.5)^2+\sin^2\omega=\frac{5}{4}-\cos\omega\geq\frac{1}{4}.$$

Therefore $|H(e^{j\omega})|\leq 2$, with equality at $\omega=0$. Its peak frequency gain is exactly 2. Since $|a|=0.5<1$, the stable linear-system energy-gain theorem applies and this peak equals the induced ℓ_2 gain from zero initial state. Thus the unrestricted scalar quadratic-storage LMI is exact here. Nonlinear QC abstractions and restrictions on the storage can introduce conservatism in other problems.

Key Papers

Paper	Venue	Contribution	Why read it
Boyd, El Ghaoui, Feron & Balakrishnan — Linear Matrix Inequalities in System and Control Theory	SIAM Studies in Applied Mathematics 15, 1994	The book that defined the toolbox: Schur complements, S-procedure, Lyapunov, Riccati and bounded-real LMIs, interior-point solution.	Free PDF. Chapter 2 for LMIs, interior-point methods, elimination and the S-procedure; the later chapters catalogue the classical control LMIs (Lyapunov, Lur'e systems, IQCs, synthesis) behind Modules 11–14.
Boyd & Vandenberghe — Convex Optimization	Cambridge University Press,	Lagrangian duality, KKT, Slater, sensitivity, SDPs	Chapter 5 is the source of Section 1; Sections 5.1–5.6 in

Paper	Venue	Contribution	Why read it
	2004	and interior-point methods.	one sitting.
Pólik & Terlaky — A Survey of the S-Lemma	SIAM Review 49(3):371–418, 2007	Complete account of the S-lemma: several proofs (including via Dines), extensions, and where it fails with two or more constraints.	The reference to open before believing any "the S-procedure is lossless here" claim.
Dines — On the mapping of quadratic forms	Bull. AMS 47(6):494–498, 1941	The joint range of two real quadratic forms is a convex cone.	A five-page note; the classical fact that makes the S-lemma proof a separation argument.
Willems — Dissipative dynamical systems part I: General theory	Arch. Rational Mech. Anal. 45(5):321–351, 1972	Storage functions, supply rates, available storage and required supply.	The origin of the vocabulary of Section 6 and of every "storage matrix" in LipKernel.
Rantzer — On the Kalman–Yakubovich–Popov lemma	Systems & Control Letters 28(1):7–10, 1996	A short, complete proof of the KYP lemma with the controllability condition made explicit.	Four pages, including the hard direction that Section 6 only sketches.
Megretski & Rantzer — System analysis via integral quadratic constraints	IEEE TAC 42(6):819–830, 1997	Unifies small-gain, passivity, circle/Popov and multiplier criteria as IQCs; stability theorem by homotopy.	The robust-control foundation of every IQC-based neural-network certificate.
Scherer — Dissipativity and Integral Quadratic Constraints: Tailored Computational Robustness Tests for Complex Interconnections	IEEE Control Systems Magazine 42(3):115–139, 2022	Tutorial: factorised multipliers $\Pi = \Psi^* P \Psi$, hard and soft IQCs, LMI tests through dissipativity plus the S-procedure.	Read before Module 14; it is the machinery Pauli, Gramlich and Scherer use for NN loops and 2-D systems.
Fazlyab, Robey, Hassani, Morari & Pappas — Efficient and Accurate Estimation of Lipschitz Constants for Deep Neural Networks (LipSDP)	NeurIPS 2019	Slope restriction as an incremental QC, S-procedure with diagonal multipliers, Lipschitz estimation as an SDP.	The destination of the walkthrough. Use the current arXiv version, which keeps only diagonal T.
Pauli, Koch, Berberich, Kohler & Allgöwer — Training robust neural	IEEE L-CSS 6:121–126,	Counterexample to coupled multipliers; Schur-complemented training	Step 6 of the walkthrough and the diagonal-multiplier caveat,

Paper	Venue	Contribution	Why read it
networks using Lipschitz bounds	2022 (also ACC 2021)	LMI with fixed T and $\alpha = 0$; ADMM training.	from the paper that established it.
Paternain, Chamon, Calvo-Fullana & Ribeiro — Constrained Reinforcement Learning Has Zero Duality Gap	NeurIPS 2019	Zero duality gap for CMDPs under Slater, via concavity of the perturbation function on the convex set of occupancy measures.	The destination of Section 1: the geometric picture of the duality gap, made rigorous for a non-convex problem.
Yin, Seiler & Arcak — Stability Analysis Using Quadratic Constraints for Systems With Neural Network Controllers	IEEE TAC 67(4):1980–1987, 2022	Lyapunov function plus local sector QCs plus a Schur-complement LMI for the region of attraction of an LTI plant with an NN controller.	Sections 3(a) and 5 in action on a real closed loop; the template Module 14 extends.
Pauli, Wang, Manchester & Allgöwer — LipKernel: Lipschitz-Bounded Convolutional Neural Networks via Dissipative Layers	Automatica 188 (2026) 112959	Layer-wise incremental dissipativity with interface matrices; telescoping yields Lipschitz-by-design CNNs with fast inference.	The composition theorem of Section 6 used at scale; note its ρ is the constant, not its square.

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Weak duality: statement and the one-line reason.

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